CVT System Model Derivation

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Abstract

This work develops a first-principles dynamic model for a mechanically actuated dry rubber V-belt continuously variable transmission (CVT) and its coupling to the vehicle drivetrain. The model is motivated by a persistent gap in the available literature: while quasi-static Baja-oriented models are common and detailed dynamic models exist for hydraulically actuated metal-belt CVTs, publicly available formulations for mechanically actuated rubber belt CVTs do not generally combine sheave inertia, flyweight actuation, torque-reactive secondary clamping, belt geometric closure, traction-limited torque transmission, and vehicle longitudinal dynamics within a single unified framework.

Starting from pulley geometry and belt-length closure, the derivation develops the CVT kinematics, the rotational dynamics of the primary and secondary pulleys, the axial equation of motion governing shift, and the torque-transfer closure required to distinguish sticking from traction-limited slip. The primary clutch model includes centrifugal flyweight actuation and spring preload, while the secondary model includes helix-generated torque feedback together with spring forces. A centrifugal belt correction is also derived to account for the reduction in effective normal force at speed. These elements are assembled into a closed nonlinear state-space model suitable for direct numerical time integration.

The resulting formulation is intended both as a simulation model and as a derivation document in which each major modeling step remains physically interpretable. Simulation results are used to examine whether the completed model behaves in mechanically sensible ways under transient launch conditions. Comparisons of launch shift trajectories, pulley axial-force balance, and torque admissibility across multiple tuning cases show a consistent mechanism chain from tuning changes, to clamping-force evolution, to stick–slip behavior, to overall launch outcome. The default tuning produces the most balanced launch response, combining strong early admissible coupling torque, controlled low-ratio retention, and a smoother main shift. More aggressive cases reveal renewed slip and visibly underdamped oscillatory transients during shift onset, highlighting both the usefulness of the model and the limitations introduced by idealized loss assumptions.

Overall, the formulation provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behavior during launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behavior remains traceable back to clear mechanical causes.

Contents

Nomenclature and Notation

All quantities are expressed in SI units unless otherwise stated. Angles are expressed in radians. Angular velocities are expressed in rad/s. Torques are expressed in N m, forces in N, lengths in m, and masses in kg.

Symbol Summary

Symbols used throughout the derivation are summarized in Table 1.

Symbol	Meaning
State and kinematics	
$\boldsymbol{x}(t)$	state vector
t	time
$\omega_p,\ \omega_s$	primary and secondary angular speed
$s,\ \dot{s},\ \ddot{s}$	shift position, velocity, and acceleration
$R,\ \dot{R}$	CVT ratio and ratio rate
v_b	belt transport speed
$v_b^*,\ T_b$	slip-branch target belt speed and belt-speed relaxation time constant
v_Δ	slip metric: primary-imposed minus secondary-imposed belt-line speed
Geometry	
$r_{p,\mathrm{outer}}(s),\ r_{s,\mathrm{outer}}(s)$	primary and secondary outer radius
$r_{p,\mathrm{eff}}(s),\ r_{s,\mathrm{eff}}(s)$	primary and secondary effective torque radius
$r_{p,\mathrm{outer,min}}$	minimum primary outer radius
$r_{s,\mathrm{outer,max}}$	maximum secondary outer radius
$s_{\mathrm{dz}},\ s_{\max}$	deadzone threshold and maximum shift
β	sheave face half-angle (radial reference)
C	pulley center distance
L_b	total belt length
ϕ	belt wrap angle used in distributed contact model
$r,\ \dot{r}$	local belt path radius and its time derivative
r_{cm}	belt element center-of-mass radius
Constraint and mapping functions	
$f(r_p, r_s, C)$	belt-length compatibility root function ($f = 0$)
$\mathrm{root}_{r_s}(\cdot)$	numerical root solve for secondary radius
$r_f(s),\ \dfrac{dr_f}{ds}$	primary flyweight radius map and gradient
$\theta_s(s),\ \dfrac{d\theta_s}{ds}$	secondary helix angle map and gradient

Symbol	Meaning
Forces and torques	
$F_{p,\text{ax}},\ F_{s,\text{ax}}$	primary and secondary axial mechanism force
$F_{c,p},\ F_{c,s}$	primary and secondary belt centrifugal axial contribution
F_{ax}	generic axial clamp force at one pulley contact
$\tau_{\text{eng}},\ \tau_{\text{load}}$	engine input torque and reflected load torque
$\tau_p,\ \tau_s$	coupling torque seen at primary and secondary
$\tau,\ \tau_c$	generic interface torque and applied coupling/friction torque
τ_{ns}	no-slip requested coupling torque
$\tau_-^{\text{stick}},\ \tau_+^{\text{stick}}$	aggregate lower and upper coupling-torque bounds for stick admissibility
$\tau_-^{\text{slip}},\ \tau_+^{\text{slip}}$	aggregate lower and upper coupling-torque bounds on slip branches
$\tau_{c,p,\pm}^{\text{stick}},\ \tau_{c,s,\pm}^{\text{stick}}$	primary/secondary pulley-wise stick-branch lower/upper bounds
$\tau_{c,p,\pm}^{\text{slip}},\ \tau_{c,s,\pm}^{\text{slip}}$	primary/secondary pulley-wise slip-branch lower/upper bounds
Mass/inertia and material parameters	
$I_p,\ I_s$	primary and secondary rotational inertia
m_f	effective flyweight mass
$m_{p,m},\ m_{s,m}$	equivalent primary and secondary translating mechanism mass
ρ_b	belt material density
A_b	belt cross-sectional area
μ	belt–pulley traction coefficient
$k_p,\ x_{p,0}$	primary spring stiffness and preload displacement
$k_{s,\theta},\ \theta_{s,0}$	secondary torsional stiffness and preload angle
$k_{s,x},\ x_{s,0}$	secondary axial stiffness and preload displacement

Table 1: Core symbols used throughout the derivation.

Constants and Parameters

All parameters in Table 2 are treated as constant values that do not vary through the model. They are either physical constants, fixed by the physical design of the CVT and vehicle or else are tuning parameters that are set and held constant during a given simulation run.

Symbol	Name	Unit	Description
Geometry and belt			
L_b	Belt length	m	Fixed total belt length
β	Sheave half-angle	rad	Cone half-angle used for radial/axial conversion
h_b	Belt thickness	m	Radial belt thickness
b_{out}	Belt outer width	m	Belt width at outer face
b_{in}	Belt inner width	m	Belt width at inner face
$r_{p,\text{outer,min}}$	Primary minimum outer radius	m	Radius at fully open primary
$r_{s,\text{outer,max}}$	Secondary maximum outer radius	m	Radius at fully closed secondary
s_{dz}	Shift deadzone threshold	m	Onset of effective radial motion

Symbol	Name	Unit	Description
$s_{\max}$	Maximum shift	m	Physical upper travel bound
Primary/secondary actuation			
m_f	Flyweight mass	kg	Effective flyweight mass
$r_{f,0}$	Initial flyweight radius	m	Flyweight radius offset at $s = 0$
k_p	Primary spring stiffness	N/m	Primary compression spring rate
$x_{p,0}$	Primary spring preload	m	Initial compression of primary spring
$k_{s,\theta}$	Secondary torsional stiffness	N m/rad	Torsional spring rate in helix assembly
$\theta_{s,0}$	Secondary torsional preload	rad	Initial torsional deflection preload
r_h	Helix reference radius	m	Effective cylindrical radius in helix kinematics
$k_{s,x}$	Secondary axial stiffness	N/m	Secondary axial compression spring rate
$x_{s,0}$	Secondary axial preload	m	Initial compression of secondary axial spring
Inertia and drivetrain			
I_{eng}	Engine inertia	kg m^2	Engine rotational inertia
I_{prim}	Primary pulley inertia	kg m^2	Primary CVT pulley inertia
I_p	Effective primary inertia	kg m^2	Implemented primary-side inertia in current setup
I_{sec}	Secondary pulley inertia	kg m^2	Secondary CVT pulley inertia
I_{gb}	Gearbox inertia	kg m^2	Gearbox/intermediate shaft inertia
I_{wheel}	Wheel inertia	kg m^2	Wheel/hub inertia about wheel axis
I_s	Effective secondary inertia	kg m^2	Lumped secondary-side inertia about secondary axis
G	Final drive ratio	1	Fixed ratio defined by $\omega_s = G\,\omega_w$
m	Vehicle mass	kg	Total vehicle mass
r_w	Wheel rolling radius	m	Effective tire rolling radius
Contact and environment			
ρ_b	Belt density	kg/m^3	Belt material density
μ_s	Belt–pulley static friction coefficient	1	Static traction coefficient used at impending slip
μ_k	Belt–pulley kinetic friction coefficient	1	Kinetic traction coefficient used on slip branches
C_{rr}	Rolling resistance coefficient	1	Tire-road rolling resistance coefficient
C_d	Drag coefficient	1	Aerodynamic drag coefficient
A_f	Frontal area	m^2	Vehicle aerodynamic reference area
ρ	Air density	kg/m^3	Ambient air density used in drag model
g	Gravitational acceleration	m/s^2	Gravity constant

Table 2: Canonical non-varying parameters used by the CVT model equations.

Chapter 1

Introduction & Background

The continuously variable transmission (CVT) is a mechanical power transmission device capable of producing a continuously variable speed ratio between an input and output shaft, in contrast to the fixed discrete ratios of a conventional stepped gearbox. In the class of CVTs relevant to this work — dry rubber V-belt CVTs with mechanically actuated conical sheave pairs — the transmission ratio is not commanded by an external actuator or electronic controller. Instead, it is an emergent dynamic response: the equilibrium of axial forces acting on the primary and secondary sheave pairs, arising from centrifugal flyweight mechanisms, torque-reactive helix and ramp assemblies, and return springs, determines the radial position at which the belt seats within each sheave groove, and therefore the effective pitch radii and transmission ratio at every instant.

This self-regulating character is simultaneously the CVT's greatest practical asset and the primary source of difficulty in modeling it rigorously. Because no actuator explicitly commands the ratio, the ratio trajectory must emerge through the evolution of an internal shift degree of freedom whose motion is set by the coupled force balance of the pulleys and belt. In practice, this means that engine speed, transmitted torque, shift position, and traction state evolve together in time, with each affecting the others through the underlying mechanics.

This class of CVT is found throughout small-displacement powersport and utility vehicles. It is the mandated transmission in the SAE Baja competition series, where a 10 hp engine is paired with a CVT that must be designed, selected, and tuned entirely by student engineering teams from first principles. Beyond Baja, mechanically actuated rubber V-belt CVTs are standard equipment in snowmobiles, all-terrain vehicles (ATVs), utility task vehicles (UTVs), and small agricultural and industrial machinery. Despite their prevalence and the practical engineering importance of understanding their behavior, the quality of publicly available engineering models for these systems is strikingly uneven. The present work was motivated directly by this gap.

1.1 Problem and Motivation

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1.2 Conceptual Overview of a Continuously Variable Transmission

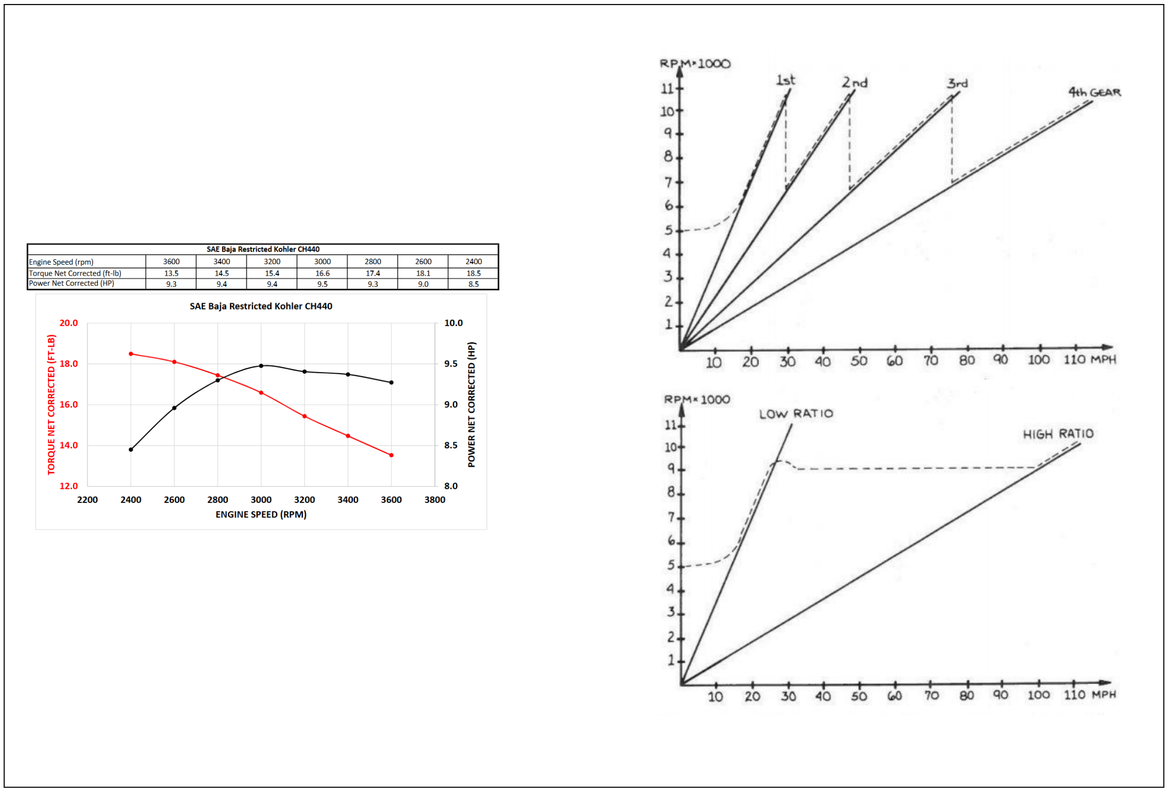
This section introduces the operating principles of the continuously variable transmission (CVT) considered in this work. The goal is not yet to derive the model equations, but to build the mechanical intuition needed to understand them. In particular, this section explains why a CVT is useful, how a rubber V-belt CVT changes ratio, why clamping and traction matter, how the primary and secondary pulleys generate their forces, and why the CVT must be understood as part of a larger drivetrain system.

1.2.1 Why Use a Continuously Variable Transmission?

A transmission is responsible for matching the operating speed of an engine to the speed required at the vehicle wheels. In a conventional stepped gearbox, this is achieved using a fixed set of discrete gear ratios. Each gear imposes a fixed relationship between engine speed and wheel speed. As the vehicle accelerates, the engine speed rises through one gear, drops when the transmission shifts, and then rises again in the next gear. This repeated sweep through the engine speed range is a direct consequence of having only a small number of available ratios.

This matters because an internal combustion engine (ICE) does not produce the same torque and power at every speed. Instead, the engine has a useful operating region where it can produce power more effectively, as shown conceptually in the left panel of Figure 1.1. A stepped gearbox can pass through this region, but it cannot generally stay there while vehicle speed continues to increase. The right panel of Figure 1.1 contrasts this stepped behavior with the smoother operating path made possible by a continuously variable transmission.

A continuously variable transmission (CVT) addresses this limitation by allowing the transmission ratio to vary smoothly within a finite range rather than changing through a fixed set of gear steps. Since the ratio can change continuously, the engine can remain closer to a desired operating region while the vehicle accelerates. This is the main benefit of a CVT: it gives the drivetrain more freedom to match engine behavior to vehicle load.



SAE Baja Restricted Kohler CH440							
Engine Speed (rpm)	3600	3400	3200	3000	2800	2600	2400
Torque Net Corrected (ft-lb)	13.5	14.5	15.4	16.6	17.4	18.1	18.5
Power Net Corrected (HP)	9.3	9.4	9.4	9.5	9.3	9.0	8.5

Figure 1.1: Conceptual motivation for a CVT. Internal combustion engines produce torque and power that vary with speed. A stepped gearbox forces the engine to move repeatedly through its speed range, while a continuously variable ratio can keep the engine closer to a useful operating region as vehicle speed changes.

1.2.2 Physical Layout and Single-Pulley Geometry

There are several forms of continuously variable transmission, including different mechanical layouts and different ways of producing the variable speed relationship. This work focuses on a belt-sheave CVT, where two variable-radius pulleys are connected by a rubber V-belt. The pulley connected to the engine side is called the primary (or driving) pulley. The pulley connected to the downstream drivetrain is called the secondary (or driven) pulley. The belt transmits motion and torque between these two pulleys.

Before considering the full two-pulley system, it is helpful to look at the geometry of a single pulley. Both the primary and secondary pulleys share the same basic anatomy. Each pulley is made from two opposing conical sheaves that form a V-shaped groove. One sheave is fixed axially to the shaft, while the other can move along the shaft. The rubber V-belt sits between the sheave faces. This basic layout is shown in Figure 1.2.

Placeholder: Single pulley anatomy

Draw one generic pulley cross-section, not specifically primary or secondary. Include and label:

- fixed sheave,
- movable sheave,
- shaft,
- V-groove,
- rubber V-belt cross-section,
- axial direction of movable-sheave motion,
- effective belt radius,
- sheave angle or cone angle, if visually useful.

The figure should make clear that both the primary and secondary pulleys use this same basic sheave-and-groove geometry.

Figure 1.2: Generic anatomy of a belt-sheave CVT pulley. Each pulley consists of opposing conical sheaves that form a V-groove. Motion of the movable sheave changes the groove width and therefore changes the radius at which the belt rides.

The key geometric idea is simple: changing the axial separation between the sheaves changes the effective radius of the belt. If the movable sheave moves toward the fixed sheave, the groove becomes narrower. Since the belt has a finite width, it cannot occupy the same radial position in the narrower groove, so it is pushed outward along the conical sheave faces. The effective belt radius therefore increases.

If the movable sheave moves away from the fixed sheave, the groove becomes wider and the belt can fall deeper between the sheaves. This reduces the effective belt radius. At this level, the mechanism is just wedge geometry: axial motion of the sheave becomes radial motion of the belt. Closing the sheaves makes the belt radius larger, while opening the sheaves makes the belt radius smaller, as shown in Figure 1.3.

Placeholder: Single pulley open, middle, and closed states

Show three cross-sections of the same generic pulley:

1. open sheaves: belt sits at small effective radius,
2. intermediate sheave spacing: belt sits at medium effective radius,
3. closed sheaves: belt is pushed outward to large effective radius.

Use the same fixed/movable sheave labels as Figure 1.2. Show the movable sheave translating axially and the belt moving radially.

Figure 1.3: Single-pulley radius change. As the sheaves close, the V-groove narrows and the belt is pushed outward to a larger effective radius. As the sheaves open, the belt can fall deeper in the groove and the effective radius decreases.

1.2.3 Two-Pulley Coupling and Ratio Change

The full CVT contains two of the variable-radius pulleys described above. The important difference is that the two pulleys are connected by the same belt. Because the belt length is approximately fixed, the effective radii of the two pulleys cannot change independently.

If the belt rides outward on the primary pulley, it must ride inward on the secondary pulley. Likewise, if the primary radius decreases, the secondary radius must increase. This opposite motion is the core geometric behavior that allows the CVT ratio to change. In the physical mechanism, the actual motion of the sheaves occurs through contact forces, springs, actuator mechanisms, and the axial force balance of the pulleys.

The CVT begins launch in a low-speed, high-reduction configuration. In this state, the primary pulley has a small effective radius and the secondary pulley has a large effective radius. This behaves like a low gear in a conventional drivetrain: the engine-side pulley turns relatively quickly while the downstream side turns more slowly, amplifying torque through the drivetrain.

As the CVT upshifts, the primary sheaves close and the primary effective radius increases. At the same time, the secondary effective radius decreases. The transmission therefore moves from its launch configuration toward a higher-speed configuration. By the end of the shift, the primary radius is larger and the secondary radius is smaller. This coupled motion is shown in Figure 1.4.

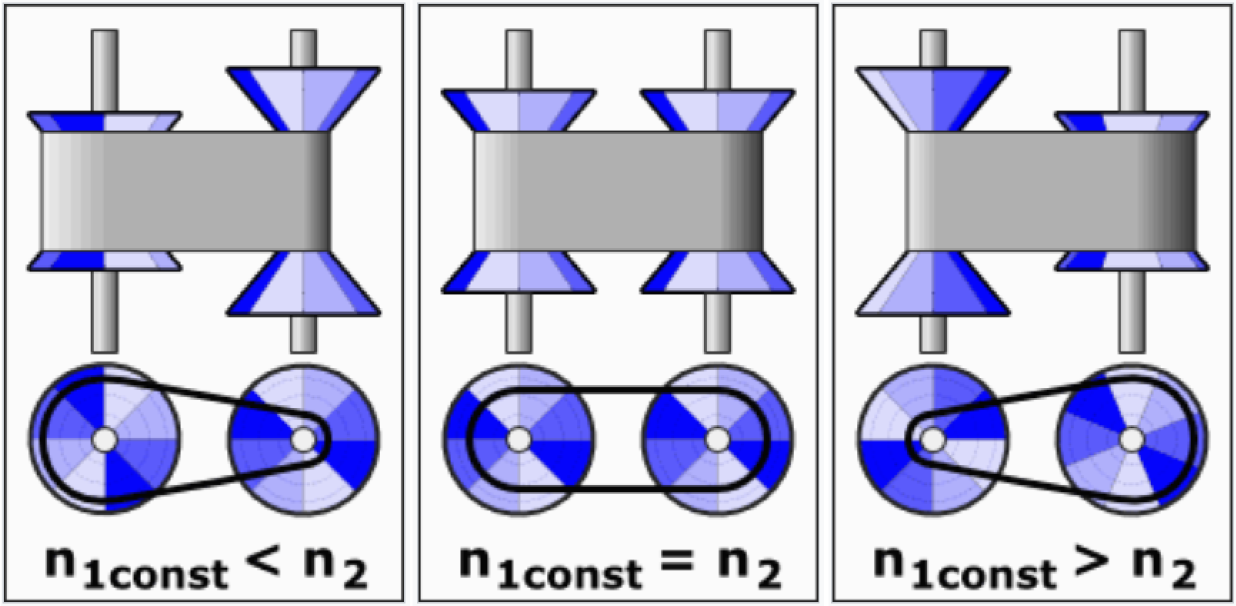


Figure 1.4: Coupled radius change in a belt-sheave CVT. Because the belt length is approximately fixed, an increase in primary effective radius is accompanied by a decrease in secondary effective radius. This coupled radius change produces the continuously variable transmission ratio.

1.2.4 Torque Transfer, Clamping Force, and Slip

The geometry described so far determines the speed relationship between the primary and secondary pulleys. However, a speed relationship alone does not guarantee that torque can be transmitted. Unlike a gear pair, the belt and pulley are not mechanically locked together by teeth. Torque is transmitted through frictional traction between the belt and the sheave faces.

For traction to develop, the belt must be physically squeezed between the sheave faces, as shown in Figure 1.5. The movable sheave presses the belt into the opposite sheave, producing a clamping force across the V-groove. This clamping force creates contact pressure on the belt faces and that contact pressure allows friction to transmit tangential force around the pulley. In practical terms, transferring more torque through the CVT requires enough clamping force to create enough available traction at the belt-pulley interface.

If the available traction is not sufficient for the torque being transferred, the belt slips relative to the pulley. Slip reduces useful torque transfer and can produce heat, wear, and efficiency loss. A working CVT must therefore do two things at once: it must allow the pulley radii to change so the ratio can shift, and it must maintain enough clamping force for the belt to transmit torque without excessive slip.

This distinction is central to the rest of the model. The pulley geometry determines how the speed ratio changes. The contact mechanics determine how much torque can actually be carried through the belt-pulley contacts. A CVT is therefore not only a variable-geometry mechanism; it is also a clamping and traction problem.

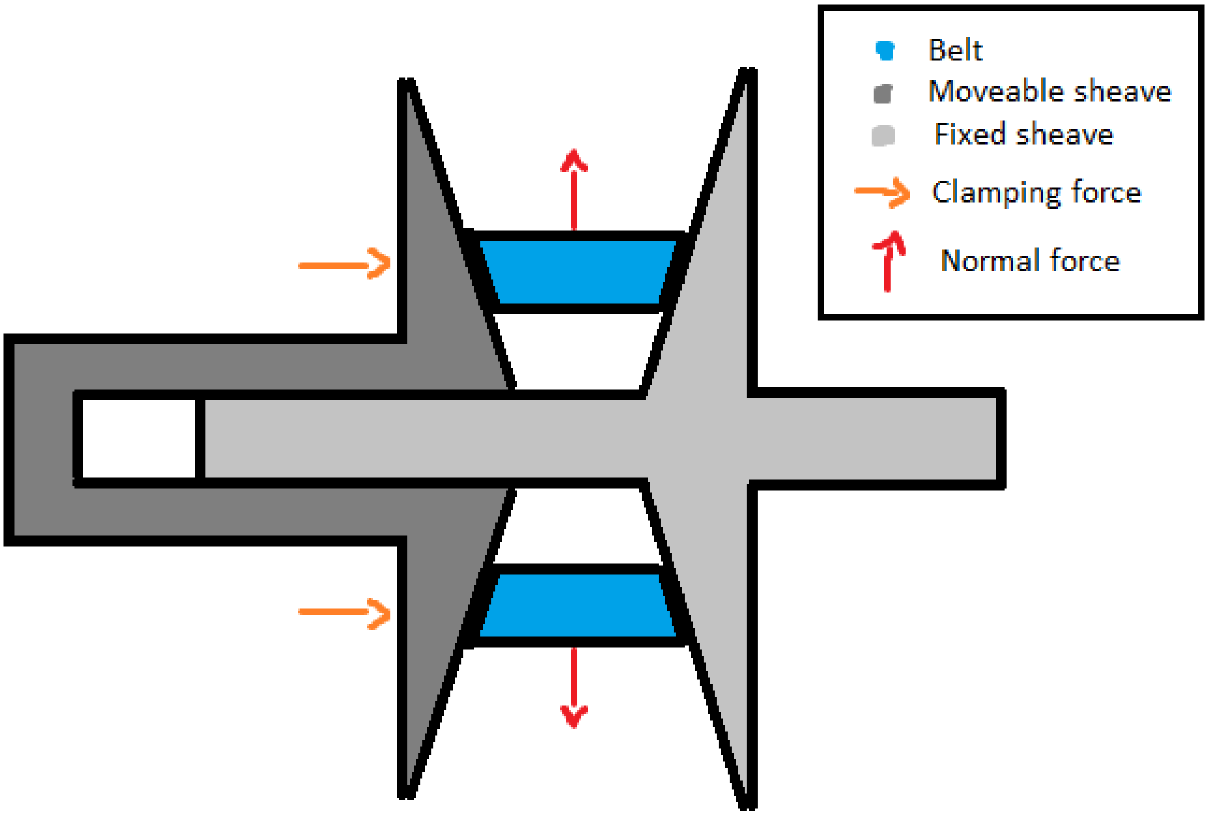


Figure 1.5: Clamping and traction in a rubber V-belt CVT. The belt must be pressed into the sheave faces to generate contact normal force. Frictional traction at this contact allows torque to be transmitted between the pulley and the belt; if the available traction is insufficient, the belt slips.

1.2.5 Mechanical Actuation in the Present CVT

Different CVTs generate radius change and clamping force in different ways. Some systems use electronic or hydraulic actuation to command pulley motion or clamping pressure directly. The system studied here is mechanically actuated: its pulley forces arise from mechanisms inside the primary and secondary pulley assemblies rather than from an externally commanded ratio.

These mechanisms are important because they determine both how the belt moves through the ratio range and how much clamping force is available for torque transfer. The primary pulley is mainly speed-sensitive, while the secondary pulley is mainly torque-reactive.

The primary pulley uses flyweights, ramps, and a spring. As engine speed increases, the flyweights move outward under centrifugal effects. The ramp surfaces convert this outward flyweight motion into a closing force on the primary sheaves. The spring resists this motion and helps set the speed at which the primary begins to close. As the primary closes, the belt is pushed outward, the primary effective radius increases, and the CVT shifts in the upshift direction introduced in Figure 1.4. A conceptual view of this mechanism is shown in Figure 1.6.

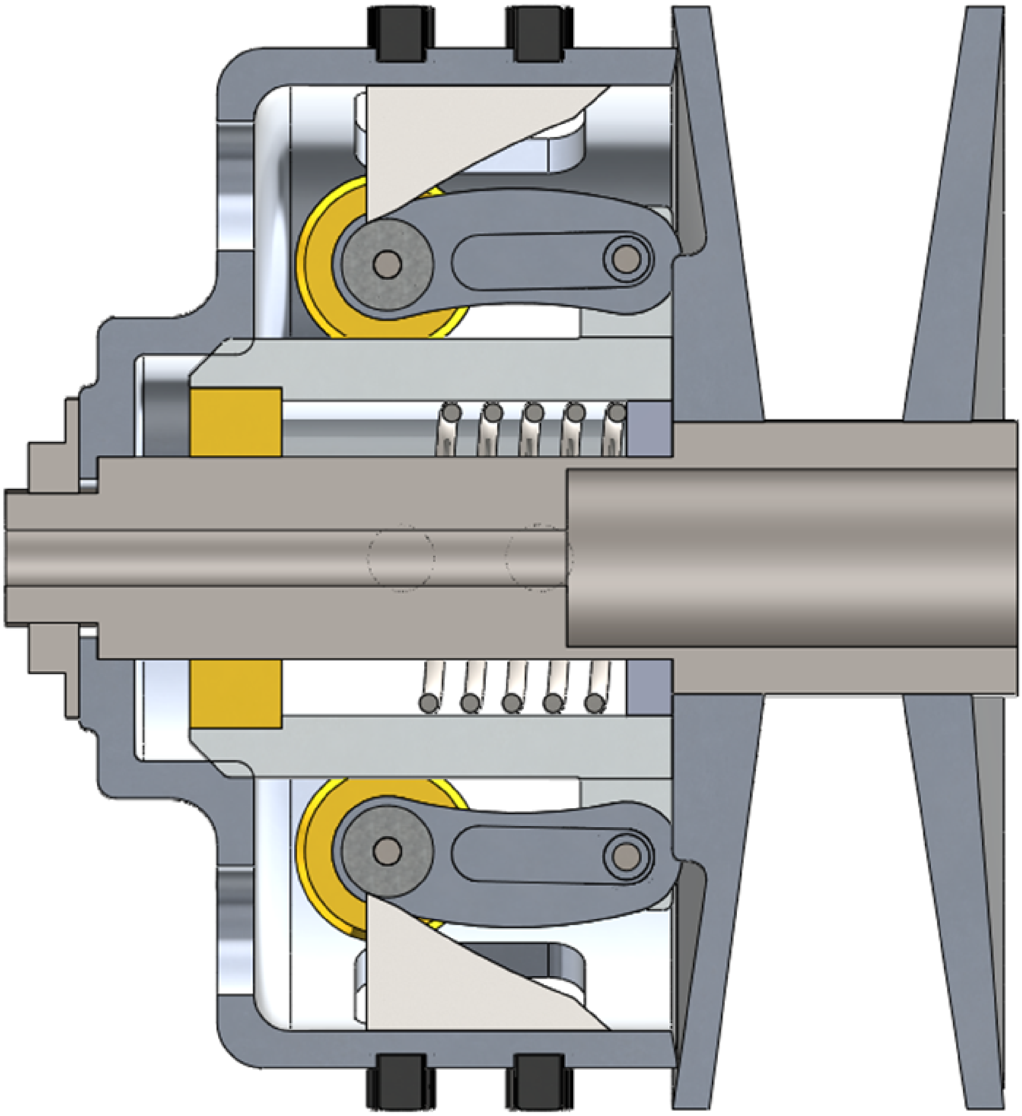


Figure 1.6: Primary pulley actuation concept. As primary speed increases, centrifugal effects drive the flyweights outward. The ramp geometry converts this motion into a closing force on the primary sheaves, while the spring resists that motion and helps set the shift behavior.

The secondary pulley uses a helix mechanism and spring loading. The helix can be understood like a screw-like interface: when torque is transmitted through the secondary, the angled helix surfaces convert part of that torque reaction into a clamping force between the sheaves. In this way, torque through the secondary helps create the clamping needed to carry torque through the belt. The spring provides preload and helps set the baseline clamping behavior.

The helix also means that secondary shift is not purely axial. As the secondary moves through its shift range, axial motion is coupled with relative rotation between secondary sheave components. This detail becomes important later when the secondary force model is developed, but the basic intuition is simply that the helix converts torque and relative rotation into axial clamping. The secondary clamping mechanism is shown conceptually in Figure 1.7.

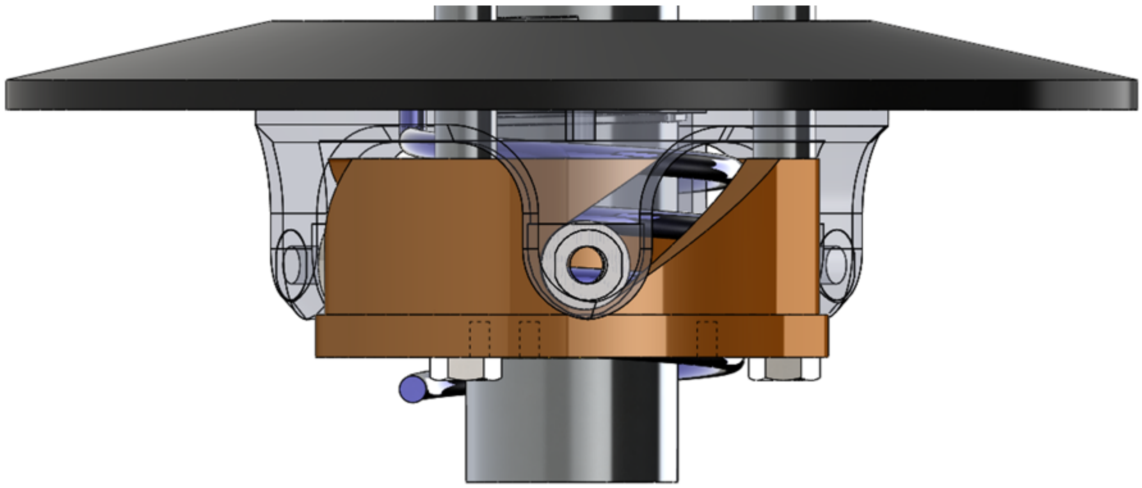


Figure 1.7: Secondary pulley actuation concept. Torque transmitted through the secondary loads the helix mechanism, which converts part of the torque reaction into clamping force between the sheaves. The helix also couples axial shift with relative rotation between secondary sheave components.

Although the primary and secondary mechanisms respond most strongly to different physical quantities, they act on the same belt and the same shift process. The primary is mainly driven by speed, the secondary is mainly driven by transmitted torque, and the resulting CVT behavior depends on their combined interaction with the belt, pulley geometry, drivetrain inertia, external load, and available clamping force.

1.2.6 Full Drivetrain Architecture

The CVT does not operate in isolation. The architecture considered in this work is shown in Figure 1.8. Engine torque enters the primary pulley, passes through the belt to the secondary pulley, travels through a fixed-ratio gearbox or final drive, and ultimately produces tractive force at the wheels. The vehicle then pushes against the ground, and the resulting load feeds back into the drivetrain through inertia, grade resistance, rolling resistance, and aerodynamic drag.

Many drivetrain architectures are possible, but this engine–CVT–gearbox– wheel layout is the system modeled in this work. This system-level view is important because the CVT ratio is not simply chosen independently of the rest of the vehicle. Engine speed affects primary actuation. Transmitted torque affects secondary clamping. Vehicle load affects the torque carried by the secondary. Belt traction determines whether that torque can be carried without slip. All of these quantities evolve together during a launch.

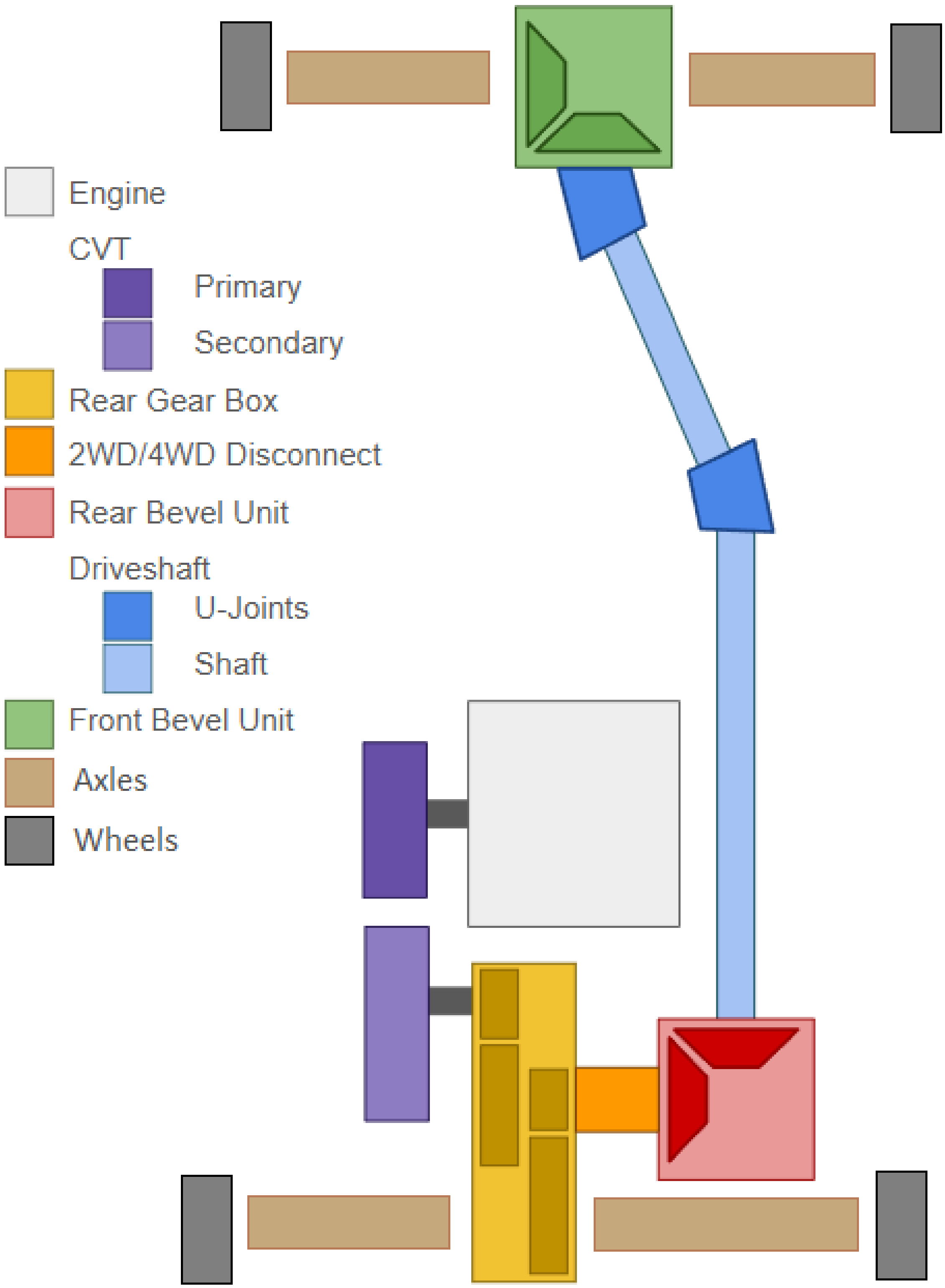


Figure 1.8: Drivetrain architecture considered in this work. Engine torque enters the primary pulley, is transmitted through the belt to the secondary pulley, passes through the downstream drivetrain, and ultimately accelerates the vehicle. The vehicle load feeds back into the CVT through the secondary side of the drivetrain.

The rest of this document turns this physical picture into a dynamic model. The geometry describes how pulley radii change during shift. The force models describe how the primary and secondary sheaves are driven. The traction model describes when torque transfer can stick and when slip occurs. The drivetrain model closes the loop between engine torque, CVT behavior, vehicle load, and vehicle acceleration. Together, these pieces explain how the CVT behaves as a coupled mechanical system during launch.

1.3 Motivation and Literature Review

CVT modelling is not a single, uniform problem. A model developed to size a variator, estimate belt efficiency, design a control strategy, or predict launch behaviour may involve the same physical components, but it will usually make different choices about which quantities are prescribed and which are solved. Some models are primarily design tools. Some focus on mechanical actuation and force balance. Some study belt forces, axial thrust, slip, or efficiency at fixed operating points. Others attempt to resolve transient shift behaviour or detailed belt-pulley contact mechanics.

The progression followed here is from design intuition, to component force balance, to mechanical actuation, to belt contact, to ratio change, to full transient launch behaviour. This ordering mirrors the way the modelling problem becomes difficult. A belt-sheave CVT can first be understood as a variable-radius transmission, then as a collection of mechanical actuators that generate axial force, then as a friction-limited belt contact problem, and finally as a coupled dynamic system whose engine speed, belt state, shift position, and vehicle motion evolve together. The literature contains strong foundations for each of these pieces, but those foundations are distributed across several modelling traditions.

1.3.1 Framework for Classifying CVT Models

A useful way to compare CVT models is to ask what part of the system is being allowed to move, and what part is being held fixed. The same physical transmission can be treated as a geometry problem, a force-balance problem, a belt-contact problem, a shift-dynamics problem, or a drivetrain problem. Each viewpoint can be useful, but each one answers a different question. This is why the literature is best read by tracking the modelling choices that define each formulation, rather than by treating all CVT models as attempts at the same final system.

One of the first choices is how the ratio is represented. In the simplest case, the transmission is evaluated at a fixed ratio. A slightly broader approach computes the ratio from a quasi-static force balance at each operating point. Other models prescribe a ratio trajectory, or use an empirical relationship for the rate of ratio change. More detailed shift models treat ratio change as a mechanical process in its own right, involving radial belt motion, pulley motion, and contact forces (Kim et al., 2002; Sorge, 2008; Cammalleri and Sorge, 2009; Sorge and Cammalleri, 2011). These approaches are not interchangeable. A fixed-ratio model can explain forces at a known condition, while a shift model tries to explain how the system moves from one condition to another.

A second choice is how axial force is produced. Some models take pulley clamping forces as known inputs. This is natural when the goal is to study belt response under specified loading, or when the variator is externally controlled. Other models include the mechanism that generates those forces, such as centrifugal weights, springs, torque-sensitive cams, hydraulic pressure, or electronic actuation (Cammalleri, 2005; Kim et al., 2002; Skinner, 2020; Sorge and Cammalleri, 2011). The important distinction is not only the actuator type, but whether clamping force is prescribed from outside the model or emerges from the same system being simulated.

A third choice is how much of the belt-pulley contact is represented. At low fidelity, the belt may be treated through a capstan-style tension relation, often adjusted for the wedge action of the V-groove. More detailed models include radial and tangential friction, sliding angle, belt penetration, bending stiffness, local normal pressure, tension variation, slip mechanisms, and efficiency losses (Gerbert, 1996; Kim and Kim, 1989; Sorge, 1996; Kong and Parker, 2005, 2006; Chen et al., 1998; Bertini et al., 2014; Messick, 2018). At still higher fidelity, the belt may be discretized into nodes or elements so that local belt motion and contact behaviour are solved directly (Juli'o and Plante, 2011; Ballew, 2015). Increasing contact fidelity can make the local mechanics more realistic, but it also changes the purpose of the model: a model built for efficiency or local belt force prediction may not be the most transparent way to study the full drivetrain response.

A fourth choice is how the rotating system is coupled. Many models prescribe one or both pulley speeds, or impose the speed ratio algebraically. This is often appropriate for steady contact analysis, efficiency prediction, or controlled operating-point studies. Other models solve rotational equations of motion so that input speed, output speed, and transmitted torque evolve together (Srivastava and Haque, 2006; Ballew, 2015; Duan et al., 2016). Extending this further, the input may be coupled to an engine model and the output may be coupled to a dynamometer or vehicle load. This determines whether the CVT is being studied as an isolated variator or as one part of a larger powertrain.

These choices build on one another. A model can have detailed belt contact but prescribed pulley speeds. Another can include vehicle dynamics but use a simplified belt law. Another can model mechanical actuation carefully but treat shifting as quasi-static. None of these choices is inherently wrong; they reflect the question being asked. The important point is that the assumptions define the model's domain of usefulness.

Launch and clutching sit near the intersection of these choices. In that regime, ratio change, actuation, belt contact, rotational dynamics, and external loading can all matter at once. This does not make launch the only goal of CVT modelling, but it does make launch a useful stress test of model closure: a formulation that prescribes too many of these quantities may still be valuable, but it is not resolving the same coupled problem. The following review uses this framework to move from practical and quasi-static design models, to mechanical actuation and shift mechanics, to belt contact and efficiency models, to transient rubber-belt formulations, and finally to related dynamic models in metal-belt and chain CVTs.

1.3.2 Practical Design and Tuning Foundations

The practical understanding of mechanically actuated rubber V-belt CVTs developed largely through machine design and tuning practice. General machine design references such as Budynas and Nisbett (2015) provide the background needed to treat the CVT as a real mechanical assembly involving belts, springs, clutches, shafts, friction interfaces, uncertainty, and design iteration. Not all of these effects are included in a compact model, but they provide useful context for deciding which behaviours are retained and which are idealized.

Practical tuning literature provides a more application-specific foundation. Aaen (1979) describes powersport CVTs using the language of engagement speed, shift speed, flyweights, primary springs, secondary springs, torque sensing, backshift, and speed variance. Its value is not that it provides a complete dynamic model, but that it builds component-level intuition from the perspective of a tuner or racer. It explains how changes to weights, springs, ramps, and helices are expected to affect behaviour, while leaving much of the coupled system response to experience and testing.

These sources are valuable because they describe the physical machine before it is reduced to equations. They also show why prediction is difficult: the same hardware change can affect engagement, shift rate, belt grip, backshift, engine speed, and vehicle acceleration at the same time. Qualitative tuning rules are therefore useful starting points, but they are not enough to determine the full system response under changing operating conditions.

1.3.3 Design-Oriented and Quasi-Static CVT Models

Design-oriented CVT models attempt to convert practical component knowledge into quantitative design procedures. Rather than asking how every transient state evolves in time, these models usually ask which geometry, actuator, belt, or tuning parameters should be selected to satisfy a desired operating requirement. They commonly relate vehicle requirements, pulley geometry, belt length, wrap angle, speed ratio, flyweight force, spring preload, helix force, belt tension, and torque capacity. Their main value is that they connect tunable hardware to expected operating behaviour.

A useful starting point is the numerical design method of Arango and Mu noz Alzate (2020), which begins from vehicle requirements, road specifications, and desired performance before working toward the CVT design. This is valuable because it places component selection in its proper context: the belt, pulley geometry, speed-ratio range, engine or motor selection, and mechanical centrifugal actuation are not independent choices, but responses to the requirements imposed by the vehicle and its operating route. In this sense, the work provides a design-level framework where system requirements drive CVT component choices.

Other design-oriented works focus more narrowly on particular parts of the variator design. Cammalleri (2005) develops a speed-torque-controlled rubber V-belt variator design using equilibrium relationships for a centrifugal driver and torque-sensitive driven pulley. La Battaglia et al. (2022) focuses on the kinematic design of motorcycle rubber V-belt CVTs, emphasizing the roller sliding profile of the front pulley as a critical design variable. Both works show how simplified mathematical models can be used to choose actuator or geometry parameters without requiring a full transient simulation of the drivetrain.

Within Baja-style applications, Skinner (2020) provides a practical force-balance and tuning-oriented model. The work emphasizes quantifiable tuning objectives such as engagement speed, shift-out speed, static RPM shifting, and efficient power transfer, and relates those objectives to the primary and secondary subsystems. It is useful as an applied tuning reference because it connects the variables adjusted by teams to predicted operating behaviour, while remaining primarily an operating-point and force-balance model.

The common strength of these works is that they make CVT design less arbitrary. They identify the parameters that matter and show how geometry, springs, flyweights, helices, belt forces, and vehicle requirements can be related. They also highlight a recurring tension in CVT modelling: models that are simple enough to support early design may omit important coupled behaviour, while models that resolve more detailed mechanics can become difficult to apply directly as design tools. In this sense, design-oriented and quasi-static models occupy an important middle ground. They provide usable relationships for component selection and tuning, but they do not generally describe the complete time history of clutching, slip, shift motion, and vehicle acceleration.

1.3.4 Mechanical Actuation and Clamping Force Generation

After the basic design variables are identified, the next modelling question is how the CVT generates the axial forces that clamp the belt and drive ratio change. In externally controlled CVTs, these forces may be treated as prescribed hydraulic or electronic inputs. In mechanically actuated rubber V-belt CVTs, however, the forces are produced by the hardware itself. The primary force is typically speed-sensitive, while the secondary force is torque-sensitive.

Cammalleri (2005) provides an important design treatment of this problem through a speed-torque-controlled rubber V-belt variator. The centrifugal driver and torque-sensitive driven pulley are treated as part of the variator design, rather than as external force inputs. This helps establish the link between actuator geometry, spring loading, pulley torque, and the intended operating behaviour of the CVT.

Kim et al. (2002) also study a rubber belt CVT with mechanical actuators, including both steady-state and transient characteristics. Their work is useful because it connects the mechanically generated driver and driven pulley forces to the resulting CVT behaviour. Although these earlier mechanical-actuator models include torque-reactive secondary behaviour, they simplify how torque is shared and reacted through the driven sheave assembly.

Sorge and Cammalleri (2011) address this secondary-side issue in more detail by studying the helical shift mechanics of rubber V-belt variators. Their work shows that the relative rotation between the driven sheaves is important to the secondary response, and that the helix should not be treated only as a simple torque-to-axial-force conversion. This makes the secondary a coupled torque-reactive mechanism whose axial force depends on helix geometry, transmitted torque, spring loading, and the internal motion of the driven pulley.

Together, these works show that mechanical actuation is more than a source of prescribed clamp force. The primary and secondary mechanisms are part of the dynamics of the variator: speed, torque, spring preload, ramp or helix geometry, and pulley motion all influence the axial forces applied to the belt. This establishes the actuator-side foundation needed before considering the belt contact and transient shift response.

1.3.5 Rubber V-Belt Contact, Slip, and Efficiency Models

Once axial force is generated by the actuation mechanisms, the belt-pulley interface determines what that force actually does. The sheaves may clamp the belt, but the resulting torque capacity, axial reaction, radial belt motion, slip, and loss all depend on how the rubber belt interacts with the pulley groove. A rubber V-belt therefore cannot be treated as an ideal kinematic constraint. It wedges into the sheaves, develops a tension distribution around the wrap, experiences radial and tangential friction, deforms under load, and dissipates energy through several mechanisms.

Classical belt theory gives the starting point: friction and wrap angle determine the maximum tension ratio that can be supported before gross sliding occurs. For V-belts, this picture is modified by the groove angle and the associated wedge effect. However, real belt slip is more complicated than a single capstan-style traction limit. Gerbert (1996) presents a unified treatment of belt slip in which creep, radial compliance, shear deflection, and seating/unseating effects all contribute to speed loss. The important point is not that every model must include all of these effects, but that "slip" is not one physical mechanism. A model should be clear about which slip and loss mechanisms it represents and which it leaves outside its scope.

The contact problem also determines the axial reaction felt by the sheaves. Kim and Kim (1989) developed a theoretical analysis of axial forces in a rubber V-belt CVT using simplified assumptions about the contact regions and belt force distribution. Sorge (1996) developed a compact model for axial thrust in V-belt drives, emphasizing radial penetration and the connection between belt forces and pulley thrust. These works are useful because they connect the belt contact state back to the sheave force balance: axial force is not only generated by the actuator, but also shaped by how the belt seats, slides, and transmits load in the groove.

A more detailed treatment is given by Kong and Parker (2005), who formulate the steady mechanics of belt-pulley systems using distributed belt equations. Kong and Parker (2006) extend this framework to pulley grooves and sliding friction. This line of work resolves local belt mechanics at much greater fidelity than simple force-balance models, including the distribution of forces and sliding behaviour around the contact arc.

Messick (2018) extends this contact-mechanics tradition for rubber V-belt CVTs by building on the Kong and Parker framework and validating axial-force and efficiency predictions experimentally. Messick's work is valuable for its contact-region formulation, belt property measurement, operating-range analysis, and dynamometer validation. It is also clear about its modelling scope: shifting is treated as a quasi-equilibrium process rather than as a transient response-time problem. In this sense, it is a strong reference for belt force and efficiency modelling, while leaving the time evolution of the complete drivetrain to a different class of model.

Efficiency-focused studies provide another view of the same interface. Chen et al. (1998) experimentally investigated speed and torque losses in a rubber V-belt CVT, while Bertini et al. (2014) developed an analytical model for rubber V-belt CVT power losses including sliding, hysteresis, and entry/exit effects. These works show that efficiency depends on multiple interacting mechanisms, not just on whether the belt remains below a gross slip threshold.

The contact and efficiency literature therefore provides the foundation for understanding how belt force, axial thrust, slip, and loss arise. It also shows why contact fidelity must be chosen carefully. A high-fidelity contact model may be necessary for efficiency prediction, axial-force prediction, or local belt mechanics. A lower-dimensional contact model may be more appropriate when the goal is to preserve physical intuition while coupling the belt to actuator dynamics, rotational dynamics, and drivetrain response.

1.3.6 Transient Rubber-Belt Models and the Launch Problem

Transient rubber-belt models address time-dependent behaviour more directly, but "transient" can mean several different levels of closure. A model may study how the belt migrates radially during shift, how local belt elements move through the pulley contact, how the variator responds to prescribed axial forces, or how the CVT interacts with an engine and vehicle load. Launch lies near the most coupled end of this spectrum. During launch, the belt may be slipping, the engine may be accelerating, clamping forces may be changing, the secondary torque reaction depends on transmitted torque, and the vehicle load evolves as the vehicle begins to move. A launch model therefore has to manage torque transfer and ratio change while the drivetrain state is still developing.

A useful bridge between steady force-balance models and fully transient belt simulations is the shift-mechanics work of Sorge (2008). Sorge shows that ratio change in a rubber belt variator is not simply a sequence of steady operating points. During shift, the belt moves radially through the sheave grooves while also travelling circumferentially and transmitting torque, so belt mass conservation, radial motion, sliding, and circumferential transport interact. Cammalleri and Sorge (2009) later develop approximate closed-form solutions for this shift-mechanics problem, making the theory more accessible for engineering calculations. These works isolate the mechanics of ratio change itself, without attempting to close the full engine–vehicle launch problem.

Juli'o and Plante (2011) developed an experimentally validated transient model of rubber-belt CVT mechanics using a discretized belt formulation. This work represents the belt as a distributed mechanical body rather than relying only on lumped contact assumptions. Its strength is that local belt motion and contact response are resolved directly. The tradeoff is that the global mechanism chain becomes less compact, and the relationship between model behaviour and a small set of practical tuning variables becomes less direct.

Ballew (2015) continues in the discretized-belt direction and expands the model to include engine torque output and vehicle loads such as rolling resistance and aerodynamic drag. Ballew also argues against prescribing input pulley speed, since doing so removes an important rotational degree of freedom. This makes the work especially relevant to drivetrain-level rubber-belt CVT simulation. At the same time, the reported simulations use a PI controller with feed-forward gain to generate primary axial force, fix the secondary axial force to simplify the simulation, and introduce damping between belt nodes to manage numerical behaviour. Thus, the model includes important engine and vehicle coupling, but it does not close the launch problem through a mechanically self-actuated flyweight and helix model.

The transient rubber-belt literature therefore provides important precedents, while also showing why the problem remains difficult. Shift-mechanics models clarify how ratio change occurs, discretized belt models resolve local belt behaviour, and drivetrain-coupled models show how engine and vehicle dynamics can be included. The remaining challenge is not simply to make the belt model more detailed, but to choose a level of fidelity that preserves the coupling between actuation, torque transfer, ratio evolution, and vehicle response without losing physical interpretability.

1.3.7 Adjacent Metal-Belt and Chain CVT Dynamic Models

The broader CVT literature contains an extensive dynamic modelling tradition for metal push-belt and chain CVTs, especially in automotive applications. These systems are not direct equivalents of dry rubber V-belt CVTs, but they are useful adjacent references because they show how other CVT architectures have handled shift dynamics, slip, clamping force, deformation, and validation.

Srivastava and Haque (2009) review the dynamics and control of belt and chain CVTs, covering modelling and control strategies for friction-limited transmissions. Their review is useful not only because it summarizes the field, but because it shows that CVT modelling has developed unevenly across architectures. Different CVT types are modelled with different assumptions, and fully dynamic, experimentally validated models remain less common than steady-state or control-oriented formulations. The review also reinforces an important distinction: rubber V-belts, metal push belts, and chains should not be treated interchangeably because their contact mechanics, actuation methods, deformation behaviour, torque capacity, and control objectives differ.

For metal belt CVTs, Carbone et al. (2005) study the influence of pulley deformation on shifting behaviour. Their model explains the distinction between creep-mode and slip-mode shifting and shows that pulley deformation can be essential to reproducing slow-shift behaviour under Coulomb-friction assumptions. Yildiz et al. (2017) later validate the Carbone–Mangialardi–Mantriota model under steady-state conditions and shift manoeuvres with torque load, supporting the usefulness of this modelling framework for controlled metal-belt or chain CVTs.

Other metal-belt studies provide useful analogues for traction-limited behaviour and ratio-change dynamics. Srivastava and Haque (2006) model the dynamics of metal V-belt CVTs, while Bonsen et al. (2004) analyze slip behaviour and the relationship between slip and transmitted torque. These works are relevant because they treat slip and dynamic ratio change as part of the variator mechanics, even though the transmission architecture differs from a dry rubber V-belt system.

Chain CVTs provide another adjacent comparison. Duan et al. (2016) develop a physics-based chain CVT model beginning from the kinematics and dynamics of an infinitesimal chain segment. Their model includes speed-dependent friction, pulley axial dynamics coupled with chain radial movement, and pulley deformation. This makes the work a useful example of a physically coupled CVT formulation in which rotational dynamics, axial dynamics, contact mechanics, and friction state are treated together.

The important distinction is that these models are usually developed for lubricated automotive metal-belt or chain systems with hydraulic or electronic clamping. Their methods are therefore valuable, but their assumptions cannot be imported directly into dry rubber V-belt CVTs with mechanical flyweights, springs, and torque-sensitive helices. They show what is possible in adjacent CVT architectures, while also reinforcing the need for a formulation matched to rubber-belt contact and mechanical actuation.

1.3.8 Synthesis of Prior Work

The literature establishes a clear set of foundations. Practical and design references introduce the tuning variables and component relationships used to describe rubber V-belt CVTs. Design-oriented models connect vehicle requirements, geometry, actuator choices, and force-balance relationships to expected operating behaviour. Mechanical actuation studies explain how primary and secondary clamping forces can be generated by the hardware itself, rather than prescribed externally. Rubber V-belt contact models explain how belt force, axial thrust, tension distribution, slip mechanisms, and efficiency losses arise at the belt-pulley interface. Transient rubber-belt models show how ratio change, local belt motion, and drivetrain coupling can be treated in time. Metal-belt and chain models show that coupled axial, rotational, and traction dynamics can be modelled rigorously in adjacent CVT architectures.

The remaining difficulty is not any single missing equation, but the closure of the whole system under assumptions appropriate to a mechanically actuated dry rubber V-belt CVT. Launch and clutching sit at the intersection of several effects: engine acceleration, primary engagement, secondary torque reaction, belt traction, shift motion, and vehicle load. Treating any one of these as prescribed can be reasonable for a focused study, but it changes the problem being solved. The challenge is therefore to choose a level of fidelity that preserves the dominant couplings without making the model too detailed to interpret or use for tuning.

Table 1.1 summarizes the main model families discussed in this review. The table is intended as a map of modelling emphasis rather than a ranking of model quality.

Table 1.1: Comparison of representative CVT model families.

Model family	**Representative works**	**Shift treatment**	**Rotational / drivetrain dynamics**	**Actuation treatment**	**Contact fidelity**	**Launch / clutching treatment**
Practical tuning and design references	Aaen (1979), Budynas and Nisbett (2015)	Qualitative	Limited	Qualitative mechanical descriptions	Conceptual or component-level	Qualitative tuning guidance
Design-oriented and quasi-static models	Arango and Mu noz Alzate (2020), Cammalleri (2005), La Battaglia et al. (2022), Skinner (2020)	Fixed, mapped, or quasi-static	Often simplified	Geometry and force-balance based	Low to medium	Usually simplified
Mechanical actuation and clamping force generation	Cammalleri (2005), Kim et al. (2002), Sorge and Cammalleri (2011)	Steady, quasi-static, or empirically transient	Often simplified	Mechanical primary and secondary actuation considered	Medium	Limited or indirect
Rubber V-belt contact, slip, and efficiency models	Gerbert (1996), Kim and Kim (1989), Sorge (1996), Kong and Parker (2005), Kong and Parker (2006), Messick (2018), Chen et al. (1998), Bertini et al. (2014)	Mostly fixed-ratio or quasi-equilibrium	Often limited	Usually prescribed operating conditions	Medium to high	Rarely central
Transient rubber-belt and shift models	Sorge (2008), Cammalleri and Sorge (2009), Juli'o and Plante (2011), Ballew (2015)	Dynamic or approximate transient shift	Varies from local belt dynamics to engine / vehicle coupling	Prescribed, controlled, simplified, or separate from full mechanical closure	Medium to high	Partial or assumption-dependent
Metal-belt and chain dynamic models	Srivastava and Haque (2009), Srivastava and Haque (2006), Bonsen et al. (2004), Carbone et al. (2005), Yildiz et al. (2017), Duan et al. (2016)	Dynamic	Often high	Hydraulic, electronic, or prescribed clamping	High for metal/chain systems	Architecture dependent
Present formulation, for comparison	This document	Dynamic shift coordinate	Engine, CVT, and vehicle coupled	Mechanical flyweight, spring, helix, and torque feedback	Lumped rubber belt with traction admissibility	Launch and clutching included

1.4 Aim, Scope, and Contributions

The literature review in Section 1.3 motivates a specific modelling objective: to build a physically interpretable dynamic formulation for a mechanically actuated dry rubber V-belt CVT coupled to a vehicle drivetrain. The model is not intended to be the highest-fidelity representation of rubber belt contact, nor a complete loss model for efficiency prediction. Its purpose is to close the dominant system-level mechanisms into a single time-domain formulation so that actuation, shift behaviour, torque transfer, and vehicle response can be studied together.

1.4.1 Aim

The technical aim of this work is to derive a coupled nonlinear dynamic model of the CVT and drivetrain from the engine-side pulley to the vehicle wheel. The model is formulated so that pulley speeds, torque transfer, axial shift position, and vehicle motion evolve from the governing equations rather than being prescribed independently. This allows launch, shift, and traction-limited behaviour to be simulated directly in time.

The pedagogical aim is equally important. The model is written as a derivation document, not only as a simulation tool. Each major modelling step is developed from physical principles so that the reader can follow how geometry, actuator forces, belt traction, rotational dynamics, and vehicle loading enter the final system. The goal is for the model to build intuition: a reader should be able to understand not only what the simulation predicts, but why the system behaves that way.

1.4.2 Scope

The system considered in this work is a mechanically actuated dry rubber V-belt CVT coupled to a simplified vehicle drivetrain. The CVT consists of a speed-sensitive primary pulley, a torque-sensitive secondary pulley, a rubber V-belt, and a downstream gearbox feeding to the wheels of the vehicle. The model focuses on launch and shift behaviour under one-dimensional longitudinal vehicle motion.

The model includes the following effects:

- variable pulley geometry and belt-length closure;
- ratio kinematics relating shift position to pulley radii, wrap angles, and transmission ratio;
- mechanically generated primary and secondary axial forces;
- spring, ramp, helix, and torque-reactive effects in the actuation mechanisms;
- dynamic axial shift motion with the effective inertia of moving CVT components;
- explicit rotational dynamics for the engine-side and vehicle-side pulleys;
- torque transfer through the belt subject to belt-pulley traction limits;
- engine torque and engine-side inertia;
- vehicle load torque from longitudinal road-load dynamics;
- coupled launch and shift response in time.

Several effects are intentionally idealized or excluded. The belt is represented through a lumped geometric and traction model rather than a full distributed rubber contact model. The model accounts for whether the belt-pulley contacts can support the required torque, but it does not separately resolve the individual slip mechanisms identified in detailed belt-slip literature, such as creep, radial compliance, shear deformation, seating and unseating, or local rubber hysteresis. Local belt deformation, detailed radial penetration, temperature effects, belt wear, sheave compliance, detailed tribology, and full three-dimensional vehicle dynamics are also not resolved.

This scope also makes the formulation extensible. Because the assumptions are stated explicitly and the model is derived step by step, later work can replace individual closures with higher-fidelity alternatives. For example, the traction model could be replaced with a distributed contact model, the belt could be made elastic, pulley deformation could be introduced, or thermal and efficiency losses could be added without changing the overall structure of the framework.

1.4.3 Contributions

The primary contribution of this work is not any single component model in isolation. Many of the individual ingredients, such as belt geometry, centrifugal actuation, torque-sensitive secondary behaviour, traction limits, and vehicle loading, build on established ideas in CVT and belt-drive modelling. The contribution is the way these ingredients are organized into a single physically interpretable dynamic formulation for a mechanically actuated dry rubber V-belt CVT.

The main contributions are as follows:

1. A coupled time-domain formulation for a mechanically actuated rubber V-belt CVT connected to engine-side and vehicle-side dynamics.
2. A geometric closure that relates axial shift position to pulley radii, wrap angles, belt length, transmission ratio, and ratio evolution.
3. A mechanical actuation formulation in which the primary and secondary axial forces are generated by the modelled hardware rather than prescribed as external inputs.
4. A dynamic treatment of ratio change in which the shift coordinate evolves as a physical state of the system rather than being imposed from a prescribed ratio map or instantaneous equilibrium condition.
5. A torque-transfer formulation that allows belt-pulley traction limits to constrain the transmitted torque when the no-slip response is not admissible.
6. An explicit coupling between CVT behaviour and drivetrain response, allowing engine speed, pulley speeds, belt torque transfer, vehicle load, and vehicle acceleration to evolve together.
7. A launch-capable system closure in which clutching, torque transfer, shift behaviour, and vehicle acceleration emerge from the interaction of the modelled mechanisms.
8. A derivation structure intended to preserve physical intuition and provide a modular foundation for future extensions, such as higher-fidelity belt contact, belt elasticity, pulley deformation, thermal effects, or efficiency losses.

The central contribution is therefore the closure of the mechanism chain. Rather than prescribing the ratio, shift rate, pulley speeds, or clamp forces independently, the formulation allows the system trajectory to emerge from mechanical actuation, belt traction, rotational inertia, and vehicle load. This makes the model useful for studying how tuning and design changes propagate through the CVT, from component geometry and spring loading, to torque transfer and shift response, to vehicle acceleration.

Chapter 2

Model Setup

2.1 System Definition

Placeholder

2.2 Modeling Assumptions

The following assumptions define the modeling scope and physical idealizations used in the CVT dynamic formulation. These assumptions establish the limits within which the derived equations remain valid.

2.2.1 Material and Structural Idealizations

Assumption 1 (Constant Material Properties)

Material properties of structural components are treated as constant. No temperature-dependent material variation is modeled. Thermal effects are not solved as part of this system.

Assumption 2 (Rigid Structural Components)

All structural components (pulleys, shafts, housing) are modeled as rigid bodies. Elastic deformation under operational loading is neglected. The belt is assumed axially rigid (no longitudinal stretch), while still conforming geometrically around pulleys.

Assumption 3 (Negligible Wear and Surface Evolution)

Component wear, belt degradation, and pulley surface evolution are neglected. The model represents short-term operational behavior.

2.2.2 Contact and Friction Modeling

Assumption 4 (Idealized Belt–Pulley Friction Law)

Belt traction is modeled using a simplified friction law (e.g., Coulomb friction with a capstan formulation and an explicit stick–slip transition rule). Microscopic contact mechanics, rubber hysteresis, lubrication films, and detailed Stribeck effects are not explicitly modeled.

Assumption 5 (Constant Effective Friction Coefficient)

The belt–pulley coefficient of friction is treated as constant. It does not evolve with temperature, wear, or surface conditioning.

Assumption 6 (Uniform Contact Distribution)

Contact pressure and traction over the pulley wrap angle are treated as effectively uniform or averaged. Localized edge loading and non-uniform stress distributions are neglected.

2.2.3 Geometric and Kinematic Idealizations

Assumption 7 (Perfect Axis Alignment)

Primary and secondary pulley axes are assumed perfectly aligned. Misalignment, shaft eccentricity, and geometric runout are neglected.

Assumption 8 (Constant Belt Length)

The belt length is treated as constant. Longitudinal elasticity, transient stretch, permanent creep, and progressive elongation are not modeled. Effective pulley radii are determined solely from geometry and belt-length constraints.

Assumption 9 (Uniform Contact Radius Around Wrap)

At any given instant, the belt contacts each pulley at a single, uniform effective radius throughout the entire wrap angle. The contact radius is constant around the circumference of contact, with no spiral or progressive radial variation along the wrap. While the effective radius may evolve with axial shift position, it remains spatially uniform at each time instant.

2.2.4 Dynamic and Inertial Modeling

Assumption 10 (Lumped Inertia Representation)

Rotational inertias of the engine, pulleys, drivetrain, and wheels are represented as lumped equivalent inertias. Distributed shaft flex and torsional compliance are neglected.

Assumption 11 (Negligible Vibrational Effects)

High-frequency vibrational effects, structural resonances, and drivetrain oscillations are neglected. The model captures dominant rigid-body dynamics only.

Assumption 12 (Negligible Parasitic Friction)

Frictional losses in non-critical components (e.g., bearings, seals, auxiliary shafts) are neglected. Only friction directly relevant to belt traction and CVT torque transfer is modeled.

Assumption 13 (Ideal CVT Power Transmission)

Power transmission through the CVT is modeled as ideal (i.e., no internal dissipation). Although belt slip is explicitly modeled and affects torque capacity and shift dynamics, the associated power loss is neglected.

2.2.5 Environmental and Operational Scope

Assumption 14 (Negligible Gravitational Influence on CVT Internals)

Gravitational forces acting on CVT internal components are negligible relative to centrifugal and contact forces at operating speeds.

Assumption 15 (Full-Throttle Engine Model)

The engine is modeled using a prescribed full-throttle torque curve. Load feedback modifies engine speed dynamically, but part-throttle dynamics and throttle transients are not explicitly modeled.

2.3 Reference Material

This section records any information for easy reference.

2.3.1 Coordinate and Sign Conventions

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions.

Vehicle Longitudinal Coordinate and Incline Angle

Vehicle motion is modeled along a single longitudinal coordinate that follows the surface of the road.

The coordinate x is defined locally along the direction of travel, tangent to the road surface. The positive direction, $+x$, corresponds to the forward direction of vehicle motion.

Because the road may be inclined relative to horizontal, the local direction of the x axis depends on the road slope. The incline angle is denoted by α and is measured between the road surface and a horizontal reference line.

$$\alpha > 0 \quad : \text{uphill in the } +x \text{ direction}$$

$$\alpha < 0 \quad : \text{downhill in the } +x \text{ direction}$$

As illustrated in Figure 2.1, the x direction always lies tangent to the road surface. Consequently, the orientation of the longitudinal axis varies with the local road incline α.

Vehicle velocity is defined along this coordinate. A velocity $v > 0$ corresponds to motion in the forward $+x$ direction along the road, while $v < 0$ corresponds to motion in the opposite direction.

These definitions establish the geometric orientation of vehicle motion and the sign convention for road incline used throughout the model.

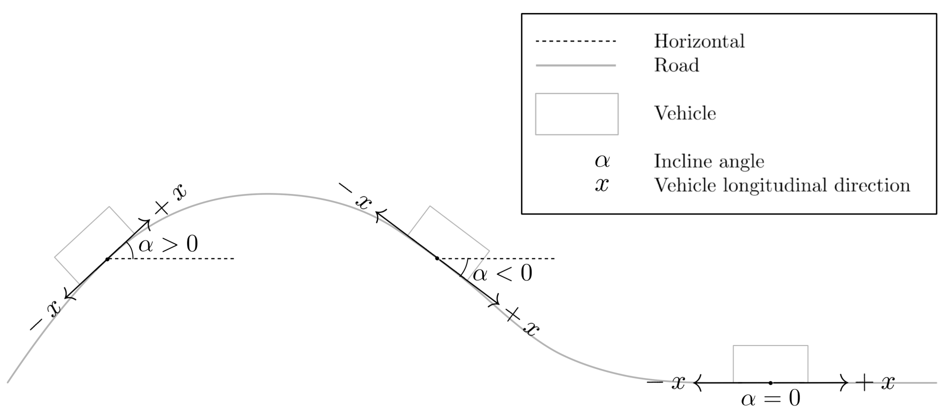


Figure 2.1: Definition of the longitudinal vehicle coordinate and incline angle. The coordinate x is defined tangent to the road surface and points in the forward direction of travel. The road incline α is measured relative to the horizontal reference line. Positive α corresponds to uphill motion in the $+x$ direction, while negative α corresponds to downhill slope.

Rotational Sign Convention

All rotating components in the drivetrain follow a consistent angular sign convention defined relative to forward vehicle motion.

The positive longitudinal direction $+x$ was defined previously as the forward direction of travel along the road surface. The rotational directions of all drivetrain components are chosen such that positive rotation produces vehicle motion in this $+x$ direction.

As illustrated in Figure 2.2, positive wheel angular velocity $\omega_w > 0$ corresponds to wheel rotation that drives the vehicle forward along $+x$.

The same convention is propagated upstream through the drivetrain. Positive rotation of the secondary pulley, $\omega_s > 0$, corresponds to the direction of rotation that produces this forward wheel motion through the final reduction gearbox and axle.

Likewise, positive rotation of the primary pulley, $\omega_p > 0$, corresponds to the direction that drives the secondary pulley in this same sense through the CVT belt.

Engine angular velocity follows the same convention. The engine rotates positively when it drives the primary pulley in the defined positive direction.

Torque follows the same sign convention. A torque is considered positive when it acts in the defined positive rotational direction and therefore tends to increase the corresponding angular velocity.

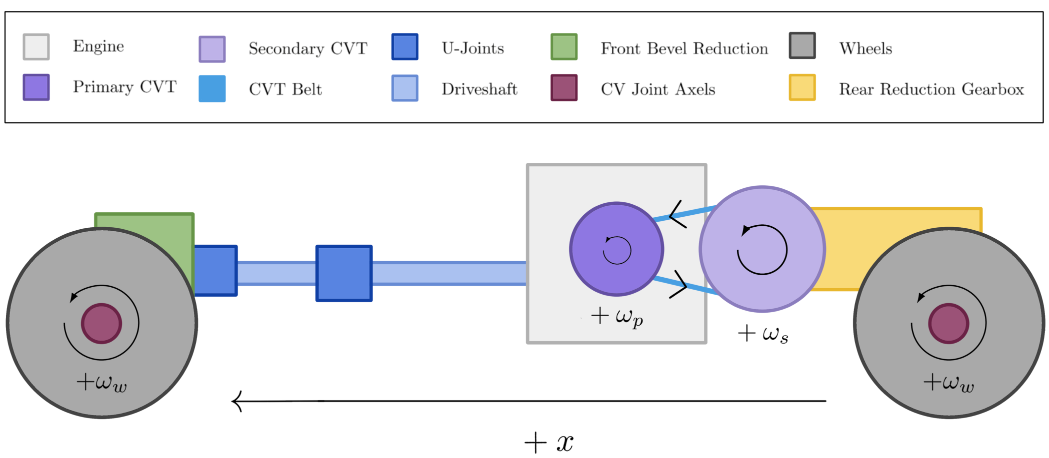


Figure 2.2: Rotational sign convention for the drivetrain. Positive angular velocities for the wheels (ω_w), secondary pulley (ω_s), and primary pulley (ω_p) are defined such that drivetrain rotation produces forward vehicle motion along the $+x$ direction.

CVT Axial Direction

The axial coordinate of the primary pulley is defined along the axis of the pulley shaft, parallel to the pulley centerline and perpendicular to the plane of belt rotation.

In the reference configuration shown in Figure 2.3 (left), the primary pulley is fully open. The movable sheave is separated from the fixed sheave by the initial gap, and this configuration defines the origin of the axial coordinate.

The axial variable s measures the translation of the movable sheave relative to this open configuration.

$$s = 0 \quad : \text{fully open configuration}$$

A positive value of s corresponds to the movable sheave translating toward the fixed sheave along the pulley axis.

$$s > 0 \quad : \text{sheave closing motion}$$

As the movable sheave closes, the belt is squeezed between the conical faces and is forced outward along the pulley ramps. This increases the effective belt radius on the primary pulley and corresponds to an upshift of the transmission.

The maximum displacement $s = S_{\max}$ corresponds to the fully closed configuration shown in Figure 2.3 (right).

Although s is defined using the primary pulley, the motion cannot occur independently. Because the belt length remains constrained, closure of the primary pulley forces the secondary pulley to open. The precise geometric relationship between these motions will be derived later.

Axial velocity follows the same sign convention: $\dot{s} > 0$ corresponds to the movable sheave translating toward the fixed sheave.

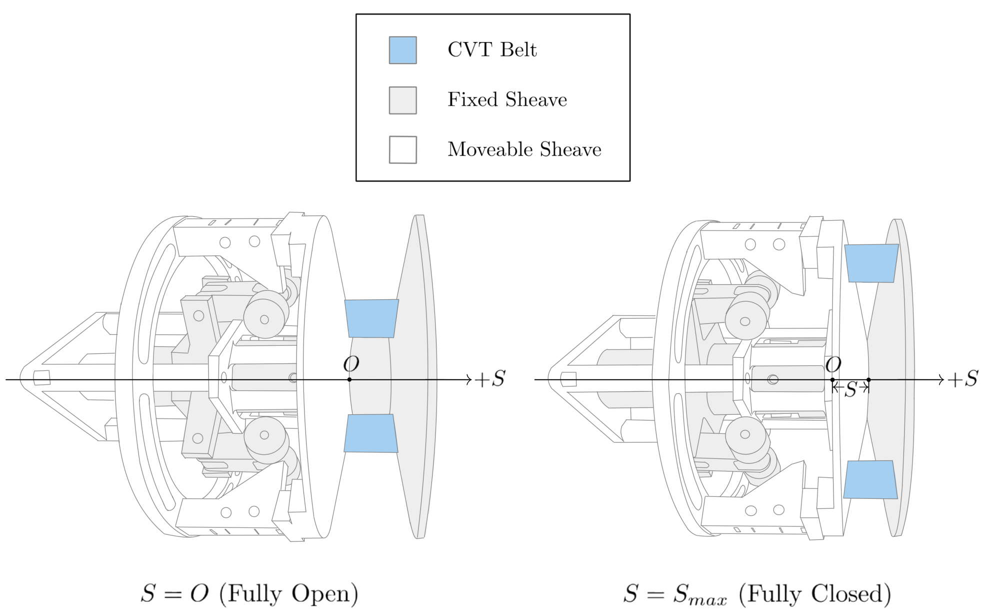


Figure 2.3: Definition of the primary axial coordinate. The system begins in an open configuration. The axial displacement s measures how much the movable sheave has closed from this initial position. Increasing s increases primary effective radius and is coupled to secondary axial motion.

Radial and Tangential Directions

Within the plane of pulley rotation, directions are defined using the standard polar coordinate convention commonly used in rotational mechanics.

At any point in the plane, two orthogonal directions are defined: the radial direction $\hat{e}_r$ and the tangential direction $\hat{e}_\theta$. These directions form a local coordinate basis centered on the pulley axis.

The radial direction lies in the plane of rotation and points directly outward from the center of the pulley. A radial force is defined as positive when it acts away from the center of rotation. Thus, $F_r > 0$ corresponds to a force directed outward along $\hat{e}_r$.

The tangential direction lies in the plane of rotation and is perpendicular to the radial direction. The positive tangential direction $\hat{e}_\theta$ is defined to be consistent with the positive rotational convention introduced in Section 2.3.1. Tangential motion in this direction corresponds to motion produced by positive angular velocity.

Figure 2.4 illustrates these directions using physical examples from the primary pulley. The centrifugal flyweights generate outward radial forces F_r, while the belt moves along the tangential direction with velocity v_t. These quantities serve as representative examples of forces and motion that act along the radial and tangential directions.

The previously defined axial direction is perpendicular to this plane of rotation and points along the pulley shaft, into the page.

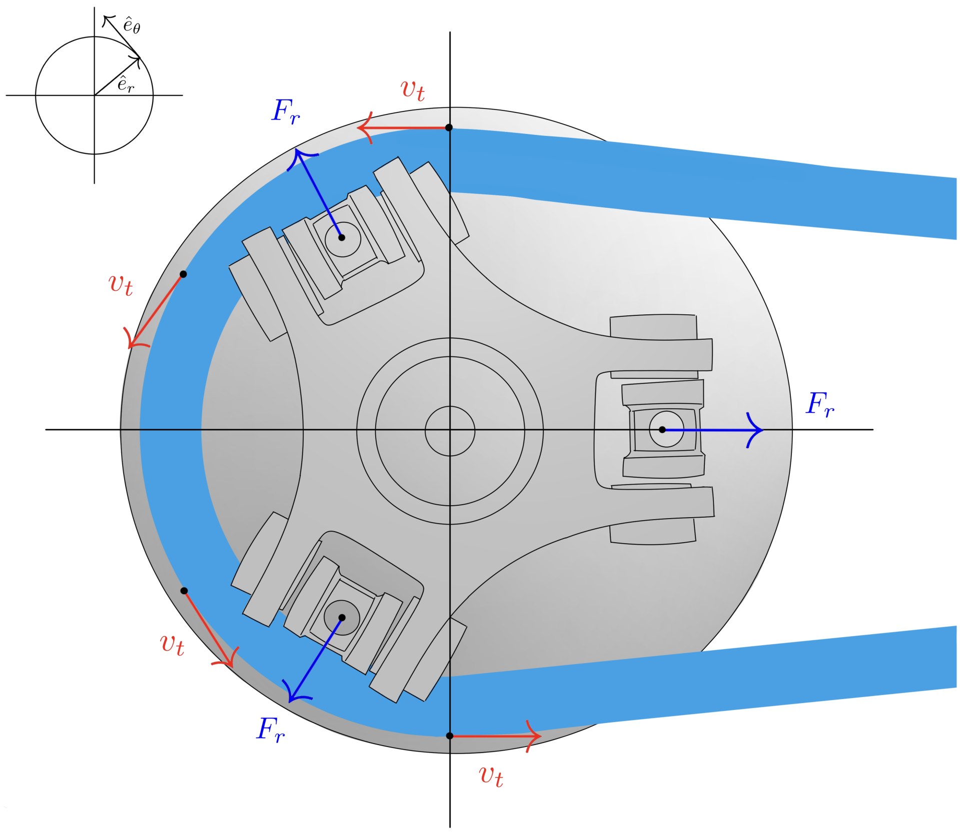


Figure 2.4: Radial and tangential directions in the plane of pulley rotation. The radial direction $\hat{e}_r$ points outward from the center of the pulley, while the tangential direction $\hat{e}_\theta$ is perpendicular to $\hat{e}_r$ and aligned with positive rotation. Example quantities illustrated include outward radial forces F_r generated by flyweights and tangential belt velocity v_t. For reference, the previously defined axial coordinate s is perpendicular to the page in this view and points along the pulley shaft, into the page.

2.4 Governing Principles

Theory 1: Newton's Second Law (Translation)

Equation

$$\sum F = ma$$

Description

- $\sum F$ is the net force (N)
- m is mass (kg)
- a is acceleration (m/s^2)

Source

Standard mechanics identity.

Derivation

From the definition of acceleration and empirical laws of motion; used here as a governing relationship.

Theory 2: Newton's Second Law (Rotation)

Equation

$$\sum \tau = I\alpha$$

Description

- $\sum \tau$ is net torque (N m)
- I is moment of inertia (kg m^2)
- α is angular acceleration (rad/s^2)

Source

Standard rigid-body dynamics identity.

Derivation

Standard rigid-body rotational dynamics relationship.

Theory 3: Torque Definition

Equation

$$\tau = Fr$$

Description

- τ is torque (N m)
- F is tangential force (N)
- r is moment arm radius (m)

Source

Standard mechanics identity.

Derivation

Defines torque as the product of tangential force and moment arm. Equivalently, $F = \tau/r$ gives the tangential force from a known torque.

Theory 4: Kinetic Energy (Translational)

Equation

$$T = \frac{1}{2}mv^2$$

Description

- T is kinetic energy (J)
- m is mass (kg)
- v is velocity (m/s)

Source

Standard mechanics identity.

Derivation

Translational kinetic energy of a point mass or rigid body moving with velocity v.

Theory 5: Centrifugal Force

Equation

$$F_c = m\omega^2 r$$

Description

- F_c is centrifugal force (N)
- m is mass (kg)
- ω is angular velocity (rad/s)
- r is radius from the axis of rotation (m)

Source

Standard circular-motion identity.

Derivation

Derived from circular motion kinematics where $a_c = \omega^2 r$, then applying $F = ma$.

Theory 6: Hooke's Law (Compressional)

Equation

$$F = k\Delta x$$

Description

- F is force (N)
- k is spring constant (N/m)
- Δx is displacement (m)

Source

Standard linear-elastic constitutive identity.

Derivation

Linear spring constitutive relationship in the elastic regime.

Theory 7: Hooke's Law (Torsional)

Equation

$$F = \frac{k\Delta\theta}{r}$$

Description

- F is force (N)
- k is torsional spring constant (N m/rad)
- $\Delta\theta$ is angular deflection (rad)
- r is torque arm radius (m)

Source

Standard torsional spring identity.

Derivation

Starting from $\tau = k\Delta\theta$ and $\tau = Fr$, then $F = \frac{k\Delta\theta}{r}$.

Theory 8: Engine Torque from Power

Equation

$$\tau = \frac{P}{\omega}$$

Description

- τ is torque (N m)
- P is power (W)
- ω is angular velocity (rad/s)

Source

Standard rotational power identity.

Derivation

From the rotational power relationship $P = \tau\omega$.

Theory 9: Coulomb Friction

Equation

$$F_f = \mu_s F_n$$

Description

- F_f is friction force (N)
- μ_s is coefficient of static friction (unitless)
- F_n is normal force (N)

Source

Standard dry-friction model identity.

Derivation

Coulomb friction model for impending slip.

Theory 10: Capstan Equation

Equation

$$T_1 = T_2 e^{\mu\theta}$$

Description

- T_1 is tight-side belt tension (N)
- T_2 is slack-side belt tension (N)
- μ is belt–pulley friction coefficient (unitless)
- θ is wrap angle (rad)

Source

ScienceDirect Topics (n.d.).

Derivation

Differential force balance on an infinitesimal belt element on a pulley yields $\frac{\mathrm{d}T}{T} = \mu \, \mathrm{d}\theta$, then integrate over wrap.

Theory 11: Density

Equation

$$\rho = \frac{m}{V}$$

Description

- ρ is density (kg/m^3)
- m is mass (kg)
- V is volume (m^3)

Source

Standard definition.

Derivation

Definition of density.

Theory 12: Air Resistance (Quadratic Drag)

Equation

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

Description

- F_d is drag force (N)
- ρ is fluid density (kg/m^3)
- v is object speed relative to fluid (m/s)
- C_d is drag coefficient (unitless)
- A is reference area (m^2)

Source

Standard aerodynamic drag model identity.

Derivation

Standard empirical/derived quadratic drag model; applicable at moderate to high Reynolds number.

Theory 13: Rolling Resistance

Equation

$$F_r = C_r N$$

Description

- F_r is rolling resistance force (N)
- C_r is coefficient of rolling resistance (unitless)
- N is normal force (N)

Source

Standard rolling-resistance model identity.

Derivation

Rolling resistance arises from deformation of contact surfaces (tire and road). For a vehicle on level ground, $N = mg$ where m is vehicle mass and g is gravitational acceleration.

Theory 14: Gravitational Force

Equation

$$F_g = mg$$

Description

- F_g is gravitational force (N)
- m is mass (kg)
- g is gravitational acceleration (m/s^2)

Source

Standard near-Earth mechanics identity.

Derivation

Near Earth's surface, $g \approx 9.81$ m/s^2. On an incline with angle θ, the component parallel to the surface is $F_g \sin\theta$.

2.5 System State Definition

The dynamic behavior of the CVT system is described using a set of *state variables*. A state variable is a quantity whose future evolution is determined by its current value and the governing differential equations of the system.

In numerical simulation, state variables are the quantities that are explicitly integrated forward in time. All other model quantities — such as torque transfer, transmission ratio, belt forces, and contact conditions — are computed algebraically from these states.

2.5.1 State Vector

State Vector **(2.5.1)**

$$\boldsymbol{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \\ v_b \end{bmatrix}$$

where:

- ω_p is the angular speed of the primary pulley (rad/s),
- ω_s is the angular speed of the secondary pulley (rad/s),
- s is the axial shift position of the primary sheave (m),
- $\dot{s}$ is the axial shift velocity (m/s).
- v_b is the belt transport speed (m/s).

2.5.2 Independent Dynamic Degrees of Freedom

The five state variables above are the dynamic degrees of freedom of the present state-space model.

The primary pulley can rotate independently under engine input and belt interaction. Its angular speed ω_p therefore represents one rotational degree of freedom.

The secondary pulley also rotates and interacts with both the belt and the external drivetrain load. Its angular speed ω_s represents a second independent rotational degree of freedom.

In addition to rotation, the CVT includes axial motion of the primary sheave. This axial displacement s determines the effective pulley geometry and therefore the transmission ratio. The axial motion is mechanically independent of pure rotation and must be modeled explicitly.

Because axial motion follows second-order dynamics, both the axial position s and its rate $\dot{s}$ are required to fully describe its evolution.

The quantity v_b is treated as an independent belt-transport degree of freedom. While detailed internal belt mechanics remain unresolved, v_b is modeled as a dynamic state whose evolution is given by an explicit equation of motion derived from torque, traction, and compatibility relations at the pulley–belt interfaces.

2.5.3 Time Integration Perspective

During numerical simulation, the state vector is advanced forward in time.

At each timestep Δt:

- The angular speeds ω_p and ω_s are updated according to their angular accelerations.
- The axial position s is updated using its current velocity $\dot{s}$.

- The axial velocity $\dot{s}$ is updated according to the axial acceleration.
- The belt transport speed v_b is updated according to its angular acceleration.

The angular and axial accelerations are determined from the governing equations derived in subsequent sections. Once the state vector has been updated, all other quantities in the model are computed directly from the current values of ω_p, ω_s, s, $\dot{s}$, and v_b.

Chapter 3

Derivation

3.1 Pulley Geometry

This section establishes the geometric relationships that govern how axial shift modifies the effective pulley radii and therefore the transmission ratio.

The analysis presented here is purely kinematic. No force balances or dynamic effects are considered. The goal is to describe how pulley geometry alone determines belt position and ratio evolution.

The objective is to derive explicit expressions for:

- Primary outer radius as a function of shift,
- Secondary outer radius as a function of shift,
- Center-to-center distance,
- CVT ratio and its rate of change.

Radius Definitions Used in This Section Because the belt has finite radial height, two geometric radii are used here: the *outer radius* (belt outer surface/contact geometry) and the *inner radius* (belt inner surface). Let h_b denote the belt height (belt radial thickness). Then,

$$r_{\text{inner}} = r_{\text{outer}} - h_b,$$

The effective torque radius used in torque and ratio relations is introduced later in Section 3.1.8.

3.1.1 Conceptual Overview of Variable Geometry

As introduced in Section 1.2, each pulley consists of two opposing conical sheaves. One sheave is fixed axially, while the other translates along the shaft.

The key controllable motion in the system is this *axial separation* between the sheaves. When the movable sheave translates toward the fixed sheave, the gap narrows and the belt is forced outward along the conical faces. When the sheaves separate, the belt settles deeper between them.

This axial motion does not directly change the pulley radius. Instead, the effective radius arises geometrically from where the belt sits within the sheave gap. In other words, axial displacement determines the belt's radial position, and therefore the effective pulley radius. The precise relationship between these quantities will be derived in the following subsection.

Because the belt length remains constrained, the effective radii of the primary and secondary pulleys cannot vary independently. An increase in the primary effective radius must therefore be accompanied by a corresponding decrease in the secondary effective radius. This geometric coupling is fundamental to CVT operation and forms the basis of the ratio model derived below.

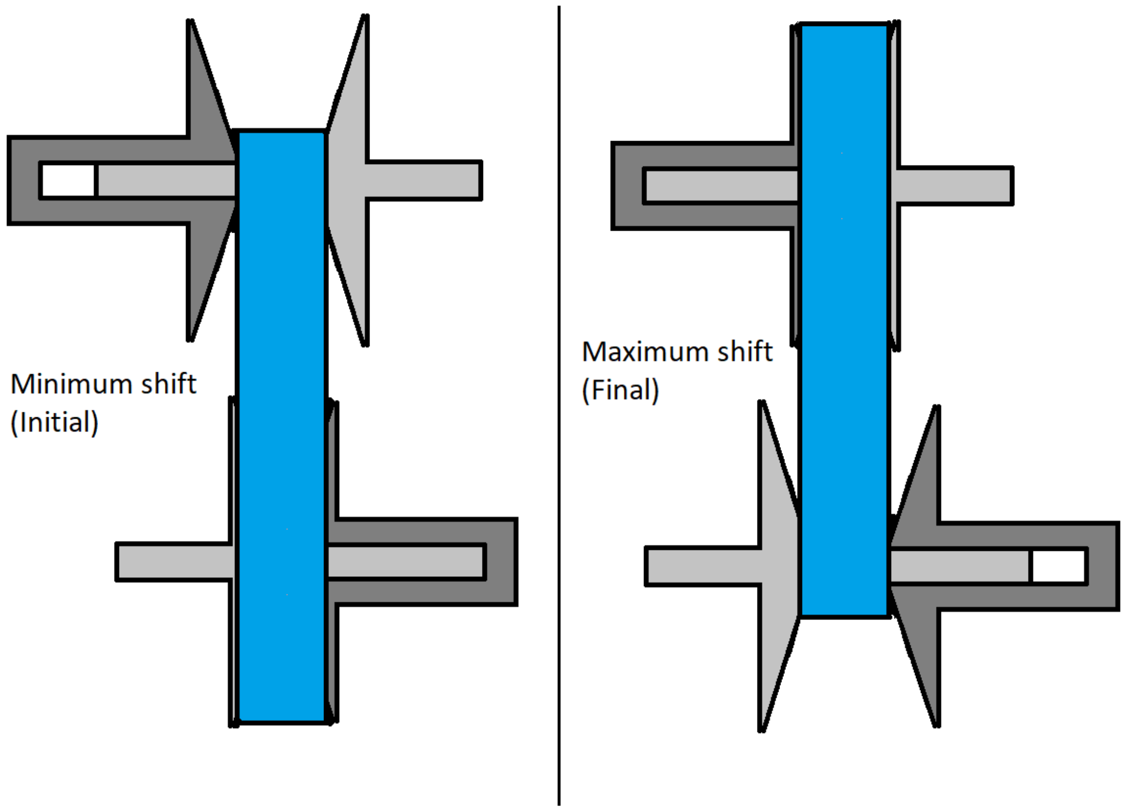


Figure 3.1: Conceptual illustration of variable pulley geometry. Changes in axial separation alter effective radii. Because belt length is constrained, an increase in primary effective radius corresponds to a decrease in secondary effective radius.

3.1.2 Definition of Axial Shift Variable

Let s denote the axial displacement of the movable primary sheave, defined according to the sign convention in Section 2.3.1.

At $s = 0$, the primary pulley is in its fully open configuration. The sheaves are at their maximum axial separation, and the belt rests at its minimum effective radius.

The shift coordinate s measures the amount of closure relative to this initial configuration. Increasing s corresponds to the movable sheave translating toward the fixed sheave, reducing the axial gap.

Deadzone and Travel Limits In practice, the initial sheave gap at $s = 0$ is larger than the belt width. As a result, small axial motion does not immediately alter the belt's contact position on the conical faces.

The interval

$$0 \leq s < s_{\text{dz}}$$

defines this *deadzone*. Within this region, the movable sheave translates axially but does not yet sufficiently compress the belt to change its effective contact radius. The belt remains at the same radial position despite the reduction in gap.

Once the sheave gap becomes small enough to fully engage the belt, further axial motion alters the belt position. For

$$s_{\text{dz}} \leq s \leq s_{\text{max}},$$

additional closure forces the belt outward along the conical faces, producing a proportional change in primary effective radius.

The shift coordinate is therefore bounded as

$$0 \leq s \leq s_{\text{max}}.$$

The upper limit s_{max} corresponds to the physical travel limit of the sheave assembly. At this position, the sheaves are fully closed and mechanically contact one another, forming a hard geometric boundary beyond which further axial motion is not possible.

3.1.3 Primary Outer Radius as a Function of Shift

When the primary sheaves close beyond the deadzone, the belt is forced outward along the conical faces. The relationship between axial closure and radial expansion follows directly from the cone geometry.

As shown in Figure 3.2, the sheave face forms a straight line in cross-section. The cone half-angle β is defined as the angle between the sheave face and the *radial direction* (As defined in Section 2.3.1).

For a small axial displacement Δs beyond the deadzone, the total closure is shared symmetrically between the two opposing sheaves, so each sheave contributes $\Delta s/2$.

From the right triangle formed by the sheave face,

$$\tan \beta = \frac{\Delta s/2}{\Delta r},$$

which gives

$$\Delta r = \frac{\Delta s}{2 \tan \beta}. \tag{3.1.1}$$

Thus, the cone angle determines how axial motion is converted into radial growth: smaller β produces greater radial change for the same axial displacement.

Taking the infinitesimal limit of Equation (3.1.1) yields the local geometric relationship between axial shift and effective radius,

$$\frac{dr}{ds} = \frac{1}{2 \tan \beta}. \tag{3.1.2}$$

Because the sheave face is straight, this derivative is constant, meaning that radial position varies linearly with axial displacement.

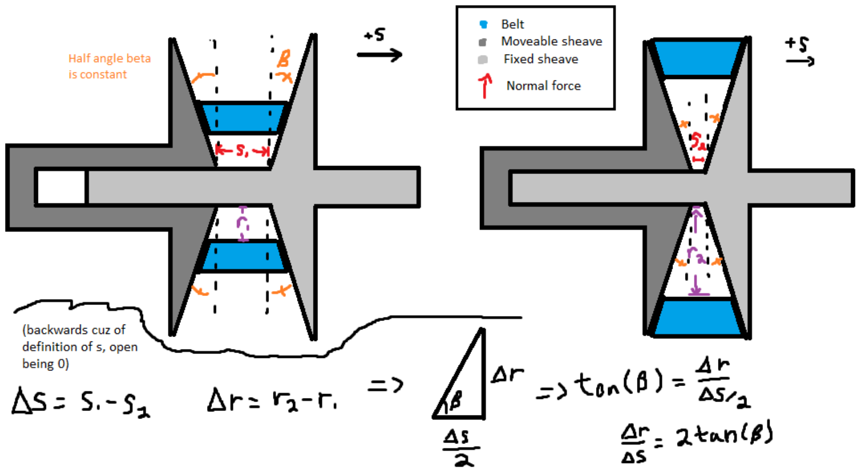


Figure 3.2: Axial–radial relationship for opposing conical sheaves. The angle β is defined between the sheave face and the radial direction.

Piecewise Radius Model Let $r_{p,\text{outer,min}}$ denote the minimum primary outer radius, corresponding to the fully open configuration ($s = 0$).

Within the deadzone, axial motion does not alter belt position. Once $s \geq s_{\text{dz}}$, only the displacement *beyond the deadzone* contributes to radial growth.

Primary Outer Radius **(3.1.3)**

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\text{outer,min}}, & 0 \leq s < s_{\text{dz}}, \\ r_{p,\text{outer,min}} + \dfrac{s - s_{\text{dz}}}{2\tan\beta}, & s_{\text{dz}} \leq s \leq s_{\max}. \end{cases}$$

3.1.4 Belt Length Constraint

In the previous subsection, the primary effective radius was expressed as a function of axial shift. The secondary radius, however, cannot be chosen independently. The belt has a fixed total length, and therefore any change in primary radius must be accompanied by a corresponding geometric adjustment of the secondary radius.

This section derives the geometric compatibility constraint imposed by constant belt length. By expressing the total belt length in terms of the primary radius r_p, secondary radius r_s, and center-to-center distance C, we obtain an implicit relationship that couples the two pulley radii.

The derivation follows directly from the belt path geometry shown in Figure 3.3. The geometric quantities used in the derivation are

- r_p — radius of the primary pulley
- r_s — radius of the secondary pulley
- C — center-to-center distance between pulley shafts
- ϕ_p — belt wrap angle on the primary pulley
- ϕ_s — belt wrap angle on the secondary pulley

- ℓ — length of a single straight tangent span

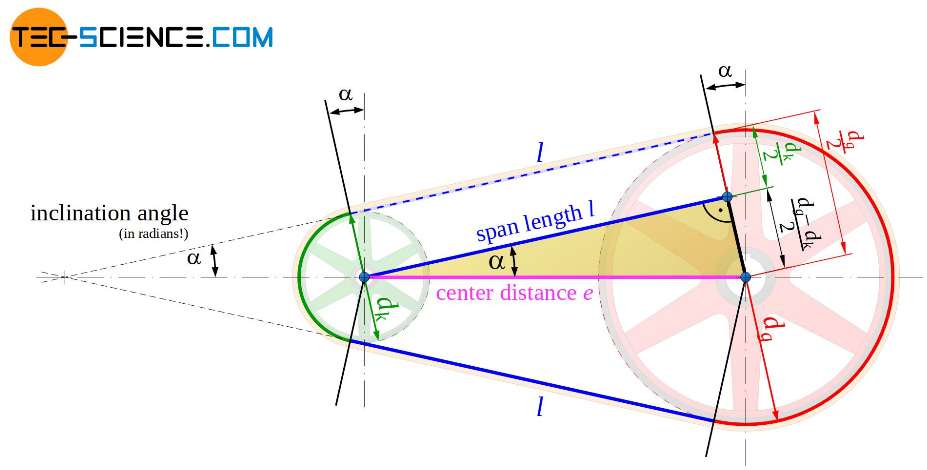


Figure 3.3: Belt geometry for two pulleys of radii r_p and r_s separated by center distance C. The belt wraps the pulleys with angles ϕ_p and ϕ_s and connects via two straight tangent spans ℓ.

Belt Length Decomposition The total belt length L_b is the sum of: (i) wrapped arc length on the primary pulley, (ii) wrapped arc length on the secondary pulley, and (iii) two straight tangent spans. Therefore,

$$L_b = r_p\phi_p + r_s\phi_s + 2\ell, \tag{3.1.4}$$

where ℓ is the length of a single straight tangent span segment.

Straight Span Length From the right-triangle geometry formed by the pulley centers and tangent points,

$$\ell = \sqrt{C^2 - (r_s - r_p)^2}. \tag{3.1.5}$$

Wrap Angles The wrap angles on the primary and secondary pulleys are

$$\phi_p = \pi - 2\lambda, \tag{3.1.6}$$
$$\phi_s = \pi + 2\lambda. \tag{3.1.7}$$

If $\lambda < 0$, these expressions automatically redistribute wrap between pulleys while preserving total belt length.

Tangency Angle (Signed Convention) Let λ denote the signed angle between the line of centers and the straight tangent belt segment (as defined in Figure 3.3). We define

$$\sin\lambda = \frac{r_s - r_p}{C}, \tag{3.1.8}$$

and equivalently,

$$\lambda = \sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{3.1.9}$$

Under this convention, λ may be negative when $r_p > r_s$. This signed definition ensures the wrap distribution automatically adjusts when the pulley ordering reverses.

Closed Form Belt-Length Expression Substituting Equation (3.1.5) and Equation (3.1.6)–Equation (3.1.7) into Equation (3.1.4) yields

$$\begin{aligned} L_b &= r_p(\pi - 2\lambda) + r_s(\pi + 2\lambda) + 2\sqrt{C^2 - (r_s - r_p)^2} \\ &= \pi(r_p + r_s) + 2\lambda(r_s - r_p) + 2\sqrt{C^2 - (r_s - r_p)^2}. \end{aligned} \tag{3.1.10}$$

Finally, substituting Equation (3.1.9) gives

$$L_b = \pi(r_p + r_s) + 2(r_s - r_p)\,\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{3.1.11}$$

Root Form (Geometric Compatibility Constraint) Rearranging Equation (3.1.11) yields

Belt Length Constraint Root Function **(3.1.12)**

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\,\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b.$$

The belt-length constraint is therefore

$$f(r_p, r_s, C) = 0. \tag{3.1.13}$$

The form of Equation (3.1.13) is unchanged if radii are defined at the belt outer surface, inner surface, or effective mid-height radius, provided all terms are referenced consistently. In particular, L_b must correspond to the same reference as r_p and r_s (e.g., outer-length with outer radii, inner-length with inner radii, or effective-length with effective radii).

Remark (Transcendental Nature and Approximation)

Equation (3.1.12) is transcendental due to the inverse-sine and square-root terms and therefore does not admit a closed-form solution.
In this work, the constraint is solved numerically using a bracketed root-finding method (e.g., bisection or Brent's method), which is deterministic and robust over the physically admissible interval.
It is common in simplified treatments to assume

$$\frac{r_s - r_p}{C} \approx 0,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this small-angle assumption, the belt-length equation reduces to a much simpler approximate form.
However, this approximation can introduce substantial geometric error when $\frac{r_s - r_p}{C}$ is not small. Appendix A quantifies this deviation and demonstrates that the approximation is not acceptable under realistic operating conditions.

3.1.5 Center-to-Center Distance Determination

In-order to make use of the belt-length constraint derived in Section 3.1.4, the center-to-center distance C must be determined.

The center-to-center distance C is a fixed geometric parameter of the assembled CVT system. Once the transmission is constructed, C does not vary during operation.

To determine C, we evaluate the belt-length constraint Equation (3.1.13) at the *known* initial configuration of the system.

Initial Geometric Configuration At startup, the CVT is assumed to be in its lowest ratio configuration:

- The primary pulley is at its minimum outer radius,

$$r_p = r_{p,\text{outer,min}}.$$

- The secondary pulley is at its maximum outer radius,

$$r_s = r_{s,\text{outer,max}}.$$

These values are determined by the physical geometry of the sheaves and represent the fully separated primary and fully closed secondary configuration.

Center Distance from Belt-Length Compatibility Substituting these known radii into Equation (3.1.12) yields

$$f(r_{p,\text{outer,min}}, r_{s,\text{outer,max}}, C) = 0. \tag{3.1.14}$$

This equation contains a single unknown, C, and is solved numerically to determine the fixed center-to-center distance consistent with the specified belt length L_b.

$$C = \underset{C}{\operatorname{root}}\ f(r_{p,\text{outer,min}}, r_{s,\text{outer,max}}, C)\,. \tag{3.1.15}$$

Remark (One-Time Geometric Solve)

The center distance C is determined once during model initialization. Thereafter, C is treated as a constant in all subsequent geometric and dynamic calculations.

3.1.6 Secondary Outer Radius as a Function of Shift

With the center-to-center distance C determined from Section 3.1.5, and the primary radius $r_{p,\text{outer}}(s)$ defined by Equation (3.1.3), the secondary outer radius is obtained by enforcing the belt-length compatibility constraint.

Geometric Coupling At any shift position s, the primary radius $r_{p,\text{outer}}(s)$ is known. The secondary radius r_s must therefore satisfy

$$f(r_{p,\text{outer}}(s), r_s, C) = 0, \tag{3.1.16}$$

where $f(\cdot)$ is defined in Equation (3.1.12).

Numerical Determination of $r_s(s)$ Because Equation (3.1.16) is transcendental in r_s, the secondary radius is obtained numerically for each value of s:

Secondary Outer Radius **(3.1.17)**

$$r_{s,\text{outer}}(s) = \underset{r_s}{\operatorname{root}}\ f(r_{p,\text{outer}}(s), r_s, C)\,.$$

Remark (Dynamic Root Solve)

Unlike the center-distance calculation in Section 3.1.5, which is performed once during initialization, this root solve is performed repeatedly as the shift coordinate s evolves during simulation. Because the physically admissible interval for r_s is bounded, a bracketed solver (e.g., Brent's method) converges rapidly.

With both $r_{p,\text{outer}}(s)$ and $r_{s,\text{outer}}(s)$ determined, all geometric quantities describing the belt path can now be expressed explicitly as functions of the shift coordinate s.

3.1.7 Wrap Angles as Functions of Shift

The wrap angles on the primary and secondary pulleys vary as the pulley radii change. Using the geometric relations from Section 3.1.4, the tangency angle becomes

$$\lambda(s) = \sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right). \tag{3.1.18}$$

Substitution into the wrap-angle definitions yields

$$\phi_p(s) = \pi - 2\sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right), \tag{3.1.19}$$

$$\phi_s(s) = \pi + 2\sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right). \tag{3.1.20}$$

3.1.8 Effective Torque Radius

All geometric relations derived thus far use the outer surface of the belt as the reference radius. This simplifies the belt-length formulation and pulley geometry.

Torque transmission, however, occurs through the belt cross-section. The appropriate moment arm for torque is therefore located approximately at the mid-height of the belt, as illustrated in Figure 3.4.

The effective torque radius for each pulley is defined as

$$r_{p,\text{eff}} = r_{p,\text{outer}} - \frac{h_b}{2}, \tag{3.1.21}$$

$$r_{s,\text{eff}} = r_{s,\text{outer}} - \frac{h_b}{2}. \tag{3.1.22}$$

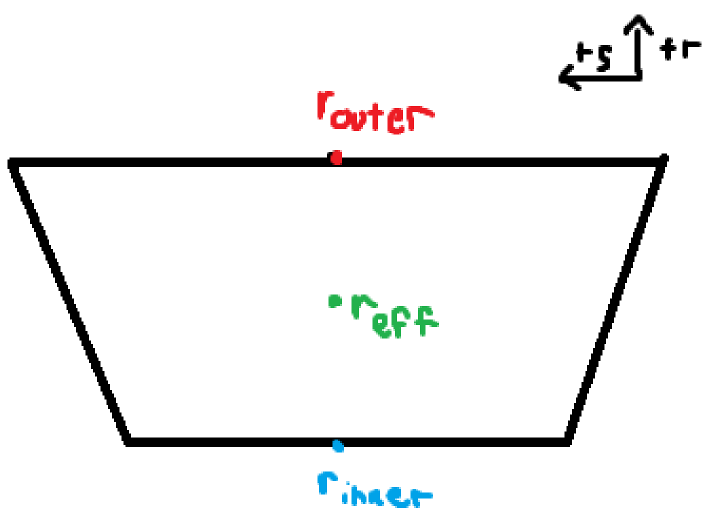


Figure 3.4: Belt cross-section showing the outer radius r_{outer}, inner radius r_{inner}, and effective torque radius r_{eff} located at the belt mid-height. The effective radius is used for all torque and power transmission calculations.

3.1.9 CVT Ratio Definition

The instantaneous transmission ratio is determined purely by pulley geometry. It is defined as the ratio of secondary to primary effective radii:

CVT Ratio **(3.1.23)**

$$R(s) = \frac{r_{s,\mathrm{eff}}(s)}{r_{p,\mathrm{eff}}(s)}.$$

Because both effective radii are functions of the shift coordinate s, the transmission ratio is likewise a function of s alone.

As the primary sheaves close, $r_{p,\mathrm{eff}}(s)$ increases. By belt-length compatibility, the secondary radius decreases. Consequently, the ratio $R(s)$ decreases with increasing s.

Throughout this document, *low ratio* and *high ratio* are named by speed-reduction behavior, not by the numerical magnitude of R. Using the no-slip relation $\omega_p = R\,\omega_s$, larger R means stronger reduction (primary faster than secondary), which we call *low ratio*; smaller R is the opposite, which we call *high ratio*.

Thus:

- Small s (minimum primary radius) corresponds to low ratio.
- Large s (maximum primary radius) corresponds to high ratio.

The CVT ratio is therefore a purely geometric quantity determined by the axial shift position.

3.1.10 Ratio Rate of Change

The CVT ratio is not only a function of shift position, but also changes in time whenever the sheaves are moving. This time variation matters because a changing ratio introduces additional kinematic coupling between the primary and secondary rotations. In other words, it is not sufficient to know the instantaneous ratio $R(s)$; the model also requires how quickly the ratio is changing.

This section derives an explicit expression for $\dot{R}$ in terms of the shift velocity $\dot{s}$ and geometric derivatives.

Start from the Ratio Definition

The ratio is defined by the effective radii as

$$R(s) = \frac{r_{s,\mathrm{eff}}(s)}{r_{p,\mathrm{eff}}(s)}. \tag{3.1.24}$$

Because R depends on time only through $s(t)$, the chain rule gives

$$\dot{R} = \frac{dR}{ds}\dot{s}. \tag{3.1.25}$$

Thus, the objective is to compute dR/ds.

Differentiate R with Respect to s

Differentiating Equation (3.1.24) with respect to s (quotient rule) gives

$$\frac{dR}{ds} = \frac{d}{ds}\left(\frac{r_{s,\text{eff}}}{r_{p,\text{eff}}}\right) = \frac{\left(\frac{dr_{s,\text{eff}}}{ds}\right) r_{p,\text{eff}} - r_{s,\text{eff}} \left(\frac{dr_{p,\text{eff}}}{ds}\right)}{r_{p,\text{eff}}^2}. \tag{3.1.26}$$

Rearranging into a form that exposes the required components:

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}} \frac{dr_{s,\text{eff}}}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \frac{dr_{p,\text{eff}}}{ds}. \tag{3.1.27}$$

At this point, the remaining task is to express the two derivatives $dr_{p,\text{eff}}/ds$ and $dr_{s,\text{eff}}/ds$ in computable form.

Primary Radius Derivative

The primary radius derivative follows directly from the sheave geometry, since the primary radius is prescribed explicitly as a function of shift. From the piecewise primary radius model Equation (3.1.3), the primary outer radius satisfies

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\text{outer,min}}, & 0 \le s < s_{\text{dz}}, \\ r_{p,\text{outer,min}} + \dfrac{s - s_{\text{dz}}}{2\tan\beta}, & s_{\text{dz}} \le s \le s_{\max}. \end{cases}$$

Differentiating with respect to s gives

Primary Radius Derivative **(3.1.28)**

$$\frac{dr_p}{ds} = \frac{dr_{p,\text{outer}}}{ds} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\ \dfrac{1}{2\tan\beta}, & s_{\text{dz}} \le s \le s_{\max}. \end{cases}$$

Because $r_{p,\text{eff}} = r_{p,\text{outer}} - h_b/2$ and h_b is constant, the same derivative applies:

$$\frac{dr_{p,\text{eff}}}{ds} = \frac{dr_p}{ds}. \tag{3.1.29}$$

Remark (At the Travel Limit)

If the model clamps s at the physical limit $s_{\max}$, then $r_{p,\text{outer}}$ (and thus $r_{p,\text{eff}}$) is constant beyond this point and the derivative is zero for $s > s_{\max}$. In this document we restrict to the admissible domain $0 \le s \le s_{\max}$.

Secondary Radius Derivative

Unlike the primary radius, the secondary radius is not prescribed explicitly as a function of shift. Instead, it is determined implicitly by the belt-length compatibility condition derived in Section 3.1.4. Thus, the secondary radius derivative must be obtained by differentiating the constraint itself.

The belt-length constraint is written as

$$f(r_p, r_s, C) = 0, \tag{3.1.30}$$

where $f(\cdot)$ is defined in Equation (3.1.12) and C is constant.

Since both $r_p = r_p(s)$ and $r_s = r_s(s)$ depend on the shift coordinate s, differentiating Equation (3.1.30) with respect to s gives, by the multivariable chain rule,

$$\frac{\partial f}{\partial r_p}\frac{dr_p}{ds} + \frac{\partial f}{\partial r_s}\frac{dr_s}{ds} = 0. \tag{3.1.31}$$

Solving Equation (3.1.31) for the secondary radius derivative yields

$$\frac{dr_s}{ds} = -\frac{\dfrac{\partial f}{\partial r_p}}{\dfrac{\partial f}{\partial r_s}}\frac{dr_p}{ds}. \tag{3.1.32}$$

Because the effective secondary radius differs from the outer radius only by the constant belt offset $h_b/2$, the same derivative applies to the effective radius:

$$\frac{dr_{s,\text{eff}}}{ds} = \frac{dr_s}{ds}. \tag{3.1.33}$$

Equation Equation (3.1.32) shows that the secondary radial motion is not independent, but is instead determined by the belt-length constraint and the primary radial motion.

At this point, the remaining task is to evaluate the two belt-constraint sensitivities $\partial f/\partial r_p$ and $\partial f/\partial r_s$, which appear in Equation (3.1.32). The following subsections derive these quantities explicitly for the belt-length function defined in Equation (3.1.12).

Partial Derivatives of the Belt Constraint

Recall the belt-length constraint written in root form as

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b = 0. \tag{3.1.34}$$

We now compute the two partial derivatives required in Equation (3.1.32), differentiating term-by-term and applying the product rule directly to the inverse-sine term.

Product rule for the inverse-sine factor Consider the factor

$$(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right).$$

Differentiating with respect to r_p (product rule) gives

$$\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \underbrace{\frac{\partial}{\partial r_p}(r_s - r_p)}_{(1)}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p)\underbrace{\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right)}_{(2)}. \tag{3.1.35}$$

We therefore compute the two sub-derivatives.

Derivative (1)

$$\frac{\partial}{\partial r_p}(r_s - r_p) = -1. \tag{3.1.36}$$

Derivative (2) Using the chain rule,

$$\frac{d}{dx}\sin^{-1}(x) = \frac{1}{\sqrt{1 - x^2}},$$

let

$$u = \frac{r_s - r_p}{C}.$$

Then

$$\frac{\partial u}{\partial r_p} = -\frac{1}{C}.$$

Therefore,

$$\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) = \frac{1}{\sqrt{1 - u^2}}\left(-\frac{1}{C}\right). \tag{3.1.37}$$

Now simplify the denominator:

$$\sqrt{1 - u^2} = \sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2} = \frac{\sqrt{C^2 - (r_s - r_p)^2}}{C}.$$

Substituting into Equation (3.1.37) yields

$$\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) = -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{3.1.38}$$

Substitute into the product rule Substituting Equation (3.1.36) and Equation (3.1.38) into Equation (3.1.35) gives

$$\begin{aligned}\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] &= (-1)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p)\left(-\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}\right) \\ &= -\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}. \end{aligned} \tag{3.1.39}$$

Derivative of the square-root term For the straight-span contribution,

$$2\sqrt{C^2 - (r_s - r_p)^2},$$

we apply the chain rule:

$$\frac{\partial}{\partial r_p}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{3.1.40}$$

Assemble $\partial f/\partial r_p$ Differentiating Equation (3.1.34) term-by-term gives

$$\begin{aligned}\frac{\partial f}{\partial r_p} &= \pi + 2\left(-\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right) + \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right).\end{aligned} \tag{3.1.41}$$

The square-root contributions cancel exactly.

Derivative with respect to r_s The same procedure yields

$$\frac{\partial}{\partial r_s}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}, \tag{3.1.42}$$

and

$$\frac{\partial}{\partial r_s}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = -\frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{3.1.43}$$

Assembling terms gives

$$\begin{aligned}\frac{\partial f}{\partial r_s} &= \pi + 2\left(\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right) - \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right).\end{aligned} \tag{3.1.44}$$

Summary of Belt Constraint Partial Derivatives From the derivations above, the partial derivatives of the belt-length constraint Equation (3.1.34) are

$$\frac{\partial f}{\partial r_p} = \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right), \tag{3.1.45}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{3.1.46}$$

These expressions are globally valid under the geometric definition of λ given in Equation (3.1.9) and require no piecewise case distinction.

Final Expression for the Secondary Radius Derivative

Equation Equation (3.1.32) expresses the secondary radius derivative in terms of the belt-constraint partial derivatives and the primary radius derivative:

$$\frac{dr_s}{ds} = -\frac{\dfrac{\partial f}{\partial r_p}}{\dfrac{\partial f}{\partial r_s}} \frac{dr_p}{ds}. \tag{3.1.47}$$

From Equation (3.1.28), the partial derivatives of the belt-length constraint Equation (3.1.12) are

$$\frac{\partial f}{\partial r_p} = \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right), \tag{3.1.48}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{3.1.49}$$

Substituting Equation (3.1.48) and Equation (3.1.49) into Equation (3.1.47) gives

$$\frac{dr_s}{ds} = -\frac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} \frac{dr_p}{ds}. \tag{3.1.50}$$

Finally, substituting the piecewise primary radius derivative Equation (3.1.28) yields

Secondary Radius Derivative **(3.1.51)**

$$\frac{dr_s}{ds} = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ -\dfrac{1}{2\tan\beta} \dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}, & s_{\mathrm{dz}} \le s \le s_{\max}. \end{cases}$$

Because $r_{s,\mathrm{eff}} = r_s - h_b/2$ with h_b constant, the same expression applies to the effective secondary radius derivative:

$$\frac{dr_{s,\mathrm{eff}}}{ds} = \frac{dr_s}{ds}. \tag{3.1.52}$$

Final Expression for dR/ds and $\dot{R}$

From Equation (3.1.27), the ratio derivative is

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}} \frac{dr_{s,\text{eff}}}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \frac{dr_{p,\text{eff}}}{ds}. \tag{3.1.53}$$

Using Equation (3.1.29) and Equation (3.1.52), this becomes

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}} \frac{dr_s}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \frac{dr_p}{ds}. \tag{3.1.54}$$

Substituting Equation (3.1.50) into Equation (3.1.54) gives

$$\frac{dR}{ds} = \left[-\frac{1}{r_{p,\text{eff}}} \frac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \right] \frac{dr_p}{ds}. \tag{3.1.55}$$

Finally, substituting the piecewise primary derivative Equation (3.1.28) yields the fully expanded result

$$\frac{dR}{ds} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\[2ex] \dfrac{1}{2\tan\beta} \left[-\dfrac{1}{r_{p,\text{eff}}} \dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} - \dfrac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \right], & s_{\text{dz}} \le s \le s_{\max}. \end{cases} \tag{3.1.56}$$

Using the chain rule Equation (3.1.25), $\dot{R} = (dR/ds)\dot{s}$. Substituting Equation (3.1.56) and rearranging gives

Rate of Change of Transmission Ratio **(3.1.57)**

$$\dot{R} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\[2ex] -\dfrac{\dot{s}}{2\tan\beta\, r_{p,\text{eff}}} \left[\dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} + \dfrac{r_{s,\text{eff}}}{r_{p,\text{eff}}} \right], & s_{\text{dz}} \le s \le s_{\max}. \end{cases}$$

3.1.11 Section Summary and Forward Connection

This section established the complete *geometric foundation* of the CVT model. The development was intentionally kinematic: the objective was to derive a closed, computable mapping from axial shift to pulley geometry and transmission ratio, without introducing force balance, slip behavior, or torque relations.

At a high level, the geometry defines the chain

$$s(t) \longrightarrow r_{p,\text{eff}}(s),\ r_{s,\text{eff}}(s) \longrightarrow R(s) \longrightarrow \dot{R}(s, \dot{s}),$$

with each intermediate relationship derived explicitly from pulley and belt geometry.

Key Results Established The following geometric components are now defined and ready for use in subsequent modeling:

1. **Primary radius law.** Equation (3.1.3) provides the piecewise relationship between axial shift s and primary outer radius, including deadzone behavior and cone-angle dependence.
2. **Belt-length compatibility constraint.** Equation (3.1.13) enforces constant belt length and defines the implicit, nonlinear coupling between r_p, r_s, and the center distance C.
3. **Center distance determination.** Section 3.1.5 fixes the assembled center-to-center distance C by evaluating the belt constraint at a known geometric configuration. This solve is performed once during initialization.
4. **Secondary radius as a function of shift.** Section 3.1.6 defines $r_s(s)$ implicitly by solving the belt constraint at each shift position, producing the required redistribution of belt radius between pulleys.
5. **Wrap angles as functions of shift.** Section 3.1.7 defines the wrap angles $\phi_p(s)$ and $\phi_s(s)$ follow directly from the tangency geometry and are available for later contact, work, and belt-distribution calculations.
6. **Effective torque radii.** Section 3.1.8 corrects outer radii to effective radii using belt thickness, ensuring subsequent torque relations use an appropriate moment arm.
7. **Ratio and ratio rate.** Equation (3.1.23) defines the geometric ratio $R(s)$, and Section 3.1.10 provides $\dfrac{dR}{ds}$ and $\dot{R}$ via Equation (3.1.56) and Equation (3.1.57), including the deadzone condition where R is constant.

Structural Role in the Full Model These results form the geometric backbone of the CVT simulation. All subsequent sections treat the relationships derived here as fixed mappings. In particular, later torque and force models will use $r_{p,\text{eff}}(s)$ and $r_{s,\text{eff}}(s)$ as inputs, and shifting kinematics will use $R(s)$ and $\dot{R}(s, \dot{s})$ to relate primary and secondary motion during ratio change.

Several key properties are now clear: (i) the geometric coupling is inherently nonlinear (through the belt-length constraint), (ii) the deadzone is explicitly represented, and (iii) within the admissible travel range the mappings are differentiable and suitable for continuous-time simulation.

Transition to Dynamics With the geometry fully specified, the next objective is to determine how the shift coordinate evolves in time. Subsequent sections introduce axial force generation (from pulley mechanisms and belt interaction) to determine $\dot{s}$ and $\ddot{s}$. Once $\dot{s}$ is known, the ratio-rate expression Equation (3.1.57) provides the corresponding ratio evolution and the required kinematic coupling between rotating components during shifting.

3.2 Pulley Axial Force Models

The geometric relations developed in the previous section define how the transmission ratio varies with axial shift. What remains is to determine *why* the shift coordinate $s(t)$ changes in time.

Axial motion of the sheaves is driven by clamping forces generated within the primary and secondary assemblies. On the primary side, centrifugal flyweights and springs produce an inward axial force that tends to close the sheaves and increase radius. On the secondary side, torque reaction and spring preload generate an opposing axial force.

The evolution of the shift coordinate is governed by the net axial force acting on the movable sheave assembly. Applying Newton's second law in the axial direction (Theory 1 (Newton's Second Law (Translation))) gives

$$m\,\ddot{s} = F_{p,\mathrm{ax}} - F_{s,\mathrm{ax}}, \tag{3.2.1}$$

where $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ denote the axial forces generated by the primary and secondary mechanisms, respectively, and m is the effective axial inertia of the moving components.

The purpose of this section is to derive expressions for $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ as functions of the current system state. Once these force models are established, the shift acceleration $\ddot{s}$ follows directly, completing the dynamic description of axial motion.

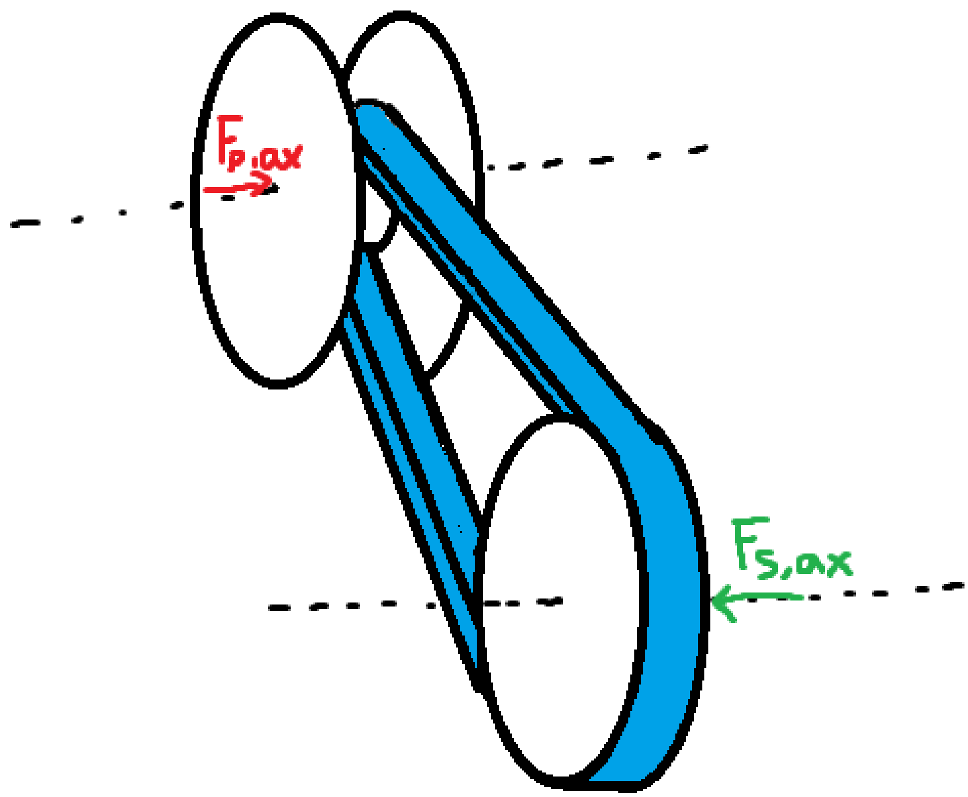


Figure 3.5: Axial force balance governing sheave motion. Primary and secondary mechanisms generate opposing axial forces. The net force determines the shift acceleration $\ddot{s}$.

Although the specific actuator mechanisms modeled here correspond to a flyweight-driven primary and a torque-reactive helix secondary, the formulation is modular: any actuator model may be substituted provided it returns the net axial force as a function of state.

3.2.1 Ramp Geometry Parameterization

Both pulley actuators redirect forces through inclined contact surfaces. As the shift coordinate s evolves, mechanical elements slide along these surfaces, converting radial or circumferential forces into axial clamping force.

To keep the formulation general and modular, the ramp geometries are parameterized directly as functions of the axial shift coordinate s. Any actuator profile may therefore be substituted provided it supplies the appropriate geometric mapping and derivative.

Primary Ramp Geometry (Axial-to-Radial Mapping) On the primary pulley, centrifugal force acting on the flyweights is redirected into axial closing force through a ramp surface (see Figure 3.6). As the movable sheave translates axially, the flyweight slides along this ramp.

We represent the ramp by an axial-to-radial profile

$$r_f = r_f(s), \tag{3.2.2}$$

where $r_f(s)$ is the effective flyweight radius.

An incremental axial displacement ds produces a radial displacement dr_f. The local slope of the ramp is therefore dr_f/ds. The instantaneous primary ramp angle is denoted by $\alpha_p(s)$ and is defined with respect to the axial direction as

$$\tan\alpha_p(s) = \frac{dr_f}{ds}, \qquad \alpha_p(s) = \tan^{-1}\left(\frac{dr_f}{ds}\right). \tag{3.2.3}$$

This definition follows directly from the geometry of the sliding motion: a steeper ramp (larger dr_f/ds) produces greater radial change per unit axial travel and therefore greater axial gain when radial force is redirected through the surface.

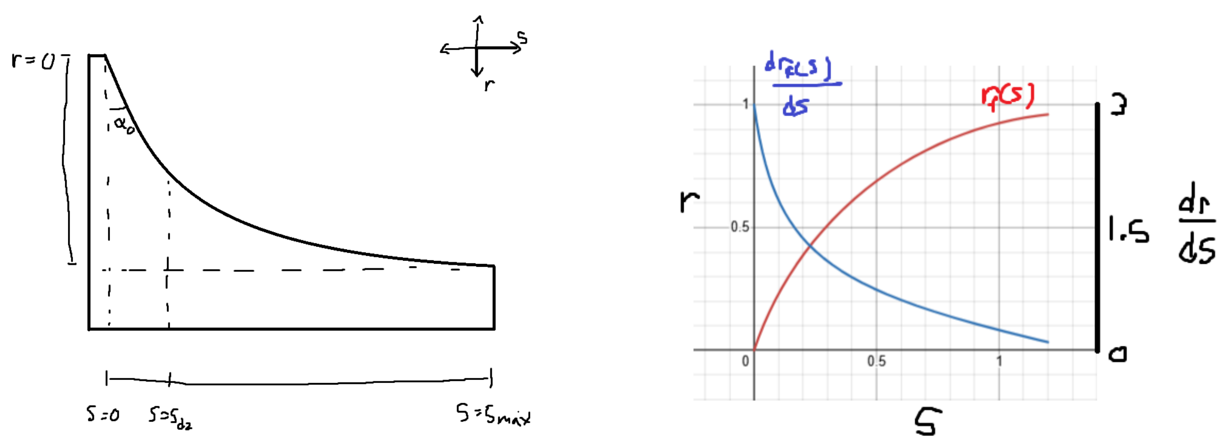


Figure 3.6: Primary ramp geometry. **Left:** As axial displacement s increases, the flyweight slides along the ramp, producing radial change dr_f. The instantaneous ramp angle satisfies $\tan\alpha_p = dr_f/ds$. **Right:** Example profiles of $r_f(s)$ and its slope dr_f/ds.

Primary Admissibility Conditions In order to be a valid primary ramp profile $r_f(s)$, it must satisfy:

1. $r_f(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{dr_f}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{dr_f}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the ramp angle $\alpha_p(s)$ is well-defined and continuous. Condition (2) guarantees that increasing axial displacement does not reduce flyweight radius. Condition (3) ensures the ramp angle remains strictly below $\pi/2$ and the axial gain $\tan\alpha_p$ remains finite.

Secondary Helix Geometry (Axial-to-Rotation Mapping) On the secondary pulley, axial travel produces relative rotation of a helical ramp mechanism, loading a torsional spring and generating torque-reactive clamping (see Figure 3.7).

We parameterize this kinematically as

$$\theta_s = \theta_s(s), \tag{3.2.4}$$

where $\theta_s(s)$ is the helix-induced rotation.

An incremental rotation $d\theta_s$ corresponds to a circumferential displacement $r_h d\theta_s$ along a cylindrical surface of effective radius r_h. The instantaneous secondary helix angle is denoted by $\alpha_s(s)$ and is defined from the local slope of the ramp surface:

$$\tan\alpha_s = \frac{\text{axial rise}}{\text{circumferential run}} = \frac{ds}{r_h d\theta_s}.$$

Rewriting in differential form gives

$$\tan\alpha_s(s) = \frac{1}{r_h \dfrac{d\theta_s}{ds}}. \tag{3.2.5}$$

Thus, a shallower helix (smaller α_s) produces greater rotation per unit axial travel, while a steeper helix produces less rotation. This inverse relationship arises naturally from the geometry of motion on a cylindrical surface.

Remark (Contrast with Primary Ramp)

The primary ramp behaves like a simple wedge: radial displacement is directly proportional to axial travel, so the ramp angle is naturally defined by $\tan\alpha_p = dr_f/ds$.
The secondary mechanism instead behaves like a power screw. Axial motion is produced through rotation around a cylindrical surface, so the relevant geometric ratio is how much rotation occurs per unit axial travel. This leads to the slope definition $ds/(r_h d\theta_s)$ and the inverse relationship in Equation (3.2.5).
In short, the primary redirects force through a wedge, while the secondary converts torque through a screw-like helix. The difference is purely geometric and reflects the distinct mechanisms by which axial motion is generated.

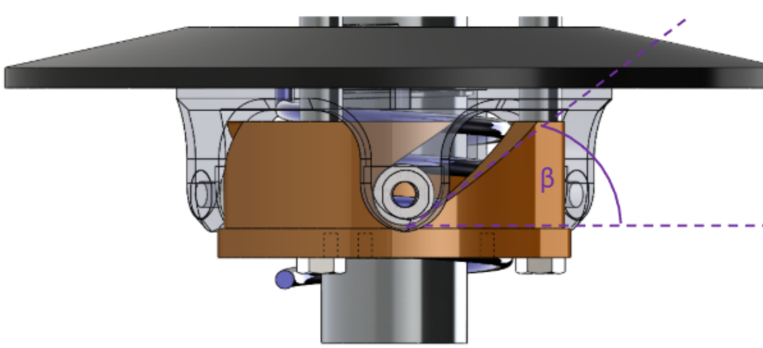


Figure 3.7: Secondary helix geometry. Axial travel s induces rotation $\theta_s(s)$ around a cylindrical surface of radius r_h. The instantaneous helix angle satisfies $\tan\alpha_s = ds/(r_h d\theta_s)$.

Secondary Admissibility Conditions The helix rotation mapping $\theta_s(s)$ must satisfy:

1. $\theta_s(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{d\theta_s}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{d\theta_s}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the torque-to-axial conversion remains continuous. Condition (2) guarantees that increasing axial displacement increases rotation. Condition (3) ensures the effective helix angle remains strictly below $\pi/2$ and the geometric gain remains finite.

Remark (Piecewise Profiles in Practice)

The C^1 requirement above is sufficient for all derivations in this document. In practice, piecewise C^1 profiles (continuous everywhere, continuously differentiable except at finitely many junction points where left and right derivatives exist) are also admissible for numerical simulation. At junction points, one-sided derivatives provide the required slope values, and the resulting force discontinuities pose no difficulty for standard ODE solvers. However, large slope discontinuities can introduce local stiffness into the system: the abrupt change in force gradients may require adaptive solvers to reduce the timestep significantly near the junction, degrading computational efficiency. In extreme cases, explicit integrators may struggle to maintain stability without impractically small step sizes, and implicit or stiff-aware solvers may be preferable.

Derivations for common ramp and helix profiles are provided in Appendix B.

3.2.2 Primary Axial Force Model

The primary axial clamping force arises from the competition between centrifugal loading of the flyweights and the restoring force of the primary compression spring.

As shown in Figure 3.8, each flyweight of mass m_f rotates with angular speed ω_p at an instantaneous radius $r_{f,0} + r_f(s)$, where $r_{f,0}$ is the flyweight radius in the reference configuration $s = 0$ and $r_f(s)$ is the additional radial displacement induced by the ramp. The resulting outward centrifugal force is redirected by the ramp geometry described in Section 3.2.1 into an axial closing force on the movable primary sheave.

Opposing this motion is the primary compression spring, characterized by stiffness k_p and preload $x_{p,0}$. As the sheaves close (s increases), the spring compression increases, producing a larger restoring force that resists further closure.

The resulting primary axial force is therefore the ramp-redirected centrifugal contribution minus the spring force. This quantity, denoted $F_{p,ax}(s, \omega_p)$, provides the primary-side contribution to the generalized axial force used in the shift equation of motion.

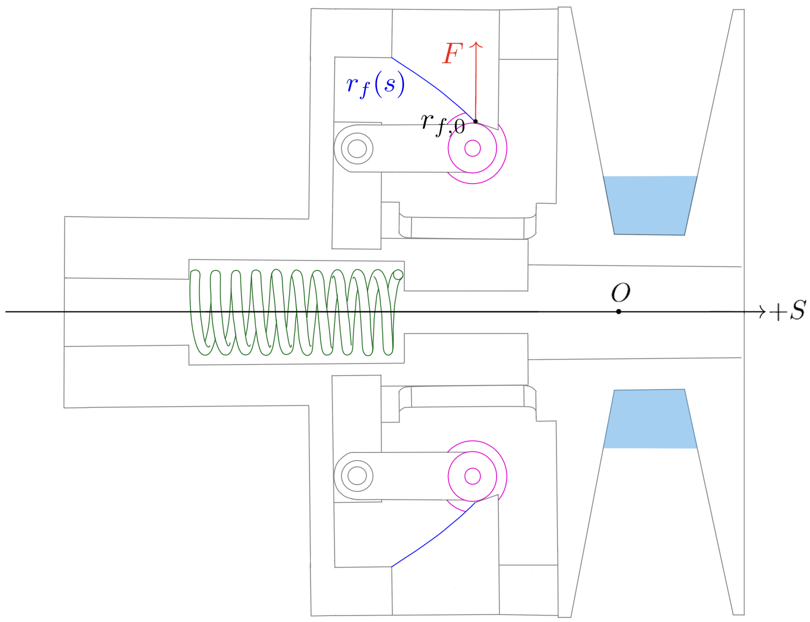


Figure 3.8: Primary CVT axial-force mechanism. Each flyweight of mass m_f rotates at angular speed ω_p and experiences an outward centrifugal force at total radius $r_{f,0} + r_f(s)$. The ramp geometry redirects this radial loading into an axial closing force on the movable primary sheave in the $+s$ direction, while the primary compression spring (k_p, $x_{p,0}$) provides an opposing restoring force. The lower flyweight–ramp pair is symmetric and shown unlabeled for clarity.

Centrifugal Loading of Flyweights From Theory 5 (Centrifugal Force),

$$F_c = m\omega^2 r.$$

Let

- m_f denote the flyweight mass,
- $r_{f,0}$ denote the initial flyweight radius at $s = 0$,
- $r_f(s)$ denote the incremental radial change produced by the ramp.

The total effective flyweight radius is therefore

$$r_{f,\mathrm{tot}}(s) = r_{f,0} + r_f(s).$$

The radial centrifugal force is

$$F_{p,\mathrm{rad}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big). \tag{3.2.6}$$

Radial-to-Axial Projection The ramp redirects radial force into axial force. As shown in Figure 3.9, the axial component of a force acting along the ramp is obtained by

$$F_{p,\text{axial component}} = F_{p,\text{rad}} \tan \alpha_p(s). \tag{3.2.7}$$

From Equation (3.2.3), the ramp angle satisfies

$$\tan \alpha_p(s) = \frac{dr_f}{ds}.$$

Substituting gives

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds}. \tag{3.2.8}$$

Thus the axial gain of centrifugal force is governed by the local ramp slope, while the centrifugal magnitude depends on the total flyweight radius.

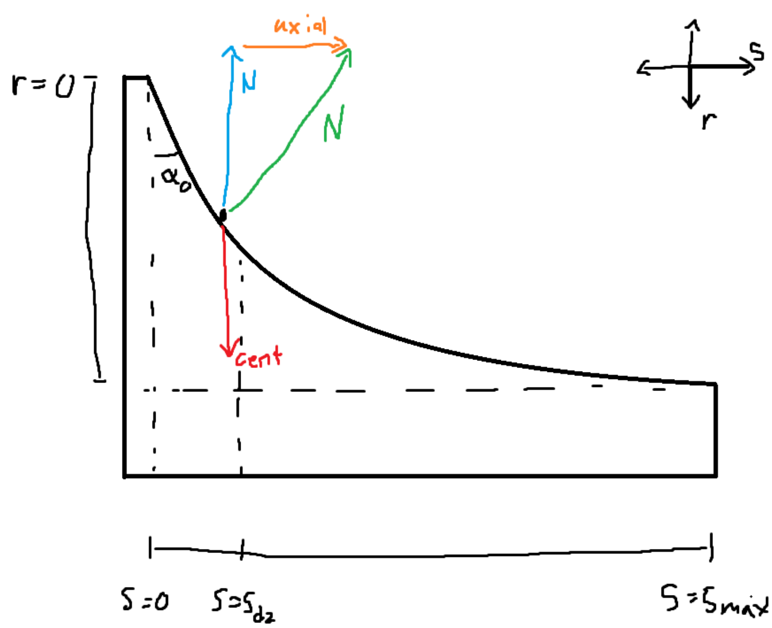


Figure 3.9: Primary ramp geometry. Radial centrifugal force is redirected into axial force through the ramp. The projection introduces a factor of $\tan \alpha_p$, which equals dr_f/ds by definition of the ramp slope.

Primary Compression Spring From Theory 6 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let k_p denote primary stiffness and $x_{p,0}$ preload.

$$F_{p,\text{spring}}(s) = k_p(x_{p,0} + s). \tag{3.2.9}$$

Net Primary Axial Force The net axial force generated by the primary mechanism is the difference between the centrifugal axial component and the spring force:

Primary Axial Force **(3.2.10)**

$$F_{p,\text{ax}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds} - k_p(x_{p,0} + s).$$

3.2.3 Secondary Axial Force Model

The secondary pulley is *torque-reactive*: its axial clamping force is generated in response to torque transmitted through the CVT. Unlike the primary, which produces clamping based on rotational speed, the secondary produces clamping primarily when torque flows through it.

The mechanism consists of a helical ramp between the two sheaves and a single spring located between them. This spring serves two roles:

- It acts axially in compression as the sheaves separate.
- It resists relative rotation torsionally as the helix turns.

Thus, axial motion and rotational motion of the helix both load the same spring.

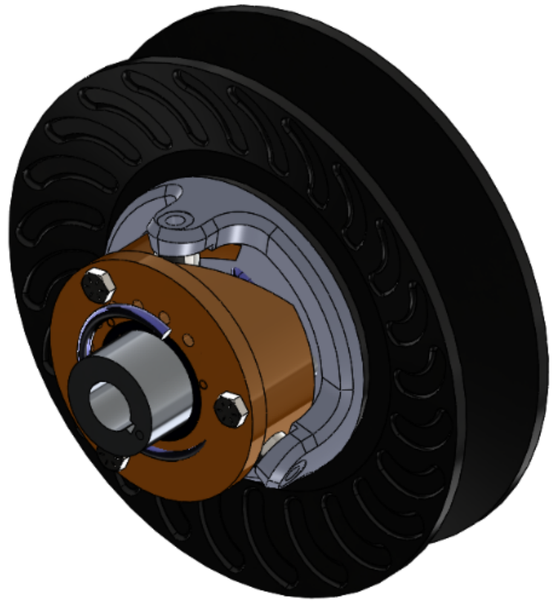


Figure 3.10: Secondary CVT assembly showing the helix mechanism and spring. Torque τ_s transmitted through the belt produces tangential force at the helix ramps, which is redirected into axial clamping force.

Physical Mechanism (Screw Analogy) The secondary helix behaves mechanically like a screw.

If you try to spin a screw in free space, nothing happens axially. If you spin a screw into a block that is not restrained, the block simply rotates with the screw—again producing no axial force. Axial force only develops when there is resistance to rotation on one side while torque is applied on the other.

In other words, a screw converts *reacted torque* into axial force.

The secondary helix operates on the same principle. The belt applies a driving torque to the secondary pulley. If the driven shaft were free, the entire assembly would rotate without developing tangential reaction force at the helix. However, when the driven load resists rotation, torque is transmitted through the helix ramps.

Those ramps are inclined surfaces—just like screw threads. As torque attempts to rotate one helix face relative to the other, contact forces develop along the inclined surfaces. Because the surfaces are angled, this tangential reaction force has an axial component.

The result is axial clamping force.

The key distinction from a conventional screw is direction: in a normal screw, torque pulls two components together. In the CVT helix, the orientation of the ramp is such that torque causes the movable sheave to be pushed outward, increasing belt clamping.

Thus, the secondary does not clamp because it spins. It clamps because torque is being transmitted and resisted.

Definition of Transmitted Torque Let τ_s denote the torque transmitted through the secondary shaft. This is the torque carried from the belt, through the helix mechanism, and into the driven load.

For the present derivation, τ_s is treated as a known input.

Torsional Spring Torque As the helix rotates relative to the opposing sheave, the spring is twisted torsionally.

From Theory 7 (Hooke's Law (Torsional)),

$$\tau = k\Delta\theta.$$

Let $k_{s,\theta}$ denote torsional stiffness and $\theta_{s,0}$ preload.

$$\tau_{s,\text{spring}}(s) = k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big). \tag{3.2.11}$$

This torsional component of the same spring contributes to helix loading even when no torque is transmitted ($\tau_s = 0$), providing baseline clamping.

Total Torque Applied to Helix

$$\tau_{s,\text{total}} = \tau_s + \tau_{s,\text{spring}}(s). \tag{3.2.12}$$

Both contributions act in the same rotational direction at the helix. The transmitted torque loads the helix through belt force, and the torsional spring resists relative rotation. Both therefore generate axial clamping through the same geometry.

Torque-to-Axial Conversion Through the Helix From Theory 3 (Torque Definition),

$$\tau = F_t r_h \quad \Rightarrow \quad F_t = \frac{\tau_{s,\text{total}}}{r_h}.$$

The tangential force F_t acts along two symmetric helix ramp faces. Each ramp carries half of the total tangential load, introducing the factor of 1/2 in the axial projection below.

From helix geometry (see Figure 3.11),

$$F_{s,\text{helix}} = \frac{F_t}{2\tan\alpha_s(s)}. \tag{3.2.13}$$

Substituting F_t and using

$$\tan\alpha_s(s) = \frac{1}{r_h \dfrac{d\theta_s}{ds}}$$

gives

$$F_{s,\text{helix}}(s, \tau_s) = \frac{\tau_{s,\text{total}}}{2}\frac{d\theta_s}{ds}. \tag{3.2.14}$$

Notably, the helix radius r_h cancels: torque is converted into axial force according to the geometric rotation-per-travel relationship $d\theta_s/ds$, consistent with screw mechanics.

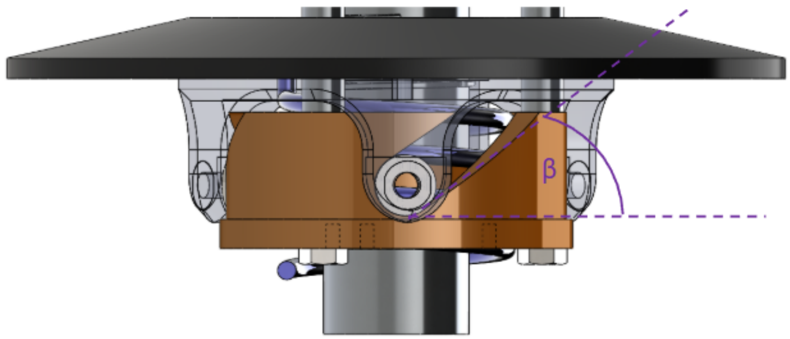


Figure 3.11: Helix free-body concept. Tangential force from shaft torque is redirected into axial force along the inclined ramps. Two symmetric ramp faces share the tangential load, producing the 1/2 factor.

Secondary Compression Spring The same spring also acts axially between the sheaves. At $s = 0$, the spring is pre-compressed by an amount $x_{s,0}$, producing a baseline separating force that assists upshift.

As the sheaves move apart (increasing s), its compression increases further.

From Theory 6 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let $k_{s,x}$ denote axial stiffness. Because increasing s increases compression, the effective compression is

$$\Delta x = x_{s,0} + s.$$

Thus,

$$F_{s,\text{comp}}(s) = k_{s,x}(x_{s,0} + s). \tag{3.2.15}$$

This axial spring force acts in the same direction as the helix force, providing additional clamping as the sheaves separate.

Net Secondary Axial Force The total secondary axial force is the sum of:

- Helix-generated torque-reactive clamping,
- Torsional spring contribution (through the helix),
- Direct axial compression spring force.

Secondary Axial Force **(3.2.16)**

$$F_{s,\mathrm{ax}}(s,\tau_s) = \underbrace{\frac{\tau_s + k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big)}{2}\frac{d\theta_s}{ds}}_{\text{torque-reactive helix (including torsional spring)}} + \underbrace{k_{s,x}(x_{s,0}+s)}_{\text{axial spring}}.$$

3.2.4 Belt Centrifugal Force on the Wrap

In addition to the actuator and spring forces derived previously, the belt itself generates centrifugal loading as it rotates around each pulley. This section derives the resulting contribution to axial clamping force.

Throughout this section, let

$$j \in \{p, s\}$$

denote either pulley, where $j = p$ corresponds to the primary and $j = s$ corresponds to the secondary.

From Theory 5 (Centrifugal Force), a mass m rotating at angular velocity ω at radius r experiences centrifugal force

$$F_c = m\omega^2 r.$$

To apply this to the belt wrapped around pulley j, three quantities must be defined:

1. the radius at which the belt mass rotates,
2. the mass of belt material carried around the wrap,
3. the angular velocity of that belt mass.

Each is developed below.

1. Radius at Which the Belt Mass Rotates Centrifugal force acts at the center of mass of the rotating belt element, not at the outer or inner belt surface. The location of this center of mass therefore determines the radius that must be used in the centrifugal force calculation.

The belt cross-section is approximated as a trapezoid (see Figure 3.12) with:

- radial thickness h_b,
- outer width b_{out},
- inner width b_{in}.

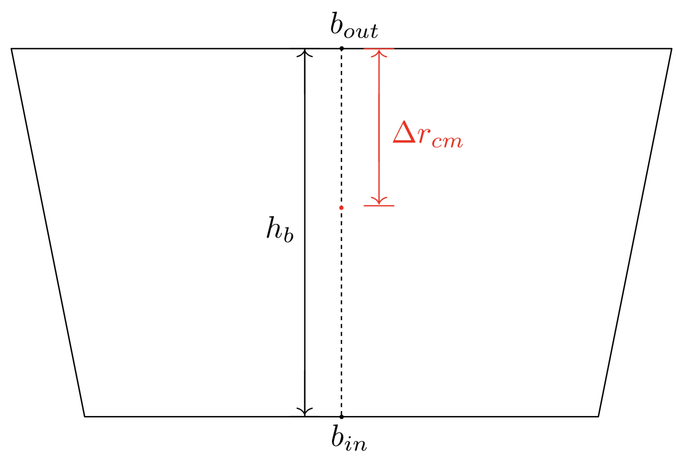


Figure 3.12: Trapezoidal belt cross-section. The outer width b_{out} corresponds to the surface away from the pulley axis; the inner width b_{in} corresponds to the surface closer to the axis. The radial thickness is h_b and the centroid is located at Δr_{cm} from the outer surface.

The cross-sectional area of the trapezoid is

$$A_b = \frac{h_b}{2} \left(b_{\text{out}} + b_{\text{in}} \right). \tag{3.2.17}$$

Let y denote radial distance measured inward from the outer surface, so that $y = 0$ at the outer surface and $y = h_b$ at the inner surface. For a trapezoid with linear sidewalls, the local width varies linearly with y:

$$b(y) = b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y. \tag{3.2.18}$$

The centroid location Δr_{cm}, measured inward from the outer surface, follows from the area-centroid definition:

$$\Delta r_{\text{cm}} = \frac{1}{A_b} \int_0^{h_b} y \, b(y) \, \mathrm{d}y. \tag{3.2.19}$$

Substituting Equation (3.2.18) into Equation (3.2.19) gives

$$
\begin{aligned}
\int_0^{h_b} y\, b(y)\ \mathrm{d}y &= \int_0^{h_b} y \left(b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y \right)\ \mathrm{d}y \\
&= \int_0^{h_b} b_{\text{out}} y\ \mathrm{d}y + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \int_0^{h_b} y^2\ \mathrm{d}y \\
&= \frac{h_b^2}{2} b_{\text{out}} + \frac{h_b^2}{3} \left(b_{\text{in}} - b_{\text{out}} \right) \\
&= \frac{h_b^2}{6} \left(b_{\text{out}} + 2 b_{\text{in}} \right). \qquad (3.2.20)
\end{aligned}
$$

Dividing by the area from Equation (3.2.17) yields

$$
\Delta r_{\text{cm}} = h_b \, \frac{b_{\text{out}} + 2 b_{\text{in}}}{3 \left(b_{\text{out}} + b_{\text{in}} \right)}. \qquad (3.2.21)
$$

For pulley j, let $r_{j,\text{out}}(s)$ denote the outer belt radius. The radius at which the belt mass rotates is therefore

$$
r_{j,\text{cm}}(s) = r_{j,\text{out}}(s) - \Delta r_{\text{cm}}. \qquad (3.2.22)
$$

2. Mass of Belt Material on the Wrap To determine the centrifugal loading on pulley j, we next determine the amount of belt mass associated with an infinitesimal portion of its wrap.

Let θ denote the angular coordinate measured from the center of the wrap, so that the wrapped region on pulley j spans

$$
\theta \in \left[-\frac{\phi_j}{2}, \ \frac{\phi_j}{2} \right],
$$

where $\phi_j(s)$ is the total wrap angle on pulley j (see Figure 3.13).

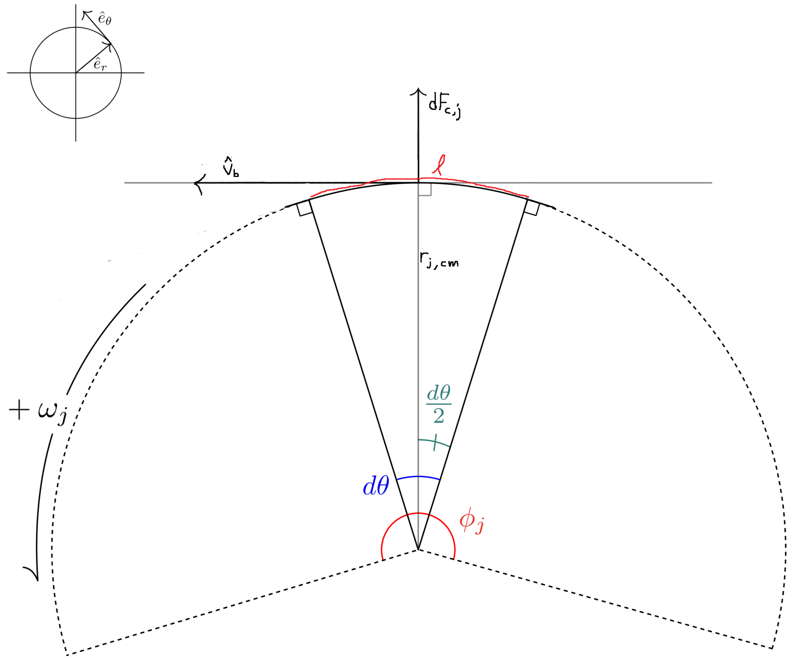


Figure 3.13: Belt wrapped around pulley j. A differential element at angle θ has arc length $\mathrm{d}\ell = r_{j,\mathrm{cm}}\ \mathrm{d}\theta$ and experiences radially outward centrifugal force.

A differential belt element subtending angle $\mathrm{d}\theta$ at the mass-centroid radius $r_{j,\mathrm{cm}}$ has arc length

$$\mathrm{d}\ell = r_{j,\mathrm{cm}}\ \mathrm{d}\theta.$$

Since the belt has cross-sectional area A_b, the differential belt element has volume

$$\mathrm{d}V_j = A_b\ \mathrm{d}\ell. \tag{3.2.23}$$

Substituting $\mathrm{d}\ell = r_{j,\mathrm{cm}}\ \mathrm{d}\theta$ gives

$$\mathrm{d}V_j = A_b\, r_{j,\mathrm{cm}}\ \mathrm{d}\theta. \tag{3.2.24}$$

Let ρ_b denote the belt material density (kg/m^3). The corresponding differential mass is then

$$\begin{aligned}\mathrm{d}m_j &= \rho_b\ \mathrm{d}V_j \\ &= \rho_b A_b\, r_{j,\mathrm{cm}}\ \mathrm{d}\theta. \end{aligned} \tag{3.2.25}$$

This gives the amount of belt material associated with an infinitesimal portion of the wrap on pulley j.

3. Angular Velocity of the Belt Mass The rotating belt mass on pulley j must also be assigned an angular speed. From Section 2.5, v_b denotes the belt transport speed. The corresponding angular velocity of the belt mass centroid on pulley j is then

$$\omega_{b,j} = \frac{v_b}{r_{j,\text{cm}}(s)}. \tag{3.2.26}$$

This relation expresses the fact that the same belt transport speed corresponds to different local angular velocities depending on the radius of motion.

Distributed Centrifugal Load Along the Wrap With the radius, differential mass, and angular velocity now defined, the centrifugal force on the differential belt element follows from Theory 5 (Centrifugal Force):

$$\begin{aligned} \mathrm{d}F_{c,j} &= \mathrm{d}m_j\, \omega_{b,j}^2\, r_{j,\text{cm}} \\ &= (\rho_b A_b\, r_{j,\text{cm}}\, \mathrm{d}\theta) \left(\frac{v_b}{r_{j,\text{cm}}}\right)^2 r_{j,\text{cm}} \\ &= \rho_b A_b\, v_b^2\, \mathrm{d}\theta. \end{aligned} \tag{3.2.27}$$

This force acts radially outward at each point along the wrap.

Total Radial Force Over the Wrap Integrating Equation (3.2.27) over the full wrap angle gives the total radially-directed centrifugal force on pulley j:

$$\begin{aligned} F_{c,j,\text{rad}} &= \int_{-\phi_j/2}^{\phi_j/2} \rho_b A_b\, v_b^2\, \mathrm{d}\theta \\ &= \rho_b A_b\, v_b^2 \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}\theta \\ &= \rho_b A_b\, v_b^2\, \phi_j(s). \end{aligned} \tag{3.2.28}$$

Radial-to-Axial Conversion Through the Sheave Cone The radial centrifugal force acts outward against the conical sheave faces. Because the sheaves are inclined at cone half-angle β, radial loading cannot occur without simultaneous axial motion of the sheaves. Thus, a purely radial force applied to the belt generates an axial reaction component that contributes to clamping.

From the pulley geometry derived previously (see Equation (3.1.2)), the relationship between radial and axial displacement in the active shift region is

$$\frac{\mathrm{d}r}{\mathrm{d}s} = \frac{1}{2\tan\beta}.$$

For an infinitesimal displacement,

$$\delta W = F_{\text{rad}}\,\mathrm{d}r = F_{\text{ax}}\,\mathrm{d}s.$$

Substituting $\mathrm{d}r = \dfrac{1}{2\tan\beta}\,\mathrm{d}s$ gives

$$F_{\text{rad}}\,\frac{1}{2\tan\beta}\,\mathrm{d}s = F_{\text{ax}}\,\mathrm{d}s.$$

Canceling $\mathrm{d}s$ yields

$$F_{\text{ax}} = \frac{F_{\text{rad}}}{2\tan\beta}. \tag{3.2.29}$$

Thus, the axial clamping contribution from belt centrifugal force on pulley j is

$$F_{c,j} = \frac{\rho_b A_b\, v_b^2\, \phi_j(s)}{2\tan\beta}. \tag{3.2.30}$$

3.2.5 Axial Shift Dynamics

Earlier (see Equation (3.2.1)), the evolution of the shift coordinate was established as

$$m\,\ddot{s} = F_{p,\text{ax}} - F_{s,\text{ax}}, \tag{3.2.31}$$

where $F_{p,\text{ax}}$ and $F_{s,\text{ax}}$ are the axial forces generated by the primary and secondary mechanisms, respectively.

We now complete this model by:

1. Expanding the force terms to include centrifugal belt loading,
2. Replacing the lumped mass m with the equivalent inertia associated with coordinated shift motion.

Equivalent Inertia Associated with s Newton's second law applies to physical masses in translation. Since s parameterizes coordinated motion of several bodies, we construct an equivalent mass m_{eq} such that the total kinetic energy associated with shift can be written in the form of Theory 4 (Kinetic Energy (Translational)):

$$T_{\text{shift}} = \frac{1}{2}\,m_{\text{eq}}\,\dot{s}^2.$$

The movable primary sheave translates with speed $\dot{s}$ in the positive coordinate direction. The movable secondary sheave translates with equal magnitude speed (but opposite physical direction), so its kinetic energy also contributes proportionally to $\dot{s}^2$.

Let $m_{p,m}$ and $m_{s,m}$ denote the masses of the movable primary and secondary sheaves.

Belt Mass Contribution The total belt mass is

$$m_b = \rho_b A_b L_b, \tag{3.2.32}$$

where A_b is the trapezoidal cross-sectional area from Equation (3.2.17) and L_b is the constant belt length.

The belt moves continuously with a single belt speed, but the axial component of that motion associated with a change in s is determined by geometry. From the axial-radial coupling in Equation (3.1.28), a change in sheave separation s is shared between the two conical faces supporting the belt wedge. Thus, the belt's axial velocity relative to the pulley axis is

$$\dot{s}_b = \frac{\dot{s}}{2}. \tag{3.2.33}$$

The kinetic energy associated with this axial motion is therefore

$$T_b = \frac{1}{2} m_b \left(\frac{\dot{s}}{2}\right)^2 = \frac{1}{2}\left(\frac{m_b}{4}\right)\dot{s}^2.$$

Expressed in the generalized coordinate s, the belt contributes an effective mass of $m_b/4$.

Total Equivalent Mass The total inertia associated with shift becomes

$$m_{\text{eq}} = m_{p,m} + m_{s,m} + \frac{m_b}{4} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}. \tag{3.2.34}$$

Net Axial Force in the s Coordinate Collecting the axial force contributions derived previously:

- $F_{p,\text{ax}}(s, \omega_p)$ from Equation (3.2.10),
- $F_{s,\text{ax}}(s, \tau_s)$ from Equation (3.2.16),
- $F_{c,p}(s, v_b)$ from Equation (3.2.30) with $j = p$,
- $F_{c,s}(s, v_b)$ from Equation (3.2.30) with $j = s$.

With the sign convention defined above, the generalized force in the positive s direction is

$$\sum F_s = \Big(F_{p,\text{ax}} + F_{c,p}\Big) - \Big(F_{s,\text{ax}} + F_{c,s}\Big). \tag{3.2.35}$$

Shift Equation of Motion Applying Newton's second law in the generalized coordinate,

$$\sum F_s = m_{\text{eq}}\,\ddot{s},$$

and substituting Equation (3.2.34) and Equation (3.2.35), we obtain

Axial Shift Acceleration **(3.2.36)**

$$\ddot{s} = \frac{\Big(F_{p,\mathrm{ax}}(s,\omega_p) + F_{c,p}(s,v_b)\Big) - \Big(F_{s,\mathrm{ax}}(s,\tau_s) + F_{c,s}(s,v_b)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}.$$

Equation (3.2.36) provides the axial shift acceleration as a function of the current state variables $(s, \dot{s}, \omega_p, \omega_s)$ and the reacted secondary torque τ_s. The remaining coupling between axial and rotational dynamics enters through τ_s, which will be determined later using Theory 2 (Newton's Second Law (Rotation)).

3.2.6 Section Summary and Forward Outlook

In this section we established a complete axial force model for the CVT mechanism and derived a closed-form equation of motion for the generalized shift coordinate $s(t)$.

At this stage, the axial subsystem is formally closed with respect to the state variables of the model, but it still depends on one remaining coupling quantity that is not determined internally by the axial force balance: the reacted secondary torque τ_s. Although τ_s appears explicitly in the secondary axial force model, its value is determined by the rotational dynamics of the drivetrain and by torque transmission through the belt. Thus, aside from the shared state variables, the axial and rotational subsystems are coupled through this torque term.

The next step is to determine how torque is transmitted across the CVT as a function of pulley radii and speed ratio. This will establish the relationship between τ_s, input torque at the primary, and the evolving ratio $R = r_s/r_p$. Once this transmission relationship is obtained, the axial and rotational equations can be combined into a unified, state-space representation of the full CVT dynamics.

With the axial force structure now complete, attention turns to the torque transmission mechanism that closes the remaining dynamical loop.

3.3 No-Slip Belt Dynamics and Torque Demand

The axial force model developed in Section 3.2 determines how the shift coordinate evolves once the primary and secondary axial forces are known. However, the secondary axial force depends on the transmitted secondary torque τ_s, which is not determined by the axial force balance itself.

This section develops the rotational and belt-dynamic relations needed to determine that torque in the no-slip case. The primary pulley, belt, and secondary pulley are treated as three coupled bodies: torque enters through the primary, is carried through belt contact, and exits through the secondary into the downstream drivetrain.

The goal is to derive the no-slip torque demand: the torque transfer that would be required if the belt remained adhered to both pulleys. This result will later be checked against the available belt–pulley traction limits to determine whether no-slip motion is physically admissible.

3.3.1 System Definition and Torque Path

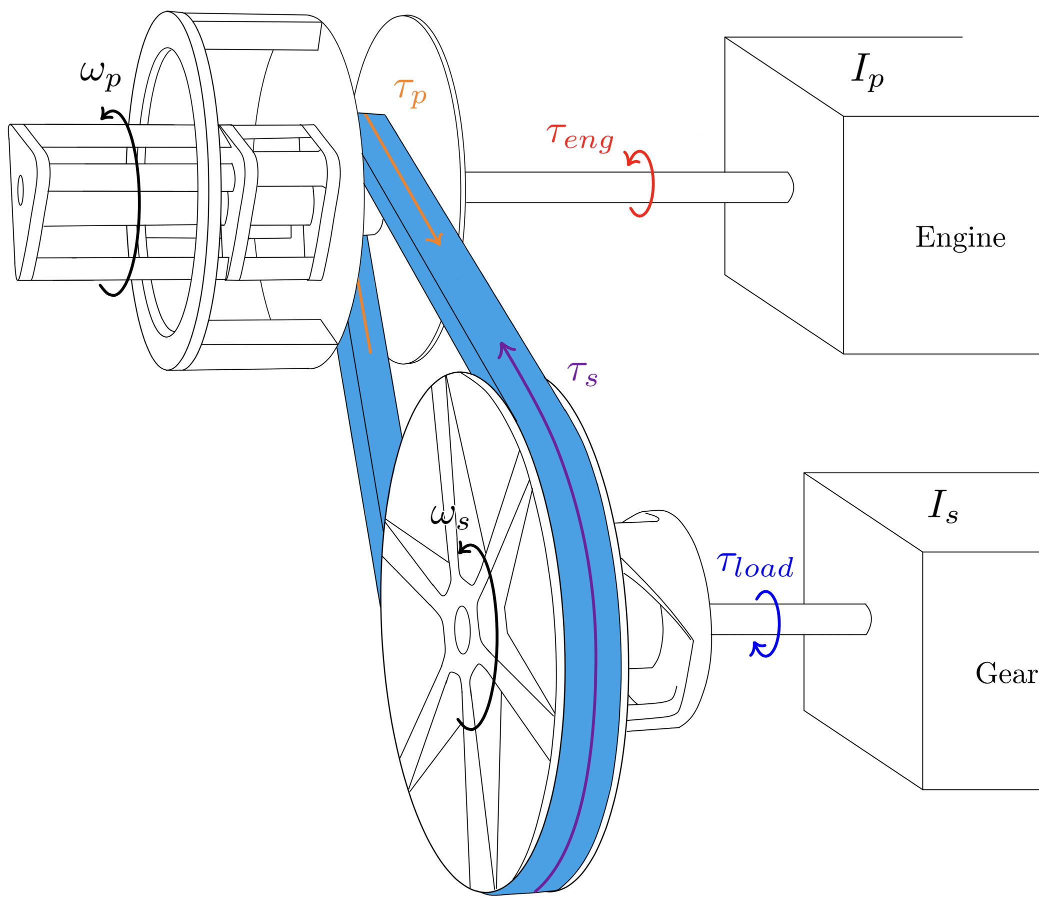


Figure 3.14: Torque path through the CVT. Engine torque τ_{eng} drives the primary pulley with angular speed ω_p. The primary pulley exchanges torque with the belt, the belt exchanges torque with the secondary pulley, and the secondary pulley drives the downstream drivetrain while opposing the reflected load torque τ_{load}.

The torque-transfer subsystem shown in Figure 3.14 is modeled using three bodies:

- the primary pulley,
- the belt,
- the secondary pulley.

The primary pulley receives the engine torque τ_{eng} and exchanges torque with the belt. Let $\tau_p(t)$ denote the torque exerted by the belt contact on the primary pulley. With the sign convention from Section 2.3.1, this torque opposes positive engine-driven rotation of the primary pulley.

The secondary pulley receives torque from the belt and delivers torque to the downstream drivetrain. Let $\tau_s(t)$ denote the torque exerted by the belt contact on the secondary pulley. This torque acts in the positive rotational direction of the secondary pulley, while the reflected load torque $\tau_{\text{load}}(t)$

opposes the motion.

Applying Theory 2 (Newton's Second Law (Rotation)) to the primary pulley gives

Primary Pulley Rotational Dynamics **(3.3.1)**

$$\tau_{\text{eng}}(t) - \tau_p(t) = I_p \dot{\omega}_p(t)$$

Similarly, applying Theory 2 (Newton's Second Law (Rotation)) to the secondary pulley gives

Secondary Pulley Rotational Dynamics **(3.3.2)**

$$\tau_s(t) - \tau_{\text{load}}(t) = I_s \dot{\omega}_s(t)$$

The belt is described by the transport speed $v_b(t)$ introduced in Section 2.5. Under Assumption 8 (Constant Belt Length), the belt moves as a continuous path with a single belt-line speed. Its angular speed around each pulley may differ because angular speed depends on local radius, but the transport speed v_b is common to the belt motion.

For this reason, the belt equation of motion is written along the belt path rather than about a pulley axis. A contact torque produces an equivalent belt-line force when divided by the effective torque radius. Using the effective radii defined in Section 3.1.8, the primary contact contributes $\tau_p/r_{p,\text{eff}}$ to the belt-line motion, while the secondary contact contributes $\tau_s/r_{s,\text{eff}}$ in the opposing direction. Applying Theory 1 (Newton's Second Law (Translation)) to the belt therefore gives

Belt Transport Dynamics **(3.3.3)**

$$m_b \dot{v}_b(t) = \frac{\tau_p(t)}{r_{p,\text{eff}}(s)} - \frac{\tau_s(t)}{r_{s,\text{eff}}(s)}$$

Here m_b is the total belt mass defined in Equation (3.2.32). Together, Equation (3.3.1), Equation (3.3.2), and Equation (3.3.3) define the equations of motion for the three bodies in the torque path.

The quantities appearing in these equations fall into two groups. The first group consists of externally supplied torques and inertial parameters:

$$\tau_{\text{eng}}, \qquad \tau_{\text{load}}, \qquad I_p, \qquad I_s, \qquad m_b.$$

These are not determined by the belt contact state itself. They are defined by the engine model, the reflected vehicle load model, and the lumped inertia and belt-mass definitions recalled in the following subsection.

The second group consists of the torque-transfer unknowns:

$$\dot{\omega}_p, \qquad \dot{\omega}_s, \qquad \dot{v}_b, \qquad \tau_p, \qquad \tau_s.$$

These five quantities describe the accelerations of the three bodies and the two contact torques exchanged through the belt. The three equations of motion above are therefore not sufficient to determine them on their own. The remaining closure comes from the belt–pulley contact state, beginning with the no-slip compatibility condition derived after the external torque and inertia models are recalled.

3.3.2 Engine Torque Model

The engine provides the driving torque $\tau_{\text{eng}}(t)$ acting on the primary pulley.

In this model, engine output torque is represented as a user-supplied function of primary angular velocity. Under the full-throttle assumption (Assumption 15 (Full-Throttle Engine Model)),

$$\tau_{\text{eng}}(t) = \tau_{\text{eng}}\big(\omega_p(t)\big), \tag{3.3.4}$$

so torque depends only on instantaneous primary speed.

This representation is consistent with standard engine characterization, where torque is measured as a function of crankshaft speed under steady full-load conditions.

Admissibility Constraints The supplied torque function must satisfy:

- It is defined over a bounded operating domain $\omega_{\text{min}} \leq \omega_p \leq \omega_{\text{max}}$.
- It is finite and real-valued over this domain.
- Outside this domain, torque is either saturated or ω_p is constrained to remain within bounds.

No specific functional form is required. The dynamical model only requires the mapping $\omega_p \mapsto \tau_{\text{eng}}$.

Example: Kohler CH440 Engine Curve As an illustrative example, Figure 3.15 shows a representative torque and power curve for the Kohler CH440 engine.

SAE Baja Restricted Kohler CH440							
Engine Speed (rpm)	3600	3400	3200	3000	2800	2600	2400
Torque Net Corrected (ft-lb)	13.5	14.5	15.4	16.6	17.4	18.1	18.5
Power Net Corrected (HP)	9.3	9.4	9.4	9.5	9.3	9.0	8.5

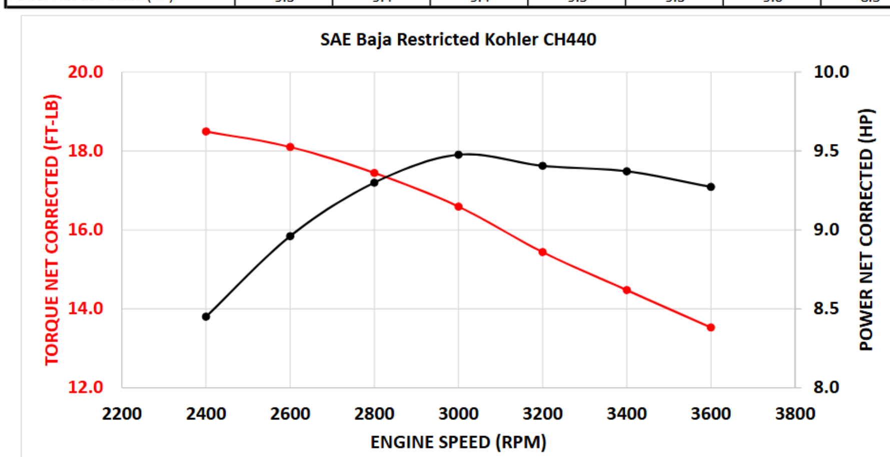


Figure 3.15: Representative torque and power curves for the Kohler CH440 engine. Torque is supplied as a function of angular velocity and serves as the input mapping in Equation (3.3.4).

In simulation, the discrete curve is interpolated to provide a continuous admissible torque function over the operating range.

3.3.3 Load Torque Model

The resisting torque $\tau_{\text{load}}(t)$ in Equation (3.3.2) represents the longitudinal road load reflected to the secondary pulley.

We proceed in three stages:

1. Express vehicle speed in terms of secondary rotation.
2. Derive the total longitudinal road force.
3. Reflect that force back to the secondary shaft.

Kinematic Relations Let r_w denote effective wheel rolling radius and $G > 0$ the fixed gear ratio defined by

$$\omega_s = G\,\omega_w.$$

Vehicle speed is therefore

$$v(t) = r_w \omega_w(t) = \frac{r_w}{G}\,\omega_s(t). \tag{3.3.5}$$

Longitudinal Road Force Using Theory 1 (Newton's Second Law (Translation)), the total longitudinal resistive force on the vehicle is modeled as

$$F_{\text{road}} = F_{\text{rr}} + F_{\text{grade}} + F_{\text{drag}}. \tag{3.3.6}$$

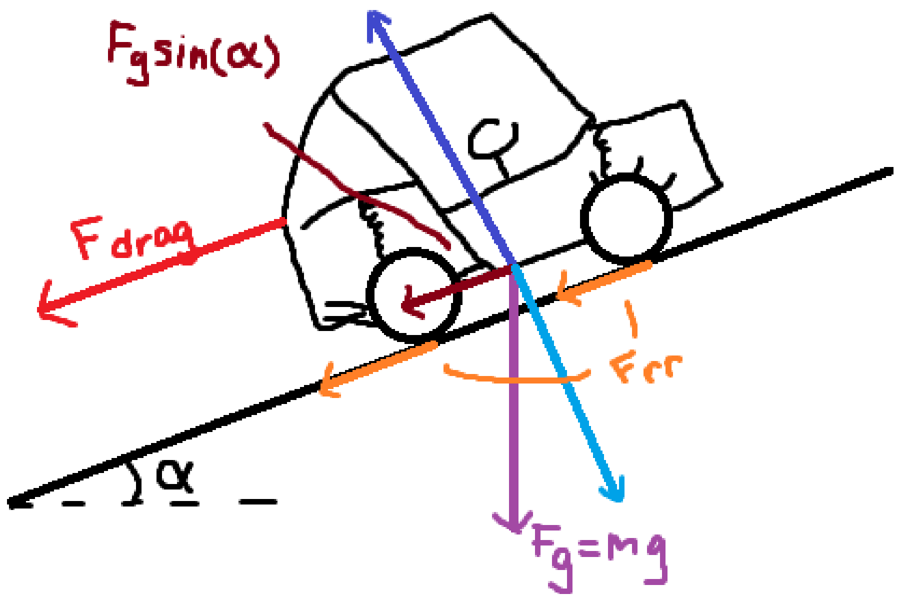


Figure 3.16: Free-body diagram of a vehicle on an incline. The weight mg resolves into $mg \sin \alpha$ along the slope and $mg \cos \alpha$ normal to the surface. Rolling resistance and aerodynamic drag oppose motion.

Rolling Resistance From Theory 13 (Rolling Resistance),

$$F_{\mathrm{rr}} = C_{\mathrm{rr}}\, N. \tag{3.3.7}$$

From the incline geometry,

$$N = mg \cos \alpha. \tag{3.3.8}$$

Thus

$$F_{\mathrm{rr}}(v, \alpha) = C_{\mathrm{rr}} mg \cos \alpha \, \mathrm{sgn}(v). \tag{3.3.9}$$

Grade Resistance From Theory 14 (Gravitational Force), the gravitational component along the slope is

$$F_{\mathrm{grade}}(\alpha) = mg \sin \alpha. \tag{3.3.10}$$

Aerodynamic Drag From Theory 12 (Air Resistance (Quadratic Drag)),

$$F_{\mathrm{drag}}(v) = \frac{1}{2} \rho C_d A_f \, v^2 \, \mathrm{sgn}(v). \tag{3.3.11}$$

Total Road Force Substituting the above components into Equation (3.3.6) yields

$$\begin{aligned}F_{\text{road}}(t) = C_{\text{rr}} mg \cos\alpha \, \text{sgn}(v(t)) \\ + mg \sin\alpha \\ + \frac{1}{2}\rho C_d A_f \, v(t)^2 \, \text{sgn}(v(t)) \,.\end{aligned} \tag{3.3.12}$$

Using Equation (3.3.5),

$$v(t) = \frac{r_w}{G}\,\omega_s(t),$$

the road force is now fully expressed in terms of the secondary state variable $\omega_s(t)$.

Reflection to Secondary Torque Wheel torque is

$$\tau_{\text{road},w}(t) = r_w F_{\text{road}}(t). \tag{3.3.13}$$

Assuming ideal power transmission (Assumption 13 (Ideal CVT Power Transmission)), the resisting torque reflected to the secondary pulley is

$$\tau_{\text{load}}(t) = \frac{1}{G}\,\tau_{\text{road},w}(t) = \frac{r_w}{G} F_{\text{road}}(t). \tag{3.3.14}$$

Substituting Equation (3.3.12) gives

$$\boxed{\tau_{\text{load}}(t) = \frac{r_w}{G}\left[C_{\text{rr}} mg \cos\alpha \, \text{sgn}\Big(\frac{r_w}{G}\omega_s(t)\Big) + mg\sin\alpha + \frac{1}{2}\rho C_d A_f \left(\frac{r_w}{G}\omega_s(t)\right)^2 \text{sgn}\Big(\frac{r_w}{G}\omega_s(t)\Big)\right].} \tag{3.3.15}$$

3.3.4 Inertia Definitions

The rotational equations Equation (3.3.1)–Equation (3.3.2) require the effective rotational inertia about each pulley axis.

Primary Pulley Inertia The primary pulley is rigidly connected to the engine. All components on this side rotate with angular velocity ω_p, so no reflection is required.

The effective inertia is therefore the direct sum

$$I_p = I_{\text{eng}} + I_{\text{prim}}, \tag{3.3.16}$$

where

- I_{eng} is the engine rotational inertia,
- I_{prim} is the primary CVT pulley inertia.

Secondary Pulley Inertia On the secondary side, all downstream components must be expressed as an equivalent inertia about the secondary axis.

Let

- I_{sec} : secondary CVT pulley inertia,
- I_{gb} : gearbox + intermediate shaft inertia,
- I_{wheel} : wheel + hub inertia about the wheel axis,
- m : total vehicle mass,
- r_w : effective wheel rolling radius,
- G : fixed gear ratio defined by $\omega_s = G\,\omega_w$.

Vehicle Mass as Equivalent Rotational Inertia The vehicle's translational kinetic energy is

$$T_{\text{trans}} = \frac{1}{2}mv^2,$$

with $v = r_w\omega_w$.

Substituting,

$$T_{\text{trans}} = \frac{1}{2}m(r_w\omega_w)^2 = \frac{1}{2}(mr_w^2)\,\omega_w^2.$$

Thus the vehicle mass behaves as an effective rotational inertia

$$I_{\text{veh,wheel}} = mr_w^2$$

about the wheel axis.

Reflection to the Secondary Axis Because $\omega_s = G\,\omega_w$, we have $\omega_w = \omega_s/G$.

Expressing the downstream kinetic energy in terms of ω_s:

$$T = \frac{1}{2}I_{\text{wheel}}\left(\frac{\omega_s}{G}\right)^2 + \frac{1}{2}(mr_w^2)\left(\frac{\omega_s}{G}\right)^2.$$

Factoring,

$$T = \frac{1}{2}\left(\frac{I_{\text{wheel}} + mr_w^2}{G^2}\right)\omega_s^2.$$

Therefore, inertias at the wheel reflect to the secondary pulley as

$$I_{\text{reflected}} = \frac{I_{\text{wheel}} + mr_w^2}{G^2}.$$

Total Secondary Inertia The total effective inertia about the secondary pulley is

$$I_s = I_{\text{sec}} + I_{\text{gb}} + \frac{I_{\text{wheel}} + m r_w^2}{G^2}. \tag{3.3.17}$$

Equation Equation (3.3.17) defines the total inertia opposing angular acceleration of the secondary pulley.

3.3.5 No-Slip Kinematic Compatibility

The equations of motion in Section 3.3.1 describe the torque-transfer subsystem before any contact constraint has been imposed. To form the no-slip candidate, we now require the belt to adhere to each pulley contact.

Let

$$j \in \{p, s\}$$

denote either the primary or secondary pulley. If the belt is adhered to pulley j, then the belt transport speed must match the tangential surface speed of that pulley at the effective torque radius. Therefore,

$$v_b(t) = r_{j,\text{eff}}(s(t))\,\omega_j(t), \qquad j \in \{p, s\}. \tag{3.3.18}$$

Equation Equation (3.3.18) is the velocity-level no-slip condition. Since the equations of motion involve accelerations, we differentiate it with respect to time. Starting from

$$v_b(t) = r_{j,\text{eff}}(s(t))\,\omega_j(t),$$

the product rule gives

$$\dot{v}_b(t) = \dot{r}_{j,\text{eff}}(t)\,\omega_j(t) + r_{j,\text{eff}}(s(t))\,\dot{\omega}_j(t). \tag{3.3.19}$$

Since the effective radius depends on time only through the shift coordinate $s(t)$,

$$\dot{r}_{j,\text{eff}}(t) = \frac{dr_{j,\text{eff}}}{ds}\dot{s}(t). \tag{3.3.20}$$

Solving Equation Equation (3.3.19) for the pulley angular acceleration gives

$$\dot{\omega}_j(t) = \frac{\dot{v}_b(t) - \dot{r}_{j,\text{eff}}(t)\,\omega_j(t)}{r_{j,\text{eff}}(s(t))}, \qquad j \in \{p, s\}. \tag{3.3.21}$$

Together with the three equations of motion from Section 3.3.1, this acceleration-level compatibility supplies the no-slip closure. The next step is to substitute Equation (3.3.21) into the pulley equations of motion and solve for the contact torque demands associated with the no-slip candidate.

3.3.6 Solving the No-Slip Torque Demand

The no-slip compatibility relation from Section 3.3.5 can now be combined with the three equations of motion from Section 3.3.1. For compactness, define

$$r_p = r_{p,\text{eff}}(s), \qquad r_s = r_{s,\text{eff}}(s),$$

with corresponding time derivatives $\dot{r}_p$ and $\dot{r}_s$.

On the no-slip candidate branch, Equation Equation (3.3.21) gives

$$\dot{\omega}_p = \frac{\dot{v}_{b,\mathrm{ns}} - \dot{r}_p \omega_p}{r_p}, \qquad \dot{\omega}_s = \frac{\dot{v}_{b,\mathrm{ns}} - \dot{r}_s \omega_s}{r_s}.$$

Substituting these relations into the primary rotational equation Equation (3.3.1) gives the primary contact torque required to maintain no slip:

Primary No-Slip Torque Demand **(3.3.22)**

$$\tau_{p,\mathrm{ns}} = \tau_{\mathrm{eng}} - I_p \left(\frac{\dot{v}_{b,\mathrm{ns}} - \dot{r}_p \omega_p}{r_p} \right)$$

Similarly, substituting into the secondary rotational equation Equation (3.3.2) gives the secondary contact torque required to maintain no slip:

Secondary No-Slip Torque Demand **(3.3.23)**

$$\tau_{s,\mathrm{ns}} = \tau_{\mathrm{load}} + I_s \left(\frac{\dot{v}_{b,\mathrm{ns}} - \dot{r}_s \omega_s}{r_s} \right)$$

At this point, both torque demands are expressed in terms of the unknown no-slip belt acceleration $\dot{v}_{b,\mathrm{ns}}$. The belt equation of motion Equation (3.3.3) provides the remaining relation:

$$m_b \dot{v}_{b,\mathrm{ns}} = \frac{\tau_{p,\mathrm{ns}}}{r_p} - \frac{\tau_{s,\mathrm{ns}}}{r_s}.$$

Substituting Equation (3.3.22) and Equation (3.3.23) gives

$$m_b \dot{v}_{b,\mathrm{ns}} = \frac{1}{r_p} \left[\tau_{\mathrm{eng}} - I_p \left(\frac{\dot{v}_{b,\mathrm{ns}} - \dot{r}_p \omega_p}{r_p} \right) \right] - \frac{1}{r_s} \left[\tau_{\mathrm{load}} + I_s \left(\frac{\dot{v}_{b,\mathrm{ns}} - \dot{r}_s \omega_s}{r_s} \right) \right].$$

Expanding and collecting the terms containing $\dot{v}_{b,\mathrm{ns}}$ gives

$$\left(m_b + \frac{I_p}{r_p^2} + \frac{I_s}{r_s^2} \right) \dot{v}_{b,\mathrm{ns}} = \frac{\tau_{\mathrm{eng}}}{r_p} - \frac{\tau_{\mathrm{load}}}{r_s} + \frac{I_p \dot{r}_p \omega_p}{r_p^2} + \frac{I_s \dot{r}_s \omega_s}{r_s^2}.$$

Solving for the no-slip belt acceleration yields

No-Slip Belt Acceleration **(3.3.24)**

$$\dot{v}_{b,\mathrm{ns}} = \frac{\dfrac{\tau_{\mathrm{eng}}}{r_p} - \dfrac{\tau_{\mathrm{load}}}{r_s} + \dfrac{I_p \dot{r}_p \omega_p}{r_p^2} + \dfrac{I_s \dot{r}_s \omega_s}{r_s^2}}{m_b + \dfrac{I_p}{r_p^2} + \dfrac{I_s}{r_s^2}}$$

Substituting Equation (3.3.24) back into Equation (3.3.22) and Equation (3.3.23) determines the corresponding no-slip torque demands at the primary and secondary contacts.

Equation (3.3.24), Equation (3.3.22), and Equation (3.3.23) therefore define the no-slip candidate torque transfer. These are the torques required to maintain adhered contact at both pulleys. They are not yet guaranteed to be physically admissible; the next section compares these demands against the available belt–pulley traction limits.

3.3.7 Transition to Traction Limits

At this point, the no-slip candidate is fully determined. For any current state, Equation (3.3.24) gives the belt acceleration required by adhered motion, and Equation (3.3.22)–Equation (3.3.23) give the corresponding primary and secondary torque demands.

These equations answer what torque transfer would be required to keep the belt stuck to both pulleys. They do not yet answer whether the contacts can actually support those torques. That question depends on the available frictional traction generated by the axial clamping forces derived in Section 3.2.

The next section derives the traction bounds for each pulley contact. The no-slip demands $\tau_{p,\mathrm{ns}}$ and $\tau_{s,\mathrm{ns}}$ can then be compared against those bounds to decide whether the no-slip candidate is admissible or whether the transmission must enter a slip branch.

3.4 Traction Limits and Slip Conditions

The previous section derived the no-slip torque demands $\tau_{p,\mathrm{ns}}$ and $\tau_{s,\mathrm{ns}}$. These are the torques that would be required at the primary and secondary belt contacts if the belt remained adhered to both pulleys.

Those demands are not automatically attainable. Torque is transmitted through frictional traction between the belt and the sheave faces, and this traction is limited by the available normal force generated by axial clamping. Therefore, each pulley contact can support only a bounded range of torque before slip must occur.

The purpose of this section is to derive those pulley-wise traction bounds. For each pulley $j \in \{p, s\}$, we first determine the maximum admissible tight–slack tension difference across the wrap, then convert that tension difference into a torque bound using the effective torque radius. These bounds will later be compared against $\tau_{p,\mathrm{ns}}$ and $\tau_{s,\mathrm{ns}}$ to decide whether the no-slip candidate is admissible.

3.4.1 Tight–Slack Tension Difference and Torque Capacity

The same traction-capacity argument applies to both pulleys. Accordingly, in this section we use the subscript

$$j \in \{p, s\},$$

where $j = p$ denotes the primary pulley and $j = s$ denotes the secondary pulley.

Consider the wrapped pulley shown in Figure 3.17. If torque is transmitted from pulley j to the belt, the belt tension cannot be the same on both spans. One side must carry a higher tension than the other. For a given transmission direction, we therefore define

$$\Delta T_j \equiv T_{j,\text{tight}} - T_{j,\text{slack}}, \tag{3.4.1}$$

where $T_{j,\text{tight}}$ and $T_{j,\text{slack}}$ are the belt tensions on the two straight spans shown in the figure.

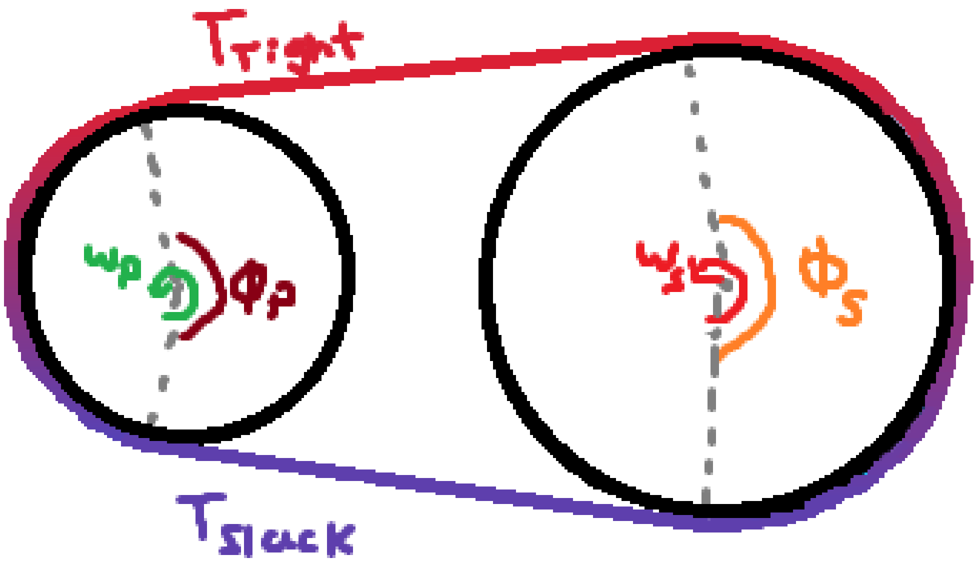


Figure 3.17: Torque transmission requires a tight–slack tension difference across the wrap.

This tension difference arises because the pulley applies tangential traction to the belt along the wrap angle ϕ_j. The cumulative effect of this traction changes the belt tension progressively from the slack side to the tight side.

At effective contact radius $r_{j,\text{eff}}$, the transmitted torque is

$$\tau_j = r_{j,\text{eff}}\,\Delta T_j, \qquad |\tau_j| = r_{j,\text{eff}}\,|\Delta T_j|. \tag{3.4.2}$$

Thus, limiting the transmitted torque is equivalent to limiting the admissible tension difference ΔT_j.

Under Assumption 4 (Idealized Belt–Pulley Friction Law) and Assumption 5 (Constant Effective Friction Coefficient), the belt–pulley interface obeys a Coulomb friction law with constant coefficient μ. The tangential traction at any point along the wrap is therefore bounded by the local normal load:

$$|F_f| \leq \mu N.$$

The largest admissible tension difference for a given pulley state and transmission direction occurs when friction is fully mobilized everywhere along the wrap, i.e. at impending slip. Beyond this point, additional tension rise cannot be supported and sliding must occur.

We therefore define a maximum supported tension difference $\Delta T_{j,\max}$, corresponding to the friction-saturated state, and the associated torque capacity

$$\tau_{j,\max} = r_{j,\text{eff}}\,\Delta T_{j,\max}. \tag{3.4.3}$$

The next subsection computes $\Delta T_{j,\max}$ by examining the incremental change in belt tension along the wrap.

Remark (Other Friction Models)

Although the present simulator adopts a Coulomb friction law for simplicity and analytic transparency, richer friction descriptions are available if greater fidelity near the stick–slip transition is desired. A common next step is a Stribeck-type law, in which the effective friction level varies with relative slip speed so that breakaway friction near zero speed can exceed the kinetic friction observed at larger slip speeds. Such a model can better represent engagement and low-slip behaviour than a purely constant Coulomb coefficient, while still remaining relatively simple to implement Armstrong-H'elouvry et al. (1994); Canudas de Wit et al. (1995).

For still higher fidelity, dynamic friction models introduce internal state variables that represent the evolving contact condition. The LuGre model is a standard example and can capture presliding displacement, hysteresis, friction lag, and smoother stick–slip transitions, while related multistate models such as the generalized Maxwell–slip model can represent more detailed microslip and memory effects Canudas de Wit et al. (1995); Al-Bender et al. (2005). These models are often more physically expressive, but they also require additional parameter identification and increase numerical and modelling complexity. For the present work, Coulomb friction is therefore retained as a deliberate low-complexity approximation, while Stribeck- or state-based friction models remain natural extensions for future refinement Armstrong-H'elouvry et al. (1994).

3.4.2 Differential Belt Element and Tangential Force Balance

To determine the maximum admissible tension difference $\Delta T_{j,\max}$ introduced in Section 3.4.1, we now examine how belt tension changes *locally* as the belt passes around a generic pulley $j \in \{p, s\}$.

This derivation uses a differential force balance on a belt element in an inertial frame of reference. Accordingly, the belt element is described through its actual acceleration, so the radial inertia appears through centripetal acceleration rather than through an explicit centrifugal force. This differs from the treatment in Section 3.2.4, where the same underlying belt inertia on the wrap was represented through an equivalent centrifugal-loading picture in order to derive an axial clamping contribution. That earlier formulation was convenient because the goal there was to obtain a net axial force on the sheaves. Here, by contrast, the goal is to determine how belt tension changes along the wrap, so a local inertial-frame force balance is the more natural description.

Specifically, we seek an expression for the incremental tension change $\mathrm{d}T_j$ over a differential wrap angle $\mathrm{d}\theta$. Once $\mathrm{d}T_j$ is related to the available friction at the interface, integration over the total wrap angle ϕ_j will yield $\Delta T_{j,\max}$.

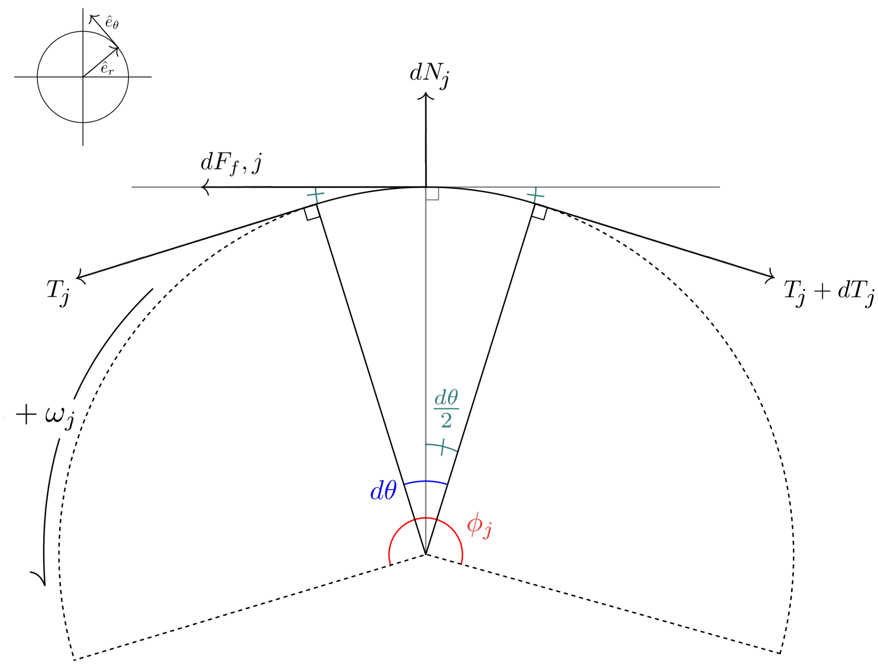


Figure 3.18: Free-body diagram of an infinitesimal belt segment in contact with pulley j, written in the local radial–tangential coordinate basis $(\hat{e}_r, \hat{e}_\theta)$ shown in the inset. The element subtends an infinitesimal contact angle $\mathrm{d}\theta$ with the total wrap angle ϕ_j. The boundary tensions T_j and $T_j + \mathrm{d}T_j$ act tangentially, while the pulley applies a normal force $\mathrm{d}N_j$ in the radial direction and a friction force $\mathrm{d}F_{f,j}$ in the tangential direction. The positive pulley rotation $+\omega_j$ is shown for reference.

Consider the differential belt element in Figure 3.18, which spans an infinitesimal wrap angle $\mathrm{d}\theta$ on pulley j. The positive tangential direction is taken along the midpoint tangent of the element, consistent with increasing wrap angle and with increasing belt tension from the slack side toward the tight side.

Tangential Newton's Second Law Applying Theory 1 (Newton's Second Law (Translation)) in the tangential direction gives

$$\sum F_t = \mathrm{d}m_j \, a_{t,j}, \tag{3.4.4}$$

where $a_{t,j}$ is the tangential acceleration of the belt material along the contact arc on pulley j.

From the element geometry, the tangential components of the boundary tensions are

$$(T_j)_t = T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right), \tag{3.4.5}$$

$$(T_j + \mathrm{d}T_j)_t = (T_j + \mathrm{d}T_j) \cos\left(\frac{\mathrm{d}\theta}{2}\right). \tag{3.4.6}$$

The left-end tension acts in the positive tangential direction, while the right-end tension acts in the negative tangential direction. Including the differential friction force $\mathrm{d}F_{f,j}$, the tangential force balance becomes

$$T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right) - (T_j + \mathrm{d}T_j)\cos\left(\frac{\mathrm{d}\theta}{2}\right) + \mathrm{d}F_{f,j} = \mathrm{d}m_j\, a_{t,j}. \tag{3.4.7}$$

Combining the two tension terms gives

$$-\,\mathrm{d}T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right) + \mathrm{d}F_{f,j} = \mathrm{d}m_j\, a_{t,j}. \tag{3.4.8}$$

Belt mass element From Equation (3.2.25), the belt mass associated with the differential contact element on pulley j is

$$\mathrm{d}m_j = \rho_b A_b r_{j,\mathrm{cm}}\, \mathrm{d}\theta. \tag{3.4.9}$$

Substituting this into Equation (3.4.8) gives

$$-\,\mathrm{d}T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right) + \mathrm{d}F_{f,j} = \rho_b A_b r_{j,\mathrm{cm}} a_{t,j}\, \mathrm{d}\theta. \tag{3.4.10}$$

Rearranging for the local tension change yields

$$\mathrm{d}T_j = \frac{\mathrm{d}F_{f,j} - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j}\, \mathrm{d}\theta}{\cos\left(\frac{\mathrm{d}\theta}{2}\right)}. \tag{3.4.11}$$

This expression shows that the local rise in belt tension is produced by the available tangential friction force, reduced by the tangential inertial force required to accelerate the belt element.

Friction limit and impending slip The quantity of interest is the *maximum* admissible tension increase. From Equation (3.4.11), this occurs when the tangential friction force reaches its largest value, since $\mathrm{d}F_{f,j}$ is the term that directly drives the increase in belt tension.

As discussed in Section 3.4.1, the maximum tension rise therefore occurs at impending slip, when friction is fully mobilized at the interface. At the differential level, the largest available friction force is

$$\mathrm{d}F_{f,j,\max} = \mu\, \mathrm{d}N_j. \tag{3.4.12}$$

Substituting Equation (3.4.12) into Equation (3.4.11) gives the maximum local tension increase that can be supported without sliding:

$$\mathrm{d}T_{j,\max} = \frac{\mu\, \mathrm{d}N_j - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j}\, \mathrm{d}\theta}{\cos\left(\frac{\mathrm{d}\theta}{2}\right)}. \tag{3.4.13}$$

3.4.3 Integration Across the Wrap

The tight–slack tension difference introduced in Section 3.4.1 is obtained by accumulating the incremental tension changes $\mathrm{d}T_j$ along the contact arc of pulley $j \in \{p, s\}$. Since the belt tension evolves continuously around the wrap,

$$\Delta T_j = T_{j,\text{tight}} - T_{j,\text{slack}} = \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}T_j. \tag{3.4.14}$$

Accordingly, the maximum admissible tension difference is obtained by integrating the local maximum increment $\mathrm{d}T_{j,\max}$ from Equation (3.4.13):

$$\Delta T_{j,\max} = \int_{-\phi_j/2}^{\phi_j/2} \frac{\mu\, \mathrm{d}N_j - \rho_b A_b r_{j,\text{cm}} a_{t,j}\, \mathrm{d}\theta}{\cos\left(\frac{\mathrm{d}\theta}{2}\right)}. \tag{3.4.15}$$

In the differential limit $\mathrm{d}\theta \to 0$, $\cos(\mathrm{d}\theta/2) \to 1$, so this reduces to

$$\Delta T_{j,\max} = \int_{-\phi_j/2}^{\phi_j/2} \mu\, \mathrm{d}N_j - \int_{-\phi_j/2}^{\phi_j/2} \rho_b A_b r_{j,\text{cm}} a_{t,j}\, \mathrm{d}\theta. \tag{3.4.16}$$

The first integral is the total normal load carried over the wrap of pulley j, which we denote by

$$N_{\phi,j} \;\equiv\; \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}N_j. \tag{3.4.17}$$

The explicit relationship between $N_{\phi,j}$ and the axial clamping force is derived in Section 3.4.5.

From Assumption 9 (Uniform Contact Radius Around Wrap), $r_{j,\text{cm}}$ and $a_{t,j}$ are uniform over the wrap, so the second integral simplifies to $\rho_b A_b r_{j,\text{cm}} a_{t,j} \phi_j$. Hence,

$$\Delta T_{j,\max} = \mu N_{\phi,j} - \rho_b A_b r_{j,\text{cm}} a_{t,j} \phi_j. \tag{3.4.18}$$

Thus, the maximum admissible tension difference for pulley j depends on two remaining quantities: the total wrap normal load $N_{\phi,j}$ and the tangential belt acceleration $a_{t,j}$.

3.4.4 Tangential Acceleration of a Belt Element on a Circular Wrap

In Equation (3.4.18), Newton's second law is applied *along the local direction of belt motion*, so we require an explicit expression for the tangential acceleration $a_{t,j}$ of the belt material on pulley j. Since the inertial term acts on the belt element mass, the relevant kinematics are those of the belt element centroid, located at radius $r_{j,\text{cm}}(t)$.

As discussed in Section 3.2.4, belt inertia on the wrap may be represented in different but equivalent ways depending on the force balance of interest. There, the goal was to obtain an axial clamping contribution, so the belt inertia was introduced through an equivalent centrifugal-loading picture. Here, by contrast, we require a local force balance on a differential belt element in an inertial frame. In this description, the belt element has an actual acceleration, whose radial component contains the usual centripetal term, while the tangential force balance depends only on the tangential component.

Goal For a belt element of differential mass $\mathrm{d}m_j$ moving along the wrap of pulley j, the tangential force balance uses

$$\sum F_t = \mathrm{d}m_j\, a_{t,j}.$$

Our goal is therefore to compute $a_{t,j}$ from the belt kinematics when both the centroid radius and the modeled belt transport speed may vary:

$$r_{j,\mathrm{cm}}(t), \qquad \dot{r}_{j,\mathrm{cm}}(t), \qquad v_b(t), \qquad \dot{v}_b(t).$$

Coordinate system and unit vectors Because we care about forces *along the belt*, we choose a basis aligned with the local wrap geometry on pulley j:

- $\mathbf{e}_{r,j}(t)$ points radially outward from the pulley center to the belt point,
- $\mathbf{e}_{\theta,j}(t)$ is the unit tangent, obtained by rotating $\mathbf{e}_{r,j}$ by $+90°$ in the direction of increasing θ_j.

The tangential direction of belt motion is therefore exactly $\mathbf{e}_{\theta,j}$.

Tangential acceleration as a component of a Tangential acceleration is the component of the acceleration vector along $\mathbf{e}_{\theta,j}$:

$$a_{t,j} \equiv \mathbf{a}_j \cdot \mathbf{e}_{\theta,j}. \tag{3.4.19}$$

We therefore compute the full acceleration $\mathbf{a}_j = \dot{\mathbf{v}}_j$ and then read off its $\mathbf{e}_{\theta,j}$ component.

Step 1: position and velocity The position of the belt element centroid relative to the pulley center is

$$\mathbf{r}_j(t) = r_{j,\mathrm{cm}}(t)\, \mathbf{e}_{r,j}(t). \tag{3.4.20}$$

The belt transport speed is the tangential speed of the belt element along the wrap. Since the modeled belt transport speed is v_b, the velocity of the belt element centroid is

$$\mathbf{v}_j = \dot{r}_{j,\mathrm{cm}}\, \mathbf{e}_{r,j} + v_b\, \mathbf{e}_{\theta,j}. \tag{3.4.21}$$

Thus, $\dot{r}_{j,\mathrm{cm}}$ is the radial speed and v_b is the tangential speed.

Step 2: time derivatives of the unit vectors Over a small angular change $\mathrm{d}\theta_j$, the unit vectors rotate rigidly by the same angle. Hence

$$\frac{\mathrm{d}\mathbf{e}_{r,j}}{\mathrm{d}\theta_j} = \mathbf{e}_{\theta,j}, \qquad \frac{\mathrm{d}\mathbf{e}_{\theta,j}}{\mathrm{d}\theta_j} = -\mathbf{e}_{r,j}. \tag{3.4.22}$$

The local angular speed of the belt element centroid on pulley j is related to the belt transport speed by

$$\dot{\theta}_j = \omega_{b,j} = \frac{v_b}{r_{j,\mathrm{cm}}}. \tag{3.4.23}$$

Using the chain rule, the unit-vector time derivatives therefore become

$$\dot{\mathbf{e}}_{r,j} = \frac{v_b}{r_{j,\mathrm{cm}}}\, \mathbf{e}_{\theta,j}, \qquad \dot{\mathbf{e}}_{\theta,j} = -\frac{v_b}{r_{j,\mathrm{cm}}}\, \mathbf{e}_{r,j}. \tag{3.4.24}$$

Step 3: acceleration and tangential component Differentiating Equation (3.4.21) gives

$$\begin{aligned}
\mathbf{a}_j &= \frac{\mathrm{d}}{\mathrm{d}t}\big(\dot{r}_{j,\mathrm{cm}}\,\mathbf{e}_{r,j}\big) + \frac{\mathrm{d}}{\mathrm{d}t}\big(v_b\,\mathbf{e}_{\theta,j}\big) \\
&= \ddot{r}_{j,\mathrm{cm}}\,\mathbf{e}_{r,j} + \dot{r}_{j,\mathrm{cm}}\,\dot{\mathbf{e}}_{r,j} + \dot{v}_b\,\mathbf{e}_{\theta,j} + v_b\,\dot{\mathbf{e}}_{\theta,j}.
\end{aligned} \tag{3.4.25}$$

Substituting Equation (3.4.24) gives

$$\begin{aligned}
\mathbf{a}_j &= \ddot{r}_{j,\mathrm{cm}}\,\mathbf{e}_{r,j} + \dot{r}_{j,\mathrm{cm}}\left(\frac{v_b}{r_{j,\mathrm{cm}}}\,\mathbf{e}_{\theta,j}\right) + \dot{v}_b\,\mathbf{e}_{\theta,j} - v_b\left(\frac{v_b}{r_{j,\mathrm{cm}}}\,\mathbf{e}_{r,j}\right) \\
&= \left(\ddot{r}_{j,\mathrm{cm}} - \frac{v_b^2}{r_{j,\mathrm{cm}}}\right)\mathbf{e}_{r,j} + \left(\dot{v}_b + \frac{\dot{r}_{j,\mathrm{cm}}}{r_{j,\mathrm{cm}}}v_b\right)\mathbf{e}_{\theta,j}.
\end{aligned} \tag{3.4.26}$$

By Equation (3.4.19), the tangential acceleration is therefore

$$a_{t,j} = \dot{v}_b + \frac{\dot{r}_{j,\mathrm{cm}}}{r_{j,\mathrm{cm}}}v_b. \tag{3.4.27}$$

Interpretation The first term, $\dot{v}_b$, is the direct time rate of change of the belt transport speed. The second term appears because the belt element is not only moving along the wrap, but also moving to a different wrap radius at the same time. As the element shifts inward or outward while continuing to travel around the pulley, the local radial direction changes, and that changing radial direction contributes a tangential component to the acceleration.

The radial component $\ddot{r}_{j,\mathrm{cm}} - v_b^2/r_{j,\mathrm{cm}}$ contains the centripetal term, but it does not enter the tangential force balance used in Section 3.4.2.

3.4.5 Normal Force from Axial Clamping

The traction bound Equation (3.4.18) depends on the total normal load over the wrap of pulley j,

$$N_{\phi,j} \equiv \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}N_j.$$

We now express $N_{\phi,j}$ in terms of the axial clamping force generated by that pulley.

Axial clamp produces radial normal load The belt is pressed between two opposing conical sheaves; the two inclined faces apply a net normal force on the belt equal to the inward radial load generated by the axial clamp. The same geometry was considered before, and we can rearrange Equation (3.2.29) to get

$$F_{\mathrm{rad},j} = 2F_{\mathrm{ax},j}\tan\beta. \tag{3.4.28}$$

Identification with the wrap-normal load The radial load $F_{\mathrm{rad},j}$ derived above represents the total inward force applied by the two sheave faces of pulley j onto the belt. That load is carried by the distributed belt–pulley contact forces $\mathrm{d}N_j$ along the wrap.

Therefore, the total normal load available for traction on pulley j is

$$\boxed{N_{\phi,j} = F_{\mathrm{rad},j} = 2F_{\mathrm{ax},j}\tan\beta.} \tag{3.4.29}$$

This result states that the total wrap normal load, and hence the normal load available for traction, is set directly by the axial clamping force through the cone geometry.

3.4.6 Torque Capacity

From Section 3.4.1, the maximum transmissible torque on pulley j is determined by the maximum admissible tension difference across the wrap:

$$\tau_{j,\max} = r_{j,\mathrm{eff}}\,\Delta T_{j,\max}. \tag{3.4.30}$$

Using the integrated result from Equation (3.4.18), together with the wrap-normal relation Equation (3.4.29) and the belt tangential acceleration from Equation (3.4.27),

$$\Delta T_{j,\max} = \mu N_{\phi,j} - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j}\phi_j, \qquad N_{\phi,j} = 2F_{\mathrm{ax},j}\tan\beta, \qquad a_{t,j} = \dot{v}_b + \frac{\dot{r}_{j,\mathrm{cm}}}{r_{j,\mathrm{cm}}}v_b,$$

we obtain

$$\Delta T_{j,\max} = 2\mu F_{\mathrm{ax},j}\tan\beta - \rho_b A_b \phi_j r_{j,\mathrm{cm}}\left(\dot{v}_b + \frac{\dot{r}_{j,\mathrm{cm}}}{r_{j,\mathrm{cm}}}v_b\right). \tag{3.4.31}$$

Substituting this into Equation (3.4.30) and re-arranging gives the explicit torque-capacity expression

Pulley Torque Capacity **(3.4.32)**

$$\tau_{j,\max} = r_{j,\mathrm{eff}}\left[2\mu F_{\mathrm{ax},j}\tan\beta - \rho_b A_b\phi_j\left(r_{j,\mathrm{cm}}\dot{v}_b + \dot{r}_{j,\mathrm{cm}}\,v_b\right)\right].$$

The remaining task is to substitute the appropriate axial clamp force $F_{\mathrm{ax},j}$ for each pulley to obtain explicit bounds on the coupling torque.

Remark (Why this differs from the classical capstan exponential)

In the classical capstan problem, the normal load is generated *by the belt tension* itself (a larger tension implies a proportionally larger contact normal), which leads to the exponential relation $T_1/T_2 = e^{\mu\phi}$.
Here, the normal load is imposed externally by the clamping mechanism: N_ϕ is determined by F_{ax} through Equation (3.4.29). Therefore, the available traction scales *linearly* with clamping force, not exponentially with belt tension. Centrifugal loading and ratio change enter additionally through the inertial term in Equation (3.4.32).

3.4.7 Primary Pulley No-Slip Admissibility

The generic capacity law Equation (3.4.32) can now be specialized to the primary pulley. At this stage, the goal is not to determine the slipping motion. The goal is only to test whether the no-slip demand derived in Section 3.3.6 can be supported by the available primary traction.

For compactness in this subsection, write

$$r_p = r_{p,\text{eff}}(s).$$

On the no-slip candidate branch, the belt acceleration is $\dot{v}_{b,\text{ns}}$, as given by Equation (3.3.24). Therefore, the primary traction capacity is evaluated using this candidate acceleration.

Applying Equation (3.4.32) with $j = p$, and using the static friction coefficient μ_s, gives

Primary No-Slip Torque Capacity **(3.4.33)**

$$\tau_{p,\text{max}}^{\text{stick}} = r_p \left[2\mu_s F_{\text{ax},p}(s, \omega_p) \tan\beta - \rho_b A_b \phi_p \left(r_{p,\text{cm}} \dot{v}_{b,\text{ns}} + \dot{r}_{p,\text{cm}}\, v_b\right)\right]$$

The primary axial force is known explicitly from Equation (3.2.10):

$$F_{\text{ax},p}(s, \omega_p) = m_f \omega_p^2 \left(r_{f,0} + r_f(s)\right) \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p\big(x_{p,0} + s\big).$$

Substituting this force model into Equation (3.4.33) gives the fully explicit primary no-slip traction capacity:

Primary No-Slip Torque Capacity (Expanded) **(3.4.34)**

$$\tau_{p,\text{max}}^{\text{stick}} = r_p \left[2\mu_s \tan\beta \left(m_f \omega_p^2 \left(r_{f,0} + r_f(s)\right) \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p\big(x_{p,0} + s\big)\right) - \rho_b A_b \phi_p \left(r_{p,\text{cm}} \dot{v}_{b,\text{ns}} + \dot{r}_{p,\text{cm}}\, v_b\right)\right]$$

This quantity is the maximum primary contact torque magnitude that can be supported while maintaining the no-slip candidate motion. The primary no-slip torque demand from Equation (3.3.22) is therefore admissible only if

$$\boxed{|\tau_{p,\text{ns}}| \le \tau_{p,\text{max}}^{\text{stick}}.} \tag{3.4.35}$$

Equivalently,

$$-\tau_{p,\text{max}}^{\text{stick}} \le \tau_{p,\text{ns}} \le \tau_{p,\text{max}}^{\text{stick}}.$$

If Equation (3.4.35) is satisfied, the primary contact can support the torque required by the no-slip candidate. If it is not satisfied, the primary contact cannot remain adhered under the demanded torque, and the final torque transfer must instead be determined by a slipping contact law.

3.4.8 Secondary Pulley No-Slip Admissibility

The secondary pulley follows the same overall procedure as the primary, but with one important difference: the secondary axial force is torque-reactive. Unlike the primary axial force, which depends only on shift position and primary speed, the secondary axial force depends on the same transmitted torque being checked for admissibility.

For compactness in this subsection, write

$$r_s = r_{s,\mathrm{eff}}(s).$$

On the no-slip candidate branch, the belt acceleration is $\dot{v}_{b,\mathrm{ns}}$, as given by Equation (3.3.24). Therefore, the secondary traction capacity is evaluated using this candidate acceleration.

Applying Equation (3.4.32) with $j = s$, and using the static friction coefficient μ_s, gives

$$\tau_{s,\max}^{\mathrm{stick}} = r_s \left[2\mu_s F_{\mathrm{ax},s}(s, \tau_s) \tan\beta - \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}} \dot{v}_{b,\mathrm{ns}} + \dot{r}_{s,\mathrm{cm}}\, v_b\right)\right]. \tag{3.4.36}$$

For the no-slip admissibility check, the torque appearing in the secondary axial force model is the no-slip demand $\tau_{s,\mathrm{ns}}$ from Equation (3.3.23). Using the secondary axial force model from Equation (3.2.16),

$$F_{\mathrm{ax},s}(s, \tau_{s,\mathrm{ns}}) = \frac{\tau_{s,\mathrm{ns}} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)}{2} \frac{\mathrm{d}\theta_s}{\mathrm{d}s} + k_{s,x}(x_{s,0} + s).$$

Substituting this into Equation (3.4.36) gives

$$\begin{aligned} \tau_{s,\max}^{\mathrm{stick}} = r_s \Bigg[&\mu_s \tan\beta \left(\tau_{s,\mathrm{ns}} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)\right) \frac{\mathrm{d}\theta_s}{\mathrm{d}s} \\ &+ 2\mu_s \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}} \dot{v}_{b,\mathrm{ns}} + \dot{r}_{s,\mathrm{cm}}\, v_b\right) \Bigg]. \end{aligned} \tag{3.4.37}$$

This expression shows why the secondary admissibility check is slightly different from the primary. The capacity depends linearly on the no-slip torque demand itself, because a larger transmitted secondary torque also increases the torque-reactive helix clamping force.

The secondary no-slip demand is admissible if

$$|\tau_{s,\mathrm{ns}}| \le \tau_{s,\max}^{\mathrm{stick}}.$$

For a positive no-slip secondary torque demand, $\tau_{s,\mathrm{ns}} \ge 0$, substituting Equation (3.4.37) gives

$$\begin{aligned} \tau_{s,\mathrm{ns}} \le r_s \Bigg[&\mu_s \tan\beta \left(\tau_{s,\mathrm{ns}} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)\right) \frac{\mathrm{d}\theta_s}{\mathrm{d}s} \\ &+ 2\mu_s \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}} \dot{v}_{b,\mathrm{ns}} + \dot{r}_{s,\mathrm{cm}}\, v_b\right) \Bigg]. \end{aligned} \tag{3.4.38}$$

Collecting the $\tau_{s,\mathrm{ns}}$ terms gives the positive-direction upper bound

Secondary No-Slip Upper Bound **(3.4.39)**

$$\tau_{s,+}^{\text{stick}} = \frac{r_s\left[\mu_s \tan\beta\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)\dfrac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu_s \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s\left(r_{s,\text{cm}}\dot{v}_{b,\text{ns}} + \dot{r}_{s,\text{cm}}\, v_b\right)\right]}{1 - r_s\mu_s \tan\beta\dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}$$

For a negative no-slip secondary torque demand, $\tau_{s,\text{ns}} \leq 0$, the admissibility condition becomes

$$-\tau_{s,\text{ns}} \leq \tau_{s,\max}^{\text{stick}}.$$

Substituting Equation (3.4.37) gives

$$\begin{aligned} -\tau_{s,\text{ns}} \leq r_s\Bigg[&\mu_s \tan\beta\left(\tau_{s,\text{ns}} + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)\right)\frac{\mathrm{d}\theta_s}{\mathrm{d}s} \\ &+ 2\mu_s \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s\left(r_{s,\text{cm}}\dot{v}_{b,\text{ns}} + \dot{r}_{s,\text{cm}}\, v_b\right)\Bigg]. \end{aligned} \tag{3.4.40}$$

Collecting the $\tau_{s,\text{ns}}$ terms gives the negative-direction lower bound

Secondary No-Slip Lower Bound **(3.4.41)**

$$\tau_{s,-}^{\text{stick}} = -\frac{r_s\left[\mu_s \tan\beta\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)\dfrac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu_s \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s\left(r_{s,\text{cm}}\dot{v}_{b,\text{ns}} + \dot{r}_{s,\text{cm}}\, v_b\right)\right]}{1 + r_s\mu_s \tan\beta\dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}$$

The secondary no-slip torque demand is therefore admissible only if

$$\boxed{\tau_{s,-}^{\text{stick}} \leq \tau_{s,\text{ns}} \leq \tau_{s,+}^{\text{stick}}.} \tag{3.4.42}$$

If Equation (3.4.42) is satisfied, the secondary contact can support the torque required by the no-slip candidate. If it is not satisfied, the secondary contact cannot remain adhered under the demanded torque, and the final torque transfer must instead be determined by a slipping contact law.

3.4.9 Section Conclusion and Transition

This section derived the traction limits needed to test the no-slip torque candidate from Section 3.3. Starting from a local belt–pulley force balance, the available tight–slack tension difference was converted into a pulley torque capacity. This produced admissibility conditions for the primary and secondary no-slip torque demands, $\tau_{p,\text{ns}}$ and $\tau_{s,\text{ns}}$.

These results determine whether the no-slip candidate can be physically maintained. If both pulley contacts satisfy their admissibility conditions, the belt can remain adhered to both pulleys and the no-slip torque transfer is accepted.

What remains is the complementary case: what happens when one or both contacts cannot remain adhered, or when relative belt–pulley motion is already present. In that situation, the torque transfer is no longer determined by the no-slip demand. Instead, the slipping contact must be governed by kinetic traction and the belt equation of motion. The next section develops the contact-state selection rules and the corresponding slip-branch dynamics.

3.5 Slip Branch Dynamics and Contact-State Selection

The previous section determined when the no-slip torque demands can be supported by the available belt–pulley traction. If the primary and secondary contacts both satisfy their admissibility conditions, the no-slip candidate is physically consistent and the belt remains adhered to both pulleys.

When one of these admissibility conditions fails, the no-slip candidate can no longer be used as the final torque-transfer law. The contact that exceeds its available traction cannot supply the demanded torque while remaining stuck, so relative motion begins at that belt–pulley interface. Once this occurs, the contact torque is no longer determined by the no-slip demand. Instead, it is set by the kinetic traction available at the slipping contact.

This section develops the rules used to select the active contact state and the corresponding equations of motion. The same three bodies introduced in Section 3.3.1 are retained: the primary pulley, the belt, and the secondary pulley. What changes from one branch to another is the contact condition imposed at each pulley. A stuck contact enforces belt-speed compatibility, while a slipping contact applies a saturated kinetic traction torque in the direction opposing relative motion.

The goal is therefore to complete the torque-transfer closure. We first define the relative belt–pulley speeds used to identify sliding, then state the rules for entering, maintaining, and exiting slip. The remaining subsections derive the branch dynamics for primary slip, secondary slip, and simultaneous slip at both contacts.

3.5.1 Relative Contact Speeds and Slip-State Logic

To determine whether a belt–pulley contact is sticking or slipping, the model compares the belt transport speed to the local surface speed imposed by each pulley.

For the primary pulley, define the relative contact speed as

$$v_{\mathrm{rel},p} = r_p\omega_p - v_b, \tag{3.5.1}$$

where $r_p = r_{p,\mathrm{eff}}(s)$. Thus, $v_{\mathrm{rel},p} > 0$ means the primary pulley surface is moving faster than the belt at the contact radius and tends to drive the belt forward.

For the secondary pulley, define

$$v_{\mathrm{rel},s} = v_b - r_s\omega_s, \tag{3.5.2}$$

where $r_s = r_{s,\mathrm{eff}}(s)$. Thus, $v_{\mathrm{rel},s} > 0$ means the belt is moving faster than the secondary pulley surface and tends to drive the secondary forward.

These definitions are chosen so that positive relative speed corresponds to the usual forward torque-transfer direction at each contact: primary-to-belt on the primary side, and belt-to-secondary on the secondary side.

A contact can remain stuck only if its relative contact speed is zero, or within a small numerical tolerance:

$$|v_{\mathrm{rel},j}| \le \varepsilon_v, \qquad j \in \{p, s\}. \tag{3.5.3}$$

If both contacts satisfy Equation (3.5.3), the model first tests the no-slip candidate derived in Section 3.3. This candidate is accepted only if both pulley contacts also satisfy the traction admissibility conditions derived in Section 3.4:

$$|\tau_{p,\mathrm{ns}}| \le \tau_{p,\max}^{\mathrm{stick}}, \tag{3.5.4}$$

and

$$\tau_{s,-}^{\mathrm{stick}} \le \tau_{s,\mathrm{ns}} \le \tau_{s,+}^{\mathrm{stick}}. \tag{3.5.5}$$

If these conditions hold, the no-slip candidate becomes the active torque-transfer branch:

$$\tau_p = \tau_{p,\mathrm{ns}}, \qquad \tau_s = \tau_{s,\mathrm{ns}}, \qquad \dot{v}_b = \dot{v}_{b,\mathrm{ns}}.$$

If either contact fails its admissibility check, that contact cannot support the required no-slip torque and must enter slip. The direction of slip is determined by the direction in which the demanded torque exceeds the admissible interval. For the primary pulley,

$$\tau_{p,\mathrm{ns}} > \tau_{p,\max}^{\mathrm{stick}} \quad \Rightarrow \quad \text{positive primary slip,}$$

while

$$\tau_{p,\mathrm{ns}} < -\tau_{p,\max}^{\mathrm{stick}} \quad \Rightarrow \quad \text{negative primary slip.}$$

For the secondary pulley,

$$\tau_{s,\mathrm{ns}} > \tau_{s,+}^{\mathrm{stick}} \quad \Rightarrow \quad \text{positive secondary slip,}$$

while

$$\tau_{s,\mathrm{ns}} < \tau_{s,-}^{\mathrm{stick}} \quad \Rightarrow \quad \text{negative secondary slip.}$$

Once a contact is already sliding, its state is determined primarily by its relative contact speed. If

$$|v_{\mathrm{rel},j}| > \varepsilon_v,$$

then contact j is treated as slipping, and its torque is set by kinetic traction rather than by the no-slip demand. The slip direction is recorded by

$$\sigma_j = \mathrm{sgn}(v_{\mathrm{rel},j}), \qquad j \in \{p, s\}. \tag{3.5.6}$$

Slip persists while the relative contact speed remains nonzero. During this time, the slipping contact applies a saturated kinetic traction torque in the direction opposing relative motion. The no-slip candidate is not reaccepted until the relative speed returns to the sticking tolerance and the static admissibility conditions can again be satisfied.

This gives three possible slipping branches:

- primary slip, where the primary contact slides and the secondary contact remains stuck;
- secondary slip, where the secondary contact slides and the primary contact remains stuck;
- two-contact slip, where both pulley contacts slide simultaneously.

The following subsections derive the equations of motion for each branch.

Remark (Power Flow During Slip)

In the no-slip branch, belt-speed compatibility constrains the primary pulley, belt, and secondary pulley to move together kinematically. Once slip occurs, that compatibility is lost at the slipping contact. The contact torque is then limited by kinetic traction, while the pulley and belt speeds evolve according to their own equations of motion.

As a result, the instantaneous mechanical power at the primary contact, $\tau_p\omega_p$, does not need to match the instantaneous mechanical power at the secondary contact, $\tau_s\omega_s$. The difference is associated with changes in stored kinetic energy of the rotating components and belt, together with slip dissipation. Thus, an apparent short-term excess of output-side power does not imply energy creation; it reflects energy being drawn from inertia while the slipping contact breaks the ideal no-slip power constraint.

3.5.2 Kinetic Traction Law for a Slipping Contact

Once a belt–pulley contact is slipping, the torque at that contact is no longer determined by the no-slip torque demand. Instead, the contact applies the largest kinetic traction torque available in the direction opposing relative motion.

The generic torque-capacity expression from Equation (3.4.32) still applies, but the static coefficient μ_s is replaced by the kinetic coefficient μ_k. For a slipping contact on pulley $j \in \{p, s\}$, define

$$\tau_{j,\max}^{\text{slip}}(\dot{v}_b) = r_j \left[2\mu_k F_{\text{ax},j} \tan\beta - \rho_b A_b \phi_j \left(r_{j,\text{cm}}\dot{v}_b + \dot{r}_{j,\text{cm}}\, v_b\right)\right], \tag{3.5.7}$$

where $r_j = r_{j,\text{eff}}(s)$. The notation $\tau_{j,\max}^{\text{slip}}(\dot{v}_b)$ emphasizes that the available slipping torque is not generally a fixed number; it depends on the belt acceleration through the inertial correction term.

The direction of the slipping torque is determined by the relative contact speed defined in Section 3.5.1. Let

$$\sigma_j = \operatorname{sgn}(v_{\text{rel},j}), \qquad j \in \{p, s\}. \tag{3.5.8}$$

With the relative-speed definitions adopted in Equation (3.5.1) and Equation (3.5.2), $\sigma_j > 0$ corresponds to the usual forward torque-transfer direction at contact j. The slipping contact torque is then modeled as the saturated kinetic traction torque

$$\tau_j = \sigma_j\, \tau_{j,\max}^{\text{slip}}(\dot{v}_b). \tag{3.5.9}$$

Equation Equation (3.5.9) should not be interpreted as a completed solution for τ_j, because the right-hand side still contains $\dot{v}_b$. In a slip branch, the belt acceleration must be solved simultaneously with the saturated contact torque using the belt equation of motion Equation (3.3.3). The following subsections carry out this solve for primary slip, secondary slip, and simultaneous slip at both contacts.

3.5.3 Primary-Slip Branch

We first consider the branch where the primary contact is slipping while the secondary contact remains stuck. In this case, the primary pulley no longer enforces belt-speed compatibility. Instead, the primary contact torque is set by the saturated kinetic traction law from Section 3.5.2.

The secondary contact, however, is still adhered. Therefore, the secondary pulley remains kinematically compatible with the belt:

$$\dot{\omega}_s = \frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s}, \tag{3.5.10}$$

Using the secondary rotational equation of motion Equation (3.3.2), the secondary contact torque is then

$$\tau_s = \tau_{\text{load}} + I_s \left(\frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s} \right). \tag{3.5.11}$$

At the slipping primary contact, let

$$\sigma_p = \operatorname{sgn}(v_{\text{rel},p}).$$

Applying Equation (3.5.9) with $j = p$ gives

$$\tau_p = \sigma_p r_p \left[2\mu_k F_{\text{ax},p} \tan\beta - \rho_b A_b \phi_p \left(r_{p,\text{cm}} \dot{v}_b + \dot{r}_{p,\text{cm}} v_b \right) \right], \tag{3.5.12}$$

where $r_p = r_{p,\text{eff}}(s)$.

The remaining unknown is the belt acceleration $\dot{v}_b$. This is found by substituting Equation (3.5.11) and Equation (3.5.12) into the belt equation of motion Equation (3.3.3):

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}. \tag{3.5.13}$$

Substituting gives

$$\begin{aligned} m_b \dot{v}_b = {} & \sigma_p \left[2\mu_k F_{\text{ax},p} \tan\beta - \rho_b A_b \phi_p \left(r_{p,\text{cm}} \dot{v}_b + \dot{r}_{p,\text{cm}} v_b \right) \right] \\ & - \frac{\tau_{\text{load}}}{r_s} - \frac{I_s}{r_s} \left(\frac{\dot{v}_b - \dot{r}_s \omega_s}{r_s} \right). \end{aligned} \tag{3.5.14}$$

Collecting the $\dot{v}_b$ terms yields the primary-slip belt acceleration:

Primary-Slip Belt Acceleration **(3.5.15)**

$$\dot{v}_{b,p-\text{slip}} = \frac{\sigma_p \left[2\mu_k F_{\text{ax},p} \tan\beta - \rho_b A_b \phi_p \dot{r}_{p,\text{cm}} v_b \right] - \dfrac{\tau_{\text{load}}}{r_s} + \dfrac{I_s \dot{r}_s \omega_s}{r_s^2}}{m_b + \sigma_p \rho_b A_b \phi_p r_{p,\text{cm}} + \dfrac{I_s}{r_s^2}}$$

Once $\dot{v}_{b,p-\text{slip}}$ is known, the branch torques follow by substitution into Equation (3.5.12) and Equation (3.5.11). The pulley accelerations are then

$$\dot{\omega}_p = \frac{\tau_{\text{eng}} - \tau_p}{I_p}, \tag{3.5.16}$$

and

$$\dot{\omega}_s = \frac{\dot{v}_{b,p-\text{slip}} - \dot{r}_s\omega_s}{r_s}. \tag{3.5.17}$$

Thus, in the primary-slip branch, the primary contact torque is set by kinetic traction, the secondary contact still enforces belt-speed compatibility, and the belt equation of motion determines the resulting belt acceleration. This branch remains valid only while the secondary contact can remain adhered. If the secondary contact also fails its sticking condition, the model must transition to the two-contact slip branch.

3.5.4 Secondary-Slip Branch

We next consider the branch where the secondary contact is slipping while the primary contact remains stuck. In this case, the secondary contact torque is set by saturated kinetic traction, while the primary contact still enforces belt-speed compatibility.

The primary contact remains adhered, so the primary pulley must satisfy

$$\dot{\omega}_p = \frac{\dot{v}_b - \dot{r}_p\omega_p}{r_p}, \tag{3.5.18}$$

where $r_p = r_{p,\text{eff}}(s)$. Using the primary rotational equation of motion Equation (3.3.1), the primary contact torque is therefore

$$\tau_p = \tau_{\text{eng}} - I_p\left(\frac{\dot{v}_b - \dot{r}_p\omega_p}{r_p}\right). \tag{3.5.19}$$

At the slipping secondary contact, let

$$\sigma_s = \text{sgn}(v_{\text{rel},s}).$$

Applying Equation (3.5.9) with $j = s$ gives

$$\tau_s = \sigma_s r_s\left[2\mu_k F_{\text{ax},s}(s,\tau_s)\tan\beta - \rho_b A_b \phi_s\left(r_{s,\text{cm}}\dot{v}_b + \dot{r}_{s,\text{cm}}v_b\right)\right], \tag{3.5.20}$$

where $r_s = r_{s,\text{eff}}(s)$.

Unlike the primary slip branch, this expression is implicit in τ_s because the secondary axial force is torque-reactive. Substituting the secondary axial force model from Equation (3.2.16) gives

$$\begin{aligned}\tau_s = \sigma_s r_s\Bigg[&\mu_k \tan\beta\left(\tau_s + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)\right)\frac{\mathrm{d}\theta_s}{\mathrm{d}s} \\ &+ 2\mu_k \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s\left(r_{s,\text{cm}}\dot{v}_b + \dot{r}_{s,\text{cm}}v_b\right)\Bigg].\end{aligned} \tag{3.5.21}$$

Collecting the τ_s terms gives

$$\tau_s = \frac{\sigma_s r_s \left[\mu_k \tan\beta\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)\dfrac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu_k \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}}\dot{v}_b + \dot{r}_{s,\mathrm{cm}} v_b\right)\right]}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}. \tag{3.5.22}$$

The remaining unknown is the belt acceleration $\dot{v}_b$. Substituting Equation (3.5.19) and Equation (3.5.22) into the belt equation of motion Equation (3.3.3),

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s},$$

and collecting the $\dot{v}_b$ terms gives the secondary-slip belt acceleration:

Secondary-Slip Belt Acceleration **(3.5.23)**

$$\dot{v}_{b,s-\mathrm{slip}} = \frac{\dfrac{\tau_{\mathrm{eng}}}{r_p} + \dfrac{I_p \dot{r}_p \omega_p}{r_p^2} - \dfrac{\sigma_s \left[\mu_k \tan\beta\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)\dfrac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu_k \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \dot{r}_{s,\mathrm{cm}} v_b\right]}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}}{m_b + \dfrac{I_p}{r_p^2} - \dfrac{\sigma_s \rho_b A_b \phi_s r_{s,\mathrm{cm}}}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}}$$

Once $\dot{v}_{b,s-\mathrm{slip}}$ is known, the branch torques follow by substitution into Equation (3.5.19) and Equation (3.5.22). The pulley accelerations are then

$$\dot{\omega}_p = \frac{\dot{v}_{b,s-\mathrm{slip}} - \dot{r}_p \omega_p}{r_p}, \tag{3.5.24}$$

and

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\mathrm{load}}}{I_s}. \tag{3.5.25}$$

Thus, in the secondary-slip branch, the secondary contact torque is set by kinetic traction, including the torque-reactive helix feedback, while the primary contact still enforces belt-speed compatibility. This branch remains valid only while the primary contact can remain adhered. If the primary contact also fails its sticking condition, the model must transition to the two-contact slip branch.

3.5.5 Two-Contact Slip Branch

Finally, consider the branch where both belt–pulley contacts are slipping. In this case, neither pulley enforces belt-speed compatibility. Both contact torques are instead determined by saturated kinetic traction, and the belt acceleration is obtained directly from the belt equation of motion.

Let

$$\sigma_p = \operatorname{sgn}(v_{\mathrm{rel},p}), \qquad \sigma_s = \operatorname{sgn}(v_{\mathrm{rel},s}).$$

At the primary contact, applying Equation (3.5.9) with $j = p$ gives

$$\tau_p = \sigma_p r_p \left[2\mu_k F_{\mathrm{ax},p} \tan\beta - \rho_b A_b \phi_p \left(r_{p,\mathrm{cm}} \dot{v}_b + \dot{r}_{p,\mathrm{cm}} v_b\right)\right], \tag{3.5.26}$$

where $r_p = r_{p,\mathrm{eff}}(s)$.

At the secondary contact, applying Equation (3.5.9) with $j = s$ gives

$$\tau_s = \sigma_s r_s \left[2\mu_k F_{\mathrm{ax},s}(s, \tau_s) \tan\beta - \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}} \dot{v}_b + \dot{r}_{s,\mathrm{cm}} v_b\right)\right], \tag{3.5.27}$$

where $r_s = r_{s,\mathrm{eff}}(s)$. As in Section 3.5.4, the secondary expression is implicit because the secondary axial force is torque-reactive. Substituting Equation (3.2.16) and collecting the τ_s terms gives

$$\tau_s = \frac{\sigma_s r_s \left[\mu_k \tan\beta\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu_k \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}} \dot{v}_b + \dot{r}_{s,\mathrm{cm}} v_b\right)\right]}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}. \tag{3.5.28}$$

The belt equation of motion is still

$$m_b \dot{v}_b = \frac{\tau_p}{r_p} - \frac{\tau_s}{r_s}.$$

Substituting Equation (3.5.26) and Equation (3.5.28), then collecting the $\dot{v}_b$ terms, gives the two-contact-slip belt acceleration:

Two-Contact-Slip Belt Acceleration **(3.5.29)**

$$\dot{v}_{b,ps-\mathrm{slip}} = \frac{\sigma_p \left[2\mu_k F_{\mathrm{ax},p} \tan\beta - \rho_b A_b \phi_p \dot{r}_{p,\mathrm{cm}} v_b\right] - \dfrac{\sigma_s \left[\mu_k \tan\beta\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu_k \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \dot{r}_{s,\mathrm{cm}} v_b\right]}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}}{m_b + \sigma_p \rho_b A_b \phi_p r_{p,\mathrm{cm}} - \dfrac{\sigma_s \rho_b A_b \phi_s r_{s,\mathrm{cm}}}{1 - \sigma_s r_s \mu_k \tan\beta \dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}}$$

Once $\dot{v}_{b,ps-\mathrm{slip}}$ is known, the branch torques follow by substitution into Equation (3.5.26) and Equation (3.5.28). Since neither contact is adhered, the pulley accelerations are obtained directly from the rotational equations of motion:

$$\dot{\omega}_p = \frac{\tau_{\mathrm{eng}} - \tau_p}{I_p}, \tag{3.5.30}$$

and

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\text{load}}}{I_s}. \tag{3.5.31}$$

Thus, in the two-contact slip branch, both pulley contacts are governed by kinetic traction, and the belt is not kinematically constrained to either pulley. The belt acceleration is determined only by the balance of the two saturated contact torques through the belt equation of motion.

3.5.6 Contact-State Selection

The preceding subsections define the possible torque-transfer closures. The active branch is selected by checking which contacts can remain adhered.

Let the primary sticking condition be

$$\mathcal{S}_p \equiv (|v_{\text{rel},p}| \le \varepsilon_v) \wedge \left(|\tau_{p,\text{ns}}| \le \tau_{p,\max}^{\text{stick}}\right),$$

and let the secondary sticking condition be

$$\mathcal{S}_s \equiv (|v_{\text{rel},s}| \le \varepsilon_v) \wedge \left(\tau_{s,-}^{\text{stick}} \le \tau_{s,\text{ns}} \le \tau_{s,+}^{\text{stick}}\right).$$

The branch closure for the belt acceleration and contact torques is then

$$(\dot{v}_b, \tau_p, \tau_s) = \begin{cases} (\dot{v}_{b,\text{ns}}, \tau_{p,\text{ns}}, \tau_{s,\text{ns}}), & \mathcal{S}_p \wedge \mathcal{S}_s, \\ (\dot{v}_{b,p-\text{slip}}, \tau_p \text{ from } \textit{Equation } (3.5.12), \tau_s \text{ from } \textit{Equation } (3.5.11)), & \neg\mathcal{S}_p \wedge \mathcal{S}_s, \\ (\dot{v}_{b,s-\text{slip}}, \tau_p \text{ from } \textit{Equation } (3.5.19), \tau_s \text{ from } \textit{Equation } (3.5.22)), & \mathcal{S}_p \wedge \neg\mathcal{S}_s, \\ (\dot{v}_{b,ps-\text{slip}}, \tau_p \text{ from } \textit{Equation } (3.5.26), \tau_s \text{ from } \textit{Equation } (3.5.28)), & \neg\mathcal{S}_p \wedge \neg\mathcal{S}_s. \end{cases} \tag{3.5.32}$$

The corresponding belt accelerations are given by Equation (3.3.24), Equation (3.5.15), Equation (3.5.23), and Equation (3.5.29), respectively.

Thus, when both contacts can remain adhered, the model uses the no-slip torque demands. When one or both contacts cannot remain adhered, the corresponding slip branch replaces the no-slip demand with saturated kinetic traction and solves the belt acceleration from the belt equation of motion.

3.5.7 Section Summary

This section completed the torque-transfer closure for cases where the no-slip candidate cannot be maintained. Relative contact speeds were introduced to identify whether each belt–pulley interface is adhered or sliding, and kinetic traction was used to define the torque applied by a slipping contact.

The resulting branch structure separates the possible contact states into no-slip, primary slip, secondary slip, and two-contact slip. In each slipping case, the saturated contact torque replaces the no-slip demand, and the belt equation of motion determines the corresponding belt acceleration.

Together with the no-slip demand from Section 3.3 and the traction admissibility conditions from Section 3.4, the branch selection law in Equation (3.5.32) provides the final torque-transfer rule needed by the complete CVT dynamic model.

3.6 Complete CVT Dynamic Model

This section collects the results of the preceding derivations into one closed nonlinear dynamical model. No new physics is introduced here. The purpose is to summarize, in one place,

1. the state variables,
2. the geometric and contact quantities computed from those states,
3. the torque-transfer closure,
4. and the final governing equations of motion.

3.6.1 State Definition

The state vector is

$$\mathbf{x}(t) = \big(\omega_p,\ \omega_s,\ s,\ \dot{s},\ v_b\big), \tag{3.6.1}$$

where

- ω_p is the primary pulley angular velocity,
- ω_s is the secondary pulley angular velocity,
- s is the axial shift coordinate,
- $\dot{s}$ is the axial shift velocity,
- v_b is the belt transport speed.

3.6.2 Geometry and Contact Kinematics

At each instant, the shift coordinate s determines the pulley geometry through the belt-length constraint and the conical kinematic relations derived in Section 3.1. In particular,

$$r_{p,\text{eff}}, \quad r_{s,\text{eff}}, \quad r_{p,\text{cm}}, \quad r_{s,\text{cm}}, \quad \phi_p, \quad \phi_s$$

are functions of s. For compactness in the complete model, write

$$r_p = r_{p,\text{eff}}(s), \qquad r_s = r_{s,\text{eff}}(s).$$

The geometric CVT ratio remains

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \tag{3.6.2}$$

Its time derivative is

$$\dot{R} = \dot{R}(s, \dot{s}), \tag{3.6.3}$$

as given by Equation (3.1.57).

The relative contact speeds at the two pulley interfaces are

$$v_{\text{rel},p} = r_p\omega_p - v_b, \qquad v_{\text{rel},s} = v_b - r_s\omega_s, \tag{3.6.4}$$

as defined in Equation (3.5.1) and Equation (3.5.2). These quantities determine whether each contact is eligible to remain adhered or must be treated as sliding.

3.6.3 Torque-Transfer Closure

The torque-transfer subsystem is closed by determining the primary contact torque τ_p, the secondary contact torque τ_s, and the belt acceleration $\dot{v}_b$. These quantities are computed algebraically from the current state and contact branch.

The no-slip candidate derived in Section 3.3.6 provides

$$\dot{v}_{b,\mathrm{ns}}, \qquad \tau_{p,\mathrm{ns}}, \qquad \tau_{s,\mathrm{ns}},$$

from Equation (3.3.24), Equation (3.3.22), and Equation (3.3.23). These are the belt acceleration and contact torque demands required for adhered motion at both pulleys.

The admissibility conditions derived in Section 3.4.7 and Section 3.4.8 determine whether those no-slip demands can be supported. The primary contact is admissible when

$$|\tau_{p,\mathrm{ns}}| \le \tau_{p,\max}^{\mathrm{stick}}, \tag{3.6.5}$$

as in Equation (3.4.35). The secondary contact is admissible when

$$\tau_{s,-}^{\mathrm{stick}} \le \tau_{s,\mathrm{ns}} \le \tau_{s,+}^{\mathrm{stick}}, \tag{3.6.6}$$

as in Equation (3.4.42).

The active contact branch is selected using the closure law from Section 3.5.6. In compact form,

$$(\dot{v}_b, \tau_p, \tau_s) = \text{the branch output selected by Equation (3.5.32).} \tag{3.6.7}$$

If both contacts remain adhered, the branch output is

$$(\dot{v}_b, \tau_p, \tau_s) = (\dot{v}_{b,\mathrm{ns}}, \tau_{p,\mathrm{ns}}, \tau_{s,\mathrm{ns}}).$$

If one or both contacts slide, the corresponding slip branch replaces the no-slip demand with saturated kinetic traction and solves $\dot{v}_b$ from the belt equation of motion. The primary-slip, secondary-slip, and two-contact-slip belt accelerations are given by Equation (3.5.15), Equation (3.5.23), and Equation (3.5.29), respectively.

3.6.4 Governing ODEs

Once the branch closure has supplied $\dot{v}_b$, τ_p, and τ_s, the primary pulley equation of motion is

$$\dot{\omega}_p = \frac{\tau_{\mathrm{eng}}(\omega_p) - \tau_p}{I_p}. \tag{3.6.8}$$

The secondary pulley equation of motion is

$$\dot{\omega}_s = \frac{\tau_s - \tau_{\mathrm{load}}(\omega_s)}{I_s}. \tag{3.6.9}$$

The shift-position kinematics are

$$\frac{\mathrm{d}}{\mathrm{d}t} s = \dot{s}. \tag{3.6.10}$$

The axial shift acceleration is obtained from Equation (3.2.36):

$$\ddot{s} = \frac{\Big(F_{p,\mathrm{ax}}(s,\omega_p) + F_{c,p}(s,v_b)\Big) - \Big(F_{s,\mathrm{ax}}(s,\tau_s) + F_{c,s}(s,v_b)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}. \tag{3.6.11}$$

Finally, the belt transport speed evolves according to the branch-selected belt acceleration:

$$\frac{\mathrm{d}}{\mathrm{d}t} v_b = \dot{v}_b, \qquad \dot{v}_b \text{ from } \textit{Equation (3.6.7)}. \tag{3.6.12}$$

3.6.5 Model Evaluation Sequence

Given the current state

$$\mathbf{x}(t) = \big(\omega_p, \omega_s, s, \dot{s}, v_b\big),$$

the model is evaluated in the following order:

1. Compute the pulley geometry from s, including r_p, r_s, $r_{p,\mathrm{cm}}$, $r_{s,\mathrm{cm}}$, ϕ_p, and ϕ_s.
2. Compute the engine torque $\tau_{\mathrm{eng}}(\omega_p)$ and the reflected load torque $\tau_{\mathrm{load}}(\omega_s)$.
3. Compute the no-slip candidate

   $$\dot{v}_{b,\mathrm{ns}}, \qquad \tau_{p,\mathrm{ns}}, \qquad \tau_{s,\mathrm{ns}},$$

   using Equation (3.3.24), Equation (3.3.22), and Equation (3.3.23).
4. Compute the primary and secondary no-slip admissibility quantities from Section 3.4.7 and Section 3.4.8.
5. Compute the relative contact speeds $v_{\mathrm{rel},p}$ and $v_{\mathrm{rel},s}$ using Equation (3.6.4).
6. Select the active torque-transfer branch using Equation (3.5.32). This determines the applied values of

   $$\dot{v}_b, \qquad \tau_p, \qquad \tau_s.$$

7. Evaluate the ODE right-hand sides Equation (3.6.8), Equation (3.6.9), Equation (3.6.10), Equation (3.6.11), and Equation (3.6.12).

3.6.6 Complete Nonlinear ODE System

The complete closed CVT model is therefore

$$
\boxed{
\begin{aligned}
\dot{\omega}_p &= \frac{\tau_{\text{eng}}(\omega_p) - \tau_p}{I_p}, \\
\dot{\omega}_s &= \frac{\tau_s - \tau_{\text{load}}(\omega_s)}{I_s}, \\
\frac{\mathrm{d}}{\mathrm{d}t} s &= \dot{s}, \\
\frac{\mathrm{d}}{\mathrm{d}t} \dot{s} &= \frac{\Big(F_{p,\text{ax}}(s, \omega_p) + F_{c,p}(s, v_b)\Big) - \Big(F_{s,\text{ax}}(s, \tau_s) + F_{c,s}(s, v_b)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}, \\
\frac{\mathrm{d}}{\mathrm{d}t} v_b &= \dot{v}_b.
\end{aligned}
}
\tag{3.6.13}
$$

with the algebraic closure

$$
(\dot{v}_b, \tau_p, \tau_s) = \text{the active branch output from Equation (3.5.32)}.
$$

In summary, the state determines the pulley geometry and relative contact speeds. The no-slip candidate determines the torque demand required for adhered motion. The traction admissibility conditions determine whether that candidate can be maintained. If not, the appropriate slip branch supplies saturated contact torques and the resulting belt acceleration. These branch-selected quantities then feed back into the rotational, belt, and axial shift dynamics, closing the complete CVT state-space model.

Chapter 4

Results and Discussion

4.1 Simulation Results

This section compares four launch simulations: a nominal baseline case, a lighter-flyweight case, a steeper-helix case with reduced secondary torque reactivity, and a combined case with both a shallower initial primary ramp and a steeper helix. The purpose of these results is not to match experimental vehicle data directly, but to show that the model produces mechanically interpretable launch behaviour and responds to tuning changes in physically sensible ways.

The results are organized around the launch shift curves of these four cases. These curves first provide an overall picture of the launch, including engagement, low-ratio operation, active upshift, and high-ratio behaviour. The ratio terminology follows Section 3.1.9: low ratio means larger R (greater speed reduction), and high ratio means smaller R. following subsections then explain those trends by examining the axial force balance across the CVT and the resulting torque-transfer and slip behaviour. In particular, the combined case is included to show a clearly slip-dominated launch, in which the model predicts loss of the low-ratio hold and sustained slip through the early portion of the launch.

Taken together, the four cases are intended to show three things: first, that the baseline tuning produces a coherent and interpretable launch trajectory; second, that changes to primary and secondary tuning shift the launch behaviour in the expected direction; and third, that when the tuning is made sufficiently unfavorable, the model captures the transition to prolonged

slipping rather than continuing to predict an unrealistic no-slip response. The implementation used to generate these simulations, along with repository, release, and data-access details, is documented in Section 4.2.

4.1.1 Simulation Cases and Phase Convention

The results section compares four full-throttle launch simulations from rest. One case is treated as the nominal baseline, while the remaining three are targeted tuning variations chosen to isolate specific changes in CVT behaviour. All cases use the same vehicle, gearbox, tire, and engine model parameters; they differ only in the clutch tuning parameters noted below.

Table 4.1: Launch cases considered in the results section. All parameters not listed remain equal to the nominal baseline.

Case	Description	Purpose in results section
D	Default tuning	Baseline reference case
F	Lighter flyweights	Shows the effect of reduced primary centrifugal clamping
H	Steeper helix	Shows the effect of reduced secondary torque reactivity
C	Shallower initial primary ramp + steeper helix	Produces a strongly slip-dominated launch

To keep the discussion consistent across the different plots, the launch is described in terms of a small number of recurring operating phases. These phases are not introduced as separate events for each case, but as common regions of behaviour used to interpret the launch trajectories and their underlying force and torque balances.

Throughout the following subsections, these phases will be indicated directly on the plots using low-opacity background shading. The same color convention is used wherever possible so that the reader can track the evolution of each case across the shift, clamping-force, and torque-transfer results.

The phase colors are

- **engagement** — light blue,
- **low-ratio operation** — light green,
- **active upshift** — light yellow, and
- **high-ratio operation** — light red.

Not every case is expected to exhibit all four phases equally clearly. The phase shading is therefore used as a visual guide to the dominant operating regime, rather than as a claim that every boundary is perfectly sharp.

4.1.2 Launch Shift Curves

The overall launch behaviour of the four tuning cases is first compared in Figure 4.1. Each trajectory is plotted in engine-speed–vehicle-speed coordinates, so that engagement, ratio change, and the final high-ratio condition can all be viewed on the same axes. All four simulations begin from the same launch initial conditions,

$$\omega_p(0) = 1800\,\text{rpm}, \qquad \omega_s(0) = 0,$$

where ω_p is the primary angular speed and ω_s is the driven angular speed corresponding to zero vehicle speed. From this state, the system is released into a full-throttle forward simulation using the complete coupled model developed in the preceding sections.

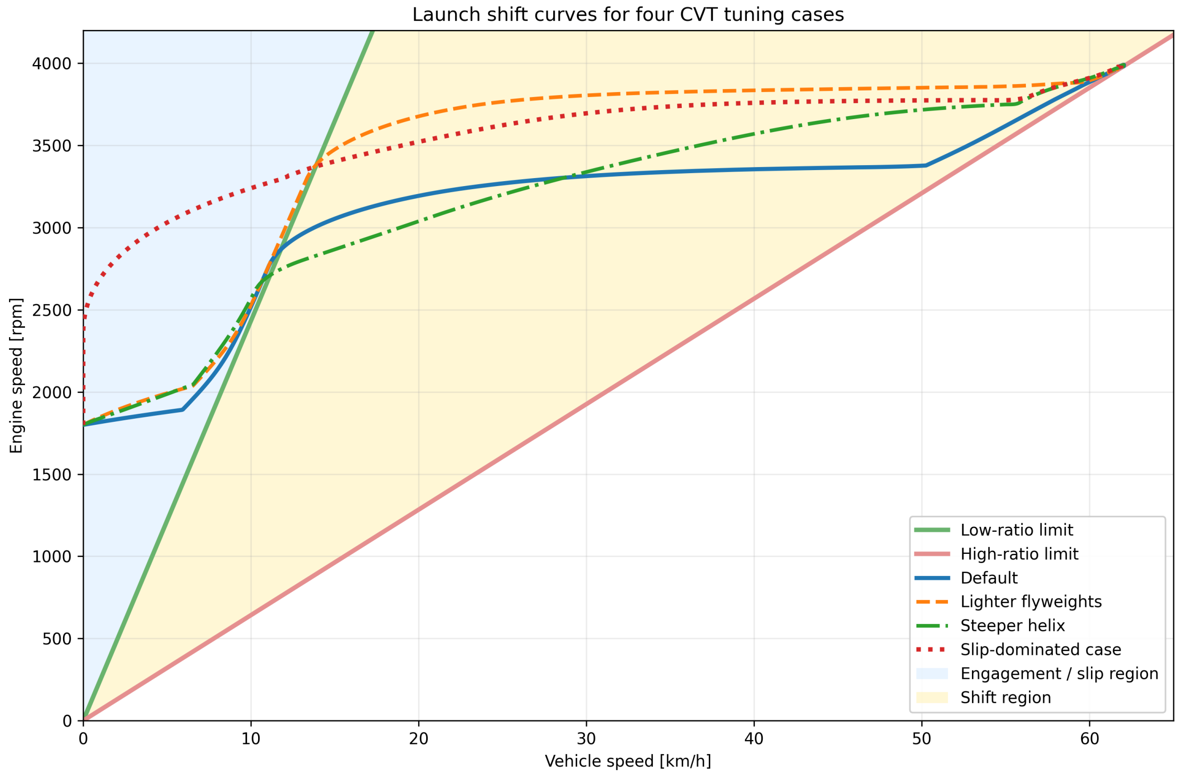


Figure 4.1: Engine-speed–vehicle-speed launch trajectories for the four tuning cases. The green line is the low-ratio limit, corresponding to a fully unshifted CVT. The red line is the high-ratio limit, corresponding to a fully shifted CVT. The blue shaded region to the left of the low-ratio line corresponds to launch engagement, where slip must be present. The yellow region between the two limit lines is the no-slip shift region.

The two reference lines are the kinematic no-slip limits of the CVT. The **low-ratio line** represents the fully unshifted transmission, where the CVT is in its torque-multiplying launch condition. The **high-ratio line** represents the fully shifted transmission, where the CVT has reached its final speed-increasing condition. Between these two limits lies the admissible shift region. A trajectory cannot move to the right of the high-ratio line, and any portion of a trajectory lying to the left of the low-ratio line must involve slip, since even the fully unshifted CVT cannot support such a low vehicle speed for the corresponding engine speed.

Read from left to right, the figure therefore shows the full launch sequence. The launch begins in the blue engagement region, where the vehicle is still too slow for no-slip operation. After crossing the low-ratio boundary, the trajectory enters the shift region. In a well-behaved launch, the CVT then tends to move approximately horizontally across this region, increasing vehicle speed while holding engine speed near a preferred operating value. This nearly horizontal portion will be referred to here as the *flat-shift region*. As the shift completes, the trajectory approaches the high-ratio line, which forms the final no-slip boundary of the launch.

With that interpretation in place, the differences between the four tuning cases are already visible. The **default** case shows the clearest baseline behaviour: a short engagement region, a brief low-ratio hold, and then a relatively flat shift through the middle of the plot before approaching the high-ratio boundary. This is the most balanced of the four trajectories, since it neither leaves low ratio too early nor rises excessively in engine speed during the main shift.

The **lighter-flyweight** case retains low ratio for longer before crossing into the shift region. Its flat-shift portion is also noticeably higher in engine speed than the default case, so the launch spends more of the main shift at elevated primary speed. The overall shape is still recognizable as a conventional launch trajectory, but the shift is delayed and carried out at a higher engine-speed level.

The **steeper-helix** case behaves differently. It leaves the low-ratio boundary much earlier, so the launch shows little sustained low-ratio hold. Its shift path is also less flat: instead of moving nearly horizontally across the shift region, it climbs upward more strongly as vehicle speed increases. This indicates that the launch does not settle into the same nearly constant engine-speed shift seen in the default case.

The **combined slip-dominated** case is the most extreme. It remains in the engagement region for much longer, showing no clear low-ratio hold before entering the shift region. Once it does cross into the no-slip range, its shift still occurs at a comparatively high engine speed, and the early launch is much more dominated by slip than by clean low-ratio operation.

At this stage, Figure 4.1 is intended to establish the overall shape of the four launches and the main qualitative differences between them. The key observations are the amount of low-ratio retention, the height and flatness of the main shift segment, and the extent of the initial slip-dominated portion of the launch. The following subsections explain these differences in terms of the underlying axial force balance and torque-transfer behaviour of the CVT.

4.1.3 Axial Force Balance Through the Launch

The shift-curve trajectories of Figure 4.1 indicate when each case remains near low ratio, when it begins to upshift, and how quickly it reaches high ratio. To explain those differences mechanically, Figure 4.2 compares the total axial force acting on the primary and secondary pulleys for each tuning case, with the same launch phases shaded in the background.

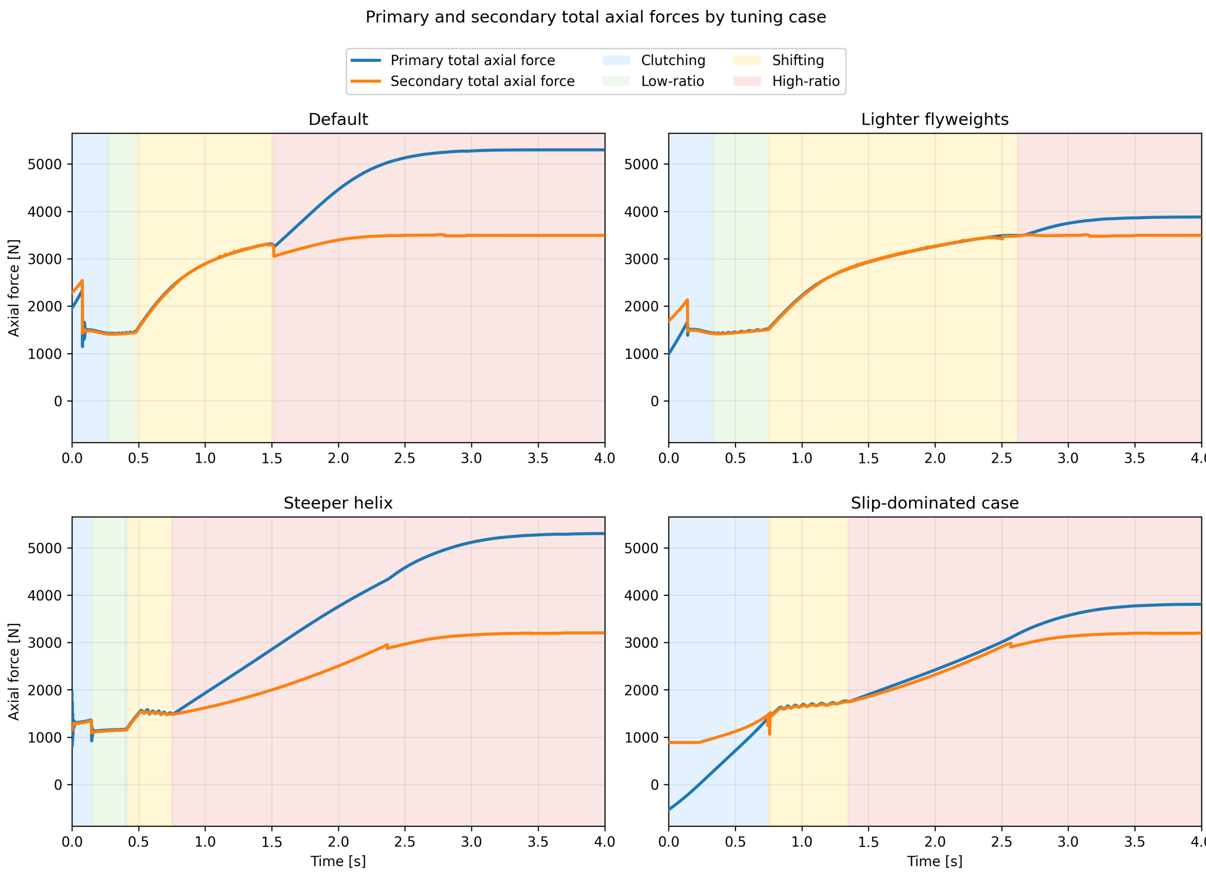


Figure 4.2: Primary and secondary total axial forces during launch for the four tuning cases. Blue curves show total primary axial force, orange curves show total secondary axial force. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as force-balance diagrams for the shift. While the secondary axial force exceeds the primary axial force, the transmission is held toward low ratio. Once the primary force overtakes the secondary, the pulley pair is driven into upshift. The timing of that crossover therefore determines how long the launch remains near low ratio and how early the transmission begins moving through its ratio range.

The **default** case shows the most balanced behaviour. During the initial clutching phase, the secondary force begins noticeably above the primary force, so the transmission is held near its low-ratio condition rather than shifting immediately. This produces the short but clear low-ratio hold seen previously in Figure 4.1. As the launch progresses, the primary axial force rises and eventually overtakes the secondary, initiating the main upshift. Importantly, the secondary force does not remain fixed while this happens. It also rises throughout the shift, so the system continues to resist ratio change even after upshift begins. This is mechanically important: if the secondary force stayed nearly constant while the primary force rose rapidly, the transmission would sweep through the shift range much more abruptly. Instead, both sides grow together, giving the default case its smoother and more controlled shift.

The **lighter-flyweight** case makes this point especially clear. Relative to the default case, the primary force is reduced in the early launch, while the secondary remains comparatively strong.

The initial force gap is therefore larger, meaning the low-ratio condition is held more firmly and for longer. This matches the longer low-ratio segment seen in the shift curves. Later in the launch, the primary still rises enough to drive upshift, but it does so later and at a higher engine-speed level than in the default case. In other words, this case does not fundamentally change the character of the shift; it mainly delays it by weakening the primary closing tendency during engagement and early launch.

The **steeper-helix** case behaves differently. Here, the initial difference between secondary and primary force is very small, and the primary overtakes the secondary much earlier. This explains why the corresponding shift curve showed little sustained low-ratio hold and entered the shifting region so quickly. We also see a differently shaped force evolution throughout the shift itself. In the default and lighter-flyweight cases, both forces continue rising together in a gradual way through the main shift region. In the steeper-helix case, both forces rise sharply early on and then level off much sooner. The two forces still remain relatively close to one another, but the overall shape they trace is different. This helps explain why the earlier shift curve did not display the same flat-shift behaviour as the default case. Rather than maintaining a long, nearly constant-speed shift, the transmission shows a more rising shift trajectory, consistent with this less gradual force evolution.

The **slip-dominated** case is the most extreme. Its early launch is characterized by a prolonged clutching phase and no meaningful low-ratio hold. This is consistent with the earlier shift-curve figure, where the trajectory spent much longer in the slip-dominated region before entering the no-slip shift region. Importantly, the beginning of geometric shift does not imply that the system has already stopped slipping. It only indicates that the shift coordinate has begun to move away from its initial value. The more important feature in this case is the low magnitude of both axial forces in the early launch. Compared with the other cases, neither pulley develops a strong axial loading during the initial portion of the event. This weak overall clamping is a key reason why this case behaves poorly, and it will be seen more directly in the following torque-transfer plots where the associated slip remains elevated for much longer.

One additional feature is visible near the beginning of each case. The early spike in secondary axial force is the torque-reactive contribution of the secondary. It appears most strongly during the initial launch when transmitted torque rises rapidly, and it provides an immediate resisting effect against premature upshift. This feature is especially useful because it shows that the secondary is responding dynamically to transmitted torque rather than acting as a static spring-only restraint. The influence of this torque reactivity will be seen more directly in the following torque-transfer results, but its signature is already visible here in the early force histories.

Taken together, Figure 4.2 provides the mechanical explanation for the differences observed in Figure 4.1. A larger initial gap between secondary and primary force corresponds to stronger low-ratio retention, as seen in the lighter-flyweight case. A very small gap corresponds to premature upshift, as seen in the steeper-helix case. The default case lies between these extremes and therefore produces the most balanced launch. Perhaps most importantly, the figure shows that the shift is not driven by the primary alone. The secondary rises with it throughout the launch, and the actual shift behaviour emerges from the evolving balance between the two rather than from either force in isolation. At the same time, this figure only explains the shift tendency and overall clamping levels; it has not yet quantified the resulting slip behaviour, which is addressed next.

4.1.4 Torque Bounds, No-Slip Demand, and Coupling Torque

The force-balance results of Figure 4.2 explain when each case tends to remain at low ratio and when it begins to shift. However, force balance alone does not determine whether the belt is actually able to transmit the torque required for sticking. To examine that question, Figure 4.3 compares four quantities during the launch: the primary and secondary traction bounds, the no-slip coupling torque τ_{ns}, and the actual transmitted coupling torque τ_c.

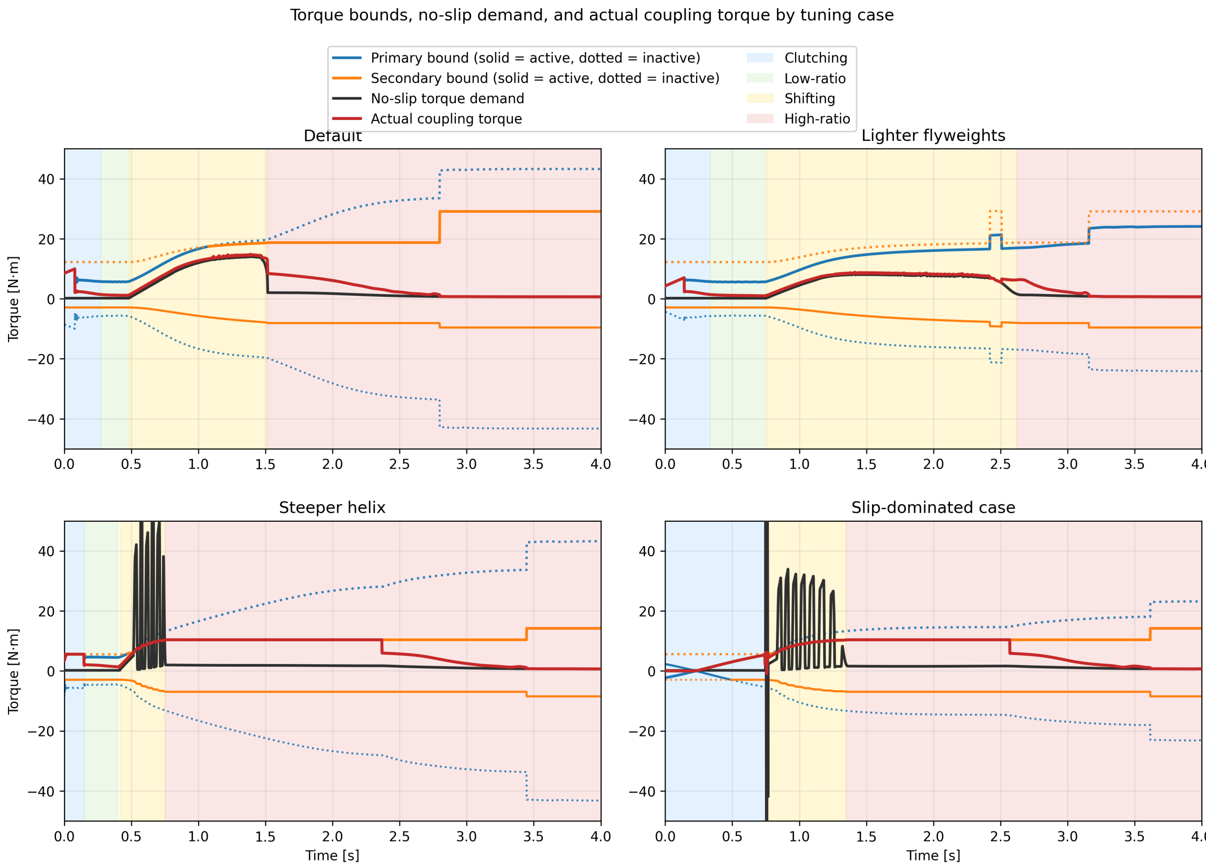


Figure 4.3: Torque bounds, no-slip demand, and actual coupling torque for the four tuning cases. Blue and orange curves show the primary and secondary bounds, with the active bound drawn solid and the inactive bound dotted. The black curve shows the no-slip torque demand τ_{ns}, and the red curve shows the actual transmitted coupling torque τ_c. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as traction-capacity diagrams. At any instant, the belt can only remain in stick if the required no-slip torque lies inside the admissible interval defined by the active upper and lower bounds. When τ_{ns} lies inside that interval, the actual coupling torque can follow the no-slip demand closely. When τ_{ns} exceeds the available bounds, sticking is no longer admissible and the transmitted torque must saturate at the corresponding active limit.

A first broad trend is immediately visible. The cases that settle out of slip early in the launch are precisely the cases with larger admissible positive torque capacity during the initial clutching and low-ratio phases. In the **default** and **lighter-flyweight** cases, the upper torque bound rises to a relatively large value early in the launch. This gives the system enough admissible coupling torque to reduce the belt-speed mismatch and recover toward sticking. By contrast, the **slip-dominated** case begins with essentially no admissible positive torque and only develops usable capacity more gradually. That lower early torque capacity directly explains why the slip-dominated case spends much longer in the clutching region.

This same distinction appears in the evolution of τ_c. In the cases that recover stick early, the actual coupling torque relaxes toward and then tracks the no-slip demand through the end of clutching and into low-ratio operation. Once low-ratio launch is established, the red and black curves remain close, indicating that the required torque stays admissible and sustained slip is not reintroduced immediately.

A second important pattern appears once the transmission begins to shift. In all four cases, the no-slip torque rises during the shift phase. This is consistent with the form of τ_{ns}, in which the ratio-rate term becomes active once the CVT begins moving away from its initial ratio. In the **default** and **lighter-flyweight** cases, this rise remains below the admissible upper torque bound, so the actual coupling torque continues to follow the no-slip demand and large renewed slip does not occur. In the **steeper-helix** and **slip-dominated** cases, however, τ_{ns} rises above the admissible upper bound during the shift. At that point, sticking is no longer feasible, the actual coupling torque saturates at the active maximum, and renewed slip appears until the state returns closer to the sticking condition.

The **steeper-helix** and **slip-dominated** cases also show visibly oscillatory behaviour in the no-slip torque during the shift onset. This is consistent with the oscillations already seen in the preceding axial-force plots. The most direct source is the ratio-rate contribution to τ_{ns}: if the shift dynamics are not smooth, then the implied no-slip torque need not evolve smoothly either. In these two poorer tuning cases, the shift motion appears more weakly damped, so the no-slip torque exhibits stronger excursions during the transition into active shifting. A plausible reason is that the present model neglects parasitic losses and other non-critical frictional effects, which in a real system would provide additional damping and suppress some of this oscillatory behaviour. This is therefore a useful limitation to note: the model captures the overall stick–slip transitions and the active limiting mechanism, but the transient oscillations in the most aggressive cases are likely amplified by the intentionally idealized loss assumptions.

Another common feature is the gradual relaxation of the actual coupling torque toward the no-slip demand rather than an instantaneous snap to it. This comes from the regularized transition law used in the simulation, discussed in Appendix C.3, which replaces a discontinuous switch with a smooth tanh-based approximation. Its purpose is numerical: it reduces chattering and improves stability near the stick–slip transition. The effect is visible here as a slight lag in the recovery of perfect sticking, but it does not change the main qualitative behaviour of the four cases. The same limiting patterns are still present, and the cases that lose admissibility remain clearly distinguishable from those that do not.

Finally, each case shows a noticeable jump in the admissible torque bound once full sticking is re-established near the end of the launch. This reflects the change in limiting behaviour used by the model after slip is removed, together with the corresponding increase in effective friction behaviour assumed in the sticking regime. The important point for the present discussion is that this jump occurs only after the system has already relaxed back toward the no-slip state; it is therefore a consequence of stick recovery rather than its cause.

Taken together, Figure 4.3 closes the interpretation begun in the previous two subsections. The shift curves showed the overall launch behaviour, and the axial-force plots explained the tendency to hold or leave low ratio. The present figure then shows whether the required coupling torque is actually admissible throughout those phases. The default and lighter-flyweight cases maintain sufficient positive torque capacity to recover stick early and remain near the no-slip demand through the main shift. The steeper-helix and slip-dominated cases do not: in both, the no-slip torque exceeds the available upper bound during the shift, forcing the actual coupling torque to saturate and reintroducing slip. In this way, the torque plots both confirm the solid patterns seen earlier and reveal one of the model's clearest shortcomings: the most weakly damped cases display exaggerated oscillatory transients due to the idealized loss assumptions.

4.1.5 Launch Performance Comparison

The preceding subsections established the mechanical differences between the four tuning cases. The shift-curve plots showed the overall launch trajectory, the axial-force plots explained the tendency to hold or leave low ratio, and the torque-bound plots showed whether the required coupling torque remained admissible throughout the launch. The final step is to compare how those differences affect overall launch performance.

Two simple metrics are used here: the time required to travel 50 m, and the time required to reach 60 km/h. These are useful because they summarize the launch in two complementary ways. The 50 m time reflects the integrated quality of the launch as a whole, while the 60 km/h time emphasizes how quickly the drivetrain can build vehicle speed through the main acceleration phase.

Table 4.2: Launch performance metrics for the four tuning cases.

Case	**Time to** 50 m [s]	**Time to** 60 km/h [s]
Default	3.93	2.27
Lighter flyweights	4.26	2.78
Steeper helix	4.19	2.82
Slip-dominated case	4.41	2.98

The default case performs best on both metrics, which is consistent with the earlier mechanism-level results. It develops enough admissible coupling torque to reduce slip early, retains low ratio long enough to launch effectively, and then transitions into a controlled shift without reintroducing large sustained slip. This is the most balanced use of the CVT: the transmission first allows a strong initial torque transfer into the driveline, and then continues accelerating the vehicle while keeping the shift progression smooth and admissible.

This point is worth emphasizing. The coupling torque plays two roles at once. It is the torque that loads the engine through the transmission, limiting how freely the primary (and therefore the engine) can accelerate, and it is also the torque delivered through the CVT that accelerates the vehicle from rest. A strong early coupling torque is therefore beneficial when it is admissible, because it quickly pulls the belt-speed mismatch down and simultaneously applies useful drive torque to the vehicle. The default case benefits from exactly this behaviour: it uses the initial slip region to establish a relatively large transmitted torque early in the launch, which helps "kick-start" vehicle acceleration before the main shift begins.

The lighter-flyweight case performs worse on both metrics, even though its shift remains comparatively well behaved. This matches the earlier interpretation that its main drawback is delayed primary loading during the early launch. By holding low ratio longer but developing less aggressive early coupling, it sacrifices initial acceleration and does not recover that loss later.

The steeper-helix case shows a different trade-off. It leaves low ratio too early and re-enters slip during the shift, which degrades the quality of the launch. However, because it still develops appreciable coupling torque early in the event, its 50 m time remains slightly better than the lighter-flyweight case. Its slower time to 60 km/h, on the other hand, is consistent with the less stable shift behaviour and the renewed slip seen in the torque-bound plots.

The slip-dominated case performs worst overall. This is the expected result from the earlier figures: its early clamping forces are weakest, its admissible torque rises too slowly, and its launch remains slip-dominated for too long. As a result, it spends too much of the start of the event establishing torque transfer rather than converting that torque cleanly into forward acceleration.

Overall, these performance metrics support the mechanism-level interpretation developed throughout the section. The best launch is not produced by simply holding low ratio as long as possible, nor by shifting as early as possible. Instead, good performance requires a balance: sufficient early admissible coupling torque to accelerate the vehicle strongly from rest, enough low-ratio retention to avoid premature upshift, and a smooth enough shift that the required no-slip torque does not repeatedly exceed the available traction limits.

4.2 Implementation and Reproducibility

The source code for the CVT simulator used in this work is available at `https://github.com/gr812b/CVT-Simulator`. This implementation includes the mathematical model developed in the main derivation sections along with the practical numerical regularization strategy described in Appendix C.3.

To support reproducibility, the exact version used to generate the figures and tables in this manuscript will be archived as a tagged release:

- repository release (to be finalized): `https://github.com/gr812b/CVT-Simulator/releases/tag/vX.Y.Z`,
- archival DOI landing page (placeholder): `https://doi.org/XX.XXXX/zenodo.XXXXXXX`.

Unless otherwise stated, simulation cases in Section 4.1 correspond to that archived release. The repository README and release notes describe the execution workflow, configuration files, and plotting scripts used to produce the reported results.

An interactive web deployment of the simulator is also available for demonstration purposes: `https://cvt.kaiarseneau.dev`. This live endpoint is provided for accessibility and does not replace the archived release as the canonical reproducibility artifact.

4.3 Conclusion

This work developed a unified first-principles dynamic model for a mechanically actuated dry rubber V-belt CVT and its coupling to the vehicle drivetrain. Starting from pulley geometry, belt-length closure, actuator mechanics, and rotational dynamics, the derivation produced a closed nonlinear state-space formulation in which ratio evolution, torque transmission, slip admissibility, and vehicle response are solved consistently in time.

The completed formulation brings together several mechanisms that strongly govern mechanically actuated CVT behaviour: primary flyweight actuation, secondary torque-reactive helix clamping, belt geometric closure, axial shift dynamics, traction-limited torque transfer, and longitudinal drivetrain coupling. The main outcome is a single dynamic framework in which these effects interact directly, while remaining mechanically interpretable in terms of the underlying pulley forces, ratio change, and torque limits.

The simulation results showed coherent launch behaviour across the tuning cases considered. The shift-curve results captured distinct differences in engagement, low-ratio retention, main-shift shape, and approach to high ratio. The axial-force results showed that these differences could be traced back to the evolving balance between primary and secondary clamping. The torque-bound results then showed when the no-slip coupling demand remained admissible and when renewed slip was forced by the available traction limits. Together, these results formed a consistent mechanism chain from tuning changes, to force balance, to torque admissibility, to overall launch behaviour.

An important outcome of the study is that the model preserved physical intuition across multiple levels. Changes in tuning did not simply produce different output curves; they produced changes that remained understandable in mechanical terms. Stronger or weaker low-ratio retention appeared through the initial clamping-force gap, earlier or later upshift appeared through the timing of force crossover, and reintroduced slip appeared when the no-slip torque rose beyond the admissible coupling bounds. This consistency gives confidence that the formulation is capturing the dominant interactions it was intended to represent.

The results also highlighted useful limitations of the present model. In the more aggressive cases, the no-slip torque demand exhibited visibly underdamped oscillatory transients during shift onset. These were consistent with irregular shift-rate evolution in the model and suggest that the idealized loss assumptions, particularly the neglect of parasitic losses and other non-critical frictional effects, leave the system with less damping than would be expected in real hardware. The formulation also remains limited to one-dimensional vehicle motion and does not yet include thermal effects, wear progression, or more detailed tribological contact evolution.

Overall, the present work provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behaviour during transient launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behaviour remains traceable back to clear mechanical causes. With further parameter identification, experimental validation, and extension to include more realistic damping and loss mechanisms, the framework developed here can serve as a useful basis for both engineering interpretation and future design-oriented CVT simulation.

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Appendix A

Belt Length Small-Angle Approximation

This appendix quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models. Using parameter values representative of the physical system considered in this work, we demonstrate that the approximation introduces substantial error in the computed center-to-center distance, then further in the ratio, which is critical for accurate dynamic modeling.

A.1 Standard Approximation

Many simplified treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this approximation, the belt-length constraint Equation (3.1.11) reduces to

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{A.1.1}$$

A.2 Derivation of the Approximate Center Distance

To determine the center-to-center distance at the initial configuration, we substitute $r_p = r_{p,\text{outer,min}}$ and $r_s = r_{s,\text{outer,max}}$ into Equation (A.1.1) and solve for C.

Step 1: Isolate the square root. Rearranging Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) = 2\sqrt{C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}. \tag{A.2.1}$$

Step 2: Square both sides.

$$\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2 = 4\left[C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2\right]. \tag{A.2.2}$$

Step 3: Solve for C^2.

$$C^2 = \frac{\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2. \tag{A.2.3}$$

Step 4: Take the square root. Expanding the squared term in the numerator:

$$\left[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\right]^2 = L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2,$$

the approximate center distance becomes

$$\boxed{C_{\text{approx}} = \sqrt{\frac{L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}.} \tag{A.2.4}$$

This closed-form expression avoids the need for numerical root finding but neglects the $\sin^{-1}$ term in the exact constraint.

A.3 Error Analysis

We now evaluate the approximation error using parameter values from the physical CVT system considered in this work.

System Parameters

Parameter	**Value**	**SI Value**
Belt length L_b	37.538 in	0.9535 m
Belt height h_b	0.613 in	0.015 57 m
Groove half-angle β	11.5°	0.2007 rad
Inner primary radius	0.8125 in	0.020 637 5 m
Primary outer radius $r_{p,\text{outer,min}}$	1.4255 in	0.036 21 m
Secondary outer radius $r_{s,\text{outer,max}}$	4.0 in	0.1016 m

The primary outer radius is computed as the inner radius plus belt height: $r_{p,\text{outer,min}} = 0.8125 + 0.613 = 1.4255$ in.

Computed Center Distances Solving the exact belt-length constraint Equation (3.1.13) numerically yields

$$C_{\text{exact}} = 0.251\,74\,\text{m} \approx 9.911\,\text{in}.$$

Evaluating the approximate formula Equation (A.2.4) yields

$$C_{\text{approx}} = 0.268\,37\,\text{m} \approx 10.566\,\text{in}.$$

Error Metrics The absolute and relative errors are:

$$\Delta C = C_{\text{approx}} - C_{\text{exact}} = 0.016\,63\,\text{m} = 0.655\,\text{in}, \tag{A.3.1}$$

$$\varepsilon_{\text{rel}} = \frac{\Delta C}{C_{\text{exact}}} = \frac{0.01663}{0.25174} = 6.61\%. \tag{A.3.2}$$

Assessment The small-angle approximation introduces a relative error of nearly 7% in the center-to-center distance for this system. This error arises because the ratio

$$\frac{r_{s,\text{outer,max}} - r_{p,\text{outer,min}}}{C} = \frac{0.1016 - 0.03621}{0.25174} = 0.260$$

is not small. The assumption $\lambda \approx 0$ is therefore not justified for this CVT geometry.

Since the center distance C propagates into all subsequent geometric calculations—wrap angles, tangent span lengths, and derived quantities such as the transmission ratio—a 7% error in C can compound into larger errors in predicted system behavior.

A.4 Propagation to Secondary Radius and Transmission Ratio

To demonstrate how the center distance error propagates into operationally relevant quantities, we now examine the secondary radius and CVT ratio at two shift positions: zero shift and maximum shift.

A.4.1 Derivation of the Approximate Secondary Radius

Given a known primary outer radius $r_{p,\text{outer}}$ and center distance C, the secondary outer radius can be obtained by solving the belt-length constraint. Under the small-angle approximation Equation (A.1.1), this yields a closed-form expression.

Step 1: Isolate the square root. From Equation (A.1.1):

$$L_b - \pi(r_{p,\text{outer}} + r_{s,\text{outer}}) = 2\sqrt{C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2}. \tag{A.4.1}$$

Step 2: Square both sides.

$$\left[L_b - \pi r_{p,\text{outer}} - \pi r_{s,\text{outer}}\right]^2 = 4\left[C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2\right]. \tag{A.4.2}$$

Step 3: Expand and collect terms. Expanding the left side:

$$(L_b - \pi r_{p,\text{outer}})^2 - 2\pi(L_b - \pi r_{p,\text{outer}})r_{s,\text{outer}} + \pi^2 r_{s,\text{outer}}^2.$$

Expanding the right side:

$$4C^2 - 4r_{s,\text{outer}}^2 + 8r_{p,\text{outer}}r_{s,\text{outer}} - 4r_{p,\text{outer}}^2.$$

Collecting powers of $r_{s,\text{outer}}$ yields a quadratic equation:

$$(\pi^2+4)r_{s,\text{outer}}^2 - 2\left[\pi L_b - \pi^2 r_{p,\text{outer}} + 4r_{p,\text{outer}}\right]r_{s,\text{outer}} + \left[(L_b - \pi r_{p,\text{outer}})^2 + 4r_{p,\text{outer}}^2 - 4C^2\right] = 0. \tag{A.4.3}$$

Step 4: Apply the quadratic formula. The physically meaningful root (smaller secondary radius for overdrive configuration) is

$$\boxed{r_{s,\text{approx}} = \frac{\pi L_b + (4-\pi^2)r_{p,\text{outer}} - 2\sqrt{(\pi^2+4)C^2 - L_b^2 + 4\pi L_b r_{p,\text{outer}} - 4\pi^2 r_{p,\text{outer}}^2}}{\pi^2+4}.} \tag{A.4.4}$$

This closed-form expression enables direct computation of the approximate secondary radius without numerical root finding.

A.4.2 Zero Shift Configuration

At zero shift, the CVT is in its lowest ratio (maximum reduction) configuration. This is also the geometry used to compute the approximate center distance C_{approx}, so much of that initial center-distance error cancels when the same configuration is used again to recover the secondary radius.

Primary Radius at Zero Shift At $s = 0$, the primary is within the deadzone, so from Equation (3.1.3):

$$r_{p,\text{outer}}(0) = r_{p,\text{outer,min}} = 0.036\,21\,\text{m}.$$

Secondary Radius Comparison Solving the exact belt-length constraint with C_{exact} yields

$$r_{s,\text{exact}} = 0.101\,60\,\text{m}.$$

Evaluating the approximate formula Equation (A.4.4) with $C_{\text{approx}} = 0.268\,37\,\text{m}$ yields

$$r_{s,\text{approx}} = 0.101\,60\,\text{m}.$$

Transmission Ratio Error Computing effective radii with $h_b/2 = 0.007\,79\,\text{m}$:

Quantity	Exact	Approximate
$r_{p,\text{eff}}$	0.0284 m	0.0284 m
$r_{s,\text{eff}}$	0.0938 m	0.0938 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	3.300	3.300

The ratio error at zero shift is

$$\varepsilon_R = \frac{|3.300 - 3.300|}{3.300} \approx 0.000\%.$$

Thus, although C_{approx} is already 6.61% too large, the induced error has little visible effect at minimum shift because the radius calculation is being evaluated at the same operating point used to construct that approximate center distance in the first place.

Figure A.1: Scaled diagram of the CVT at zero shift ($s = 0$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The two solutions nearly overlap at this operating point, indicating negligible ratio error in this case.

A.4.3 Maximum Shift Configuration

At maximum shift, the primary and secondary pulleys are closest in size, representing a near-crossover configuration.

Primary Radius at Maximum Shift Using Equation (3.1.3) with $s = s_{\max} = 0.75\,\text{in} = 0.019\,05\,\text{m}$ and deadzone $s_{\text{dz}} = 0.002\,489\,2\,\text{m}$:

$$\begin{aligned}
r_{p,\text{outer}}(s_{\max}) &= r_{p,\text{outer,min}} + \frac{s_{\max} - s_{\text{dz}}}{2\tan\beta} \\
&= 0.03621 + \frac{0.01905 - 0.0024892}{2 \times \tan(11.5°)} \\
&= 0.03621 + \frac{0.01656}{0.4070} \\
&= 0.0769\,\text{m}.
\end{aligned} \tag{A.4.5}$$

Secondary Radius Comparison Solving the exact belt-length constraint Equation (3.1.13) numerically with $C_{\text{exact}} = 0.251\,74\,\text{m}$ yields

$$r_{s,\text{exact}} = 0.066\,73\,\text{m}.$$

Evaluating the approximate formula Equation (A.4.4) with $C_{\text{approx}} = 0.268\,37\,\text{m}$ yields

$$r_{s,\text{approx}} = 0.056\,26\,\text{m}.$$

Transmission Ratio Error The CVT ratio is defined using effective radii, which account for belt seating depth (Equation (3.1.23)):

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}.$$

With $h_b/2 = 0.007\,79\,\text{m}$:

Quantity	**Exact**	**Approximate**
$r_{p,\text{eff}}$	0.0691 m	0.0691 m
$r_{s,\text{eff}}$	0.0589 m	0.0485 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	0.853	0.701

Both models predict overdrive ($R < 1$), but the approximation substantially underestimates the ratio. The ratio error is

$$\varepsilon_R = \frac{|0.701 - 0.853|}{0.853} = 17.8\%.$$

Unlike the minimum-shift case, this operating point does not benefit from the same cancellation, so the center-distance error propagates into a much larger error in secondary radius and transmission ratio.

Figure A.2: Scaled diagram of the CVT at maximum shift ($s = s_{\max}$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The approximation underestimates the secondary radius, leading to a lower predicted ratio than the exact model.

A.5 Summary of Approximation Errors

Configuration	R_{exact}	R_{approx}	**Ratio Error**
Zero shift ($s = 0$)	3.300	3.300	0.000%
Maximum shift ($s = s_{\max}$)	0.853	0.701	17.8%

The small-angle approximation error is strongly operating-point dependent. At zero shift, most of the center-distance error cancels because the approximate center distance was itself constructed from that same geometry. At maximum shift, that cancellation is lost and the ratio error becomes substantial.

These errors arise from the initial 6.61% error in center distance, which propagates nonlinearly through the belt-length geometry in a state-dependent way. Since the transmission ratio directly enters the rotational equations of motion (Section 3.6), this approximation can still produce materially inaccurate dynamics near maximum shift.

This analysis justifies the use of exact numerical solutions for all geometric computations in this work.

Appendix B

Common Ramp Profile Designs

This appendix presents common ramp profile parameterizations for CVT primary pulley actuation. From Equation (3.2.8), the centrifugal axial force is

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds}, \tag{B.0.1}$$

where m_f is the flyweight mass, $r_{f,0}$ is the initial flyweight radius, and $r_f(s)$ is the radial displacement along the ramp as a function of axial shift s.

The profile $r_f(s)$ must satisfy the admissibility conditions from Section 3.2: continuously differentiable, non-negative slope, and finite slope throughout $[0, s_{\max}]$.

B.1 Linear Ramp

A linear ramp with slope k_r (radial displacement per unit axial shift):

$$r_f(s) = k_r \cdot s, \qquad \frac{dr_f}{ds} = k_r = \text{const.} \tag{B.1.1}$$

Substituting into Equation (B.0.1):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 k_r (r_{f,0} + k_r s) = m_f \omega_p^2 k_r r_{f,0} + m_f \omega_p^2 k_r^2 s. \tag{B.1.2}$$

At constant ω_p, axial force increases linearly with s because the growing radius is multiplied by a constant slope. Simple to manufacture, but may cause aggressive shifting at high ratio.

B.2 Circular Arc Ramp

A circular arc ramp is designed to reduce force variation across the shift range by achieving *partial* cancellation. The key insight is to parameterize the arc by desired slope angles α_1 and α_2 at the start and end of the segment, where $|dr_f/ds| = \tan\alpha$ specifies the force multiplier. This provides direct control over actuation behavior at engagement (high slope) versus overdrive (low slope).

For a circle of radius R_c centered at (s_c, r_c), the ramp profile satisfies:

$$(s - s_c)^2 + (r_f - r_c)^2 = R_c^2. \tag{B.2.1}$$

Solving for $r_f(s)$ on the upper arc:

$$r_f(s) = r_c + \sqrt{R_c^2 - (s - s_c)^2}. \tag{B.2.2}$$

Implicit differentiation yields:

$$\frac{dr_f}{ds} = \frac{-(s - s_c)}{\sqrt{R_c^2 - (s - s_c)^2}} = \frac{-(s - s_c)}{r_f - r_c}. \tag{B.2.3}$$

Parameterization from slope angles. At any point on the arc where the local slope angle is α (i.e. $dr_f/ds = \tan\alpha$), the circle constraint combined with Equation (B.2.3) gives the parametric form:

$$s = s_c - R_c \sin\alpha, \qquad\qquad r_f = r_c + R_c \cos\alpha. \tag{B.2.4}$$

The slope angle α decreases monotonically from α_1 (entry) to $\alpha_2 < \alpha_1$ (exit) as s increases along the ramp. Applying Equation (B.2.4) at both boundary shifts s_1 and s_2 and subtracting:

$$s_2 - s_1 = R_c(\sin\alpha_1 - \sin\alpha_2). \tag{B.2.5}$$

With $\Delta s = s_2 - s_1$ as the total axial span of the ramp segment, the arc radius and center are determined entirely by the two slope angles:

$$R_c = \frac{\Delta s}{\sin\alpha_1 - \sin\alpha_2}, \qquad s_c = s_1 + R_c \sin\alpha_1. \tag{B.2.6}$$

The boundary condition $r_f(s_1) = 0$ (flyweights at their initial radius at ramp entry) then fixes the radial center:

$$r_c = -R_c \cos\alpha_1. \tag{B.2.7}$$

Substituting into Equation (B.0.1):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f) \cdot \frac{-(s - s_c)}{r_f - r_c}. \tag{B.2.8}$$

To see the cancellation explicitly, split $r_{f,0} + r_f = (r_{f,0} + r_c) + (r_f - r_c)$:

$$\begin{aligned} F_{p,\text{cent}} &= m_f \omega_p^2 \left[\underbrace{(r_f - r_c)}_{\text{cancels}} \cdot \frac{-(s - s_c)}{r_f - r_c} + (r_{f,0} + r_c) \cdot \frac{-(s - s_c)}{r_f - r_c} \right] \\ &= -m_f \omega_p^2 \, (s - s_c) \left[1 + \frac{r_{f,0} + r_c}{r_f - r_c} \right]. \end{aligned} \tag{B.2.9}$$

This form reveals the structure clearly. The dominant factor is $-(s - s_c)$, which is linear in shift: the $(r_f - r_c)$ terms cancel exactly in the first part of the split. The bracket $\left[1 + \frac{r_{f,0} + r_c}{r_f - r_c} \right]$ is a nonlinear correction multiplier. Using the parametric form Equation (B.2.4) with Equation (B.2.7), the bracket evaluates to

$$1 + \frac{r_{f,0} + r_c}{r_f - r_c} = 1 + \frac{r_{f,0} - R_c \cos\alpha_1}{R_c \cos\alpha}, \tag{B.2.10}$$

where α is the local slope angle at shift s, sweeping from α_1 at entry to α_2 at exit. This makes the effective proportionality between force and $(s - s_c)$ position-dependent.

The numerator $r_{f,0} - R_c \cos\alpha_1$ is the key residual: it is controlled entirely by the choice of α_1, α_2, and Δs through $R_c = \Delta s/(\sin\alpha_1 - \sin\alpha_2)$. Increasing either α_1 or Δs raises R_c, making $R_c \cos\alpha_1$ larger and the residual smaller (or negative). The entry force scale, the exit force scale, and the degree of position-dependence between them are therefore all coupled to the same two angle choices. There is no freedom to set the overall force scale independently of the nonlinearity — changing α_1 and α_2 adjusts both simultaneously.

The condition $r_{f,0} = R_c \cos\alpha_1$ makes the residual zero, so the bracket equals 1 everywhere and force is exactly proportional to shift. This is the angle configuration at which the circular arc best approximates the hyperbolic ideal — it gives a practical design rule for tuning.

The practical consequence is important: at a fixed engine speed ω_p, the centrifugal force is approximately proportional to $(s - s_c)$, growing linearly with shift. This directly matches the linear restoring force of the primary spring (force $\propto k_s \cdot s$). A spring-loaded primary can therefore be tuned to balance the centrifugal flyweights at a target shift, and the linear character of the circular arc ensures the equilibrium is stable across the shift range rather than concentrated at a single point.

Circular arcs are widely used in practice due to ease of manufacture and sufficient compensation. The nonlinear residual is small when the ramp center offset $(r_{f,0} + r_c)$ is small relative to the arc radius R_c, but it cannot be eliminated entirely. The following section shows how to *exactly* achieve the linear target by choosing a different ramp geometry.

B.3 Hyperbolic Ramp (Perfect Cancellation)

From Equation (B.2.10), the circular arc force-to-shift proportionality constant is:

$$\frac{F_{p,\text{cent}}}{-m_f \omega_p^2 (s - s_c)} = 1 + \frac{r_{f,0} - R_c \cos\alpha_1}{R_c \cos\alpha}. \tag{B.3.1}$$

This varies with position because α changes along the arc. The natural question is: what ramp profile makes this ratio a *true constant*? That constant is what we call K, and the profile that achieves it satisfies:

$$(r_{f,0} + r_f(s))\frac{dr_f}{ds} = Ks, \tag{B.3.2}$$

where $K > 0$. This produces centrifugal force exactly proportional to shift:

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 K s. \tag{B.3.3}$$

K is not a free parameter any more than R_c is for the circular arc: just as R_c is uniquely determined by α_1, α_2, and Δs (Equation (B.2.6)), K is uniquely determined by α_1, α_2, s_1, and s_2 (derived in the parameterization below). What differs is not the design inputs, but that the hyperbolic profile makes K exactly constant along the entire ramp, whereas the circular arc's equivalent ratio changes with position.

Deriving the profile. Equation (B.3.2) is separable. Substituting $u = r_{f,0} + r_f$, so $du/ds = dr_f/ds$:

$$u\,du = Ks\,ds. \tag{B.3.4}$$

Integrating both sides and writing the constant of integration as $\frac{1}{2}Ks_0^2$:

$$\tfrac{1}{2}u^2 = \tfrac{1}{2}K(s^2 + s_0^2). \tag{B.3.5}$$

Substituting back $u = r_{f,0} + r_f$:

$$\boxed{(r_{f,0} + r_f(s))^2 = K(s^2 + s_0^2),} \tag{B.3.6}$$

giving the explicit profile:

$$r_f(s) = \sqrt{K(s^2 + s_0^2)} - r_{f,0}. \tag{B.3.7}$$

In the (s, r_f) plane, the locus $(r_{f,0} + r_f)^2 = Ks^2$ (taking $s_0 = 0$) is a branch of a *hyperbola* — hence the name. Unlike the circular arc, where the residual multiplier in Equation (B.2.9) could not be removed, this profile forces the product $(r_{f,0} + r_f) \cdot dr_f/ds = Ks$ exactly for all s.

Local slope and its variation. To see how K relates to actual ramp angles, differentiate Equation (B.3.6) implicitly:

$$\frac{dr_f}{ds} = \frac{Ks}{r_{f,0} + r_f} = \frac{\sqrt{K}\, s}{\sqrt{s^2 + s_0^2}}. \tag{B.3.8}$$

Since $dr_f/ds = \tan\alpha(s)$ by definition of the ramp slope angle, the local angle satisfies:

$$\tan\alpha(s) = \frac{\sqrt{K}\, s}{\sqrt{s^2 + s_0^2}}. \tag{B.3.9}$$

Two observations follow. First, the slope is zero at $s = 0$ and grows monotonically toward the asymptote $\sqrt{K}$ as $s \to \infty$. This is the opposite trend from a circular arc, where the slope decreases from entry to exit. The reason is that the growing lever arm $(r_{f,0} + r_f)$ at large shift would otherwise amplify force super-linearly; a rising slope exactly compensates to keep the product Ks linear. Second, the asymptotic slope $\lim_{s\to\infty} \tan\alpha(s) = \sqrt{K}$ gives a simple interpretation: $K = \tan^2\alpha_\infty$ where α_∞ is the slope the profile approaches at large shift.

Parameterization by entry and exit angles. Following the same convention as Appendix B.2, specify desired slope angles α_1 and α_2 at entry shift s_1 and exit shift s_2. From Equation (B.3.9):

$$\tan^2\alpha_1 = \frac{Ks_1^2}{s_1^2 + s_0^2}, \qquad \tan^2\alpha_2 = \frac{Ks_2^2}{s_2^2 + s_0^2}. \tag{B.3.10}$$

Dividing the two equations eliminates K:

$$\frac{\tan^2\alpha_1}{\tan^2\alpha_2} = \frac{s_1^2\,(s_2^2 + s_0^2)}{s_2^2\,(s_1^2 + s_0^2)}. \tag{B.3.11}$$

Solving for s_0^2:

$$s_0^2 = \frac{s_1^2 s_2^2\,(\tan^2\alpha_2 - \tan^2\alpha_1)}{\tan^2\alpha_1\, s_2^2 - \tan^2\alpha_2\, s_1^2}. \tag{B.3.12}$$

Then K follows from either condition in Equation (B.3.10):

$$K = \frac{\tan^2\alpha_1 \, (s_1^2 + s_0^2)}{s_1^2}. \tag{B.3.13}$$

For $s_0^2 > 0$ a consistent solution requires the numerator and denominator of Equation (B.3.12) to have the same sign. Since the hyperbolic slope increases with s (Eq. (B.3.8)), the physically consistent case is $\alpha_2 > \alpha_1$ (exit angle larger than entry), with $\tan^2\alpha_1 s_2^2 > \tan^2\alpha_2 s_1^2$ ensuring a positive denominator. No solution exists when the two angle conditions give a degenerate system ($\tan^2\alpha_1 s_2^2 = \tan^2\alpha_2 s_1^2$).

Practical impact. In most cases the difference between circular and hyperbolic profiles is negligible. For the specific ramps implemented in this work, the integrated force deviation across the full shift range is less than 1%. The hyperbolic derivation is nonetheless useful: it establishes the theoretical ideal and shows that the circular arc's partial cancellation is not a fundamental limitation but rather a small correction from the constant offset $(r_{f,0} + r_c)$.

Limitations. Both profiles address only the centrifugal force term. The secondary pulley torque feedback ($F_{s,\mathrm{ax}}(s, \tau_s)$ in Equation (3.6.11)) introduces load-dependent nonlinear terms that no ramp geometry alone can cancel. Circular arcs therefore remain the practical standard—simpler to manufacture, well-understood, and sufficient given these irreducible secondary-side effects.

B.4 Piecewise Ramps and Smoothing

Any of the profiles above can be combined in a piecewise fashion: multiple linear segments, multiple circular arcs with different angle pairs, or a mix of profile types joined at transition shifts $s_1, s_2, \ldots, s_{n-1}$. This allows region-specific tuning — steeper slopes at engagement, moderate for cruise, gentler near overdrive — without committing to a single global geometry.

Admissibility at junctions. The admissibility conditions from Section 3.2 require $r_f(s)$ to be continuously differentiable (C^1). A piecewise ramp satisfies C^0 by construction (the profile value is continuous if segments are joined at a common point), but slope continuity may be violated: if the left-segment slope at s_k differs from the right-segment slope, Eq. (B.0.1) gives a finite force jump at that point. No kinematic singularity occurs, but the force has a step discontinuity.

Numerical stiffness considerations. A sharp slope discontinuity is not the same as a smooth rapid transition: many real implementations use a finite but steep blend region, which the ODE solver sees as an extremely large second derivative $d^2 r_f/ds^2$ near the junction. This can create a locally stiff region and force an adaptive integrator to take very small steps near the knot, even if the discontinuity is physically mild. If simulation performance is a concern, slope transitions should be blended — a short circular arc segment tangent to both adjoining profiles is sufficient, as it restores C^1 continuity and bounds $d^2 r_f/ds^2$ throughout the ramp.

Appendix C

Numerical Methods Summary

This section documents the numerical algorithms used for belt-geometry root finding and dynamic simulation, including the specific tolerances, bracketing strategies, and solver configurations employed in the implementation.

C.1 Root Finding for Belt-Length Constraint

Given a known primary outer radius r_1 and fixed center-to-center distance C, the secondary outer radius r_2 is the unique root of the belt-length constraint. This scalar root problem is obtained by rearranging the exact belt length closure from Equation (3.1.13) into the residual form:

$$g(r_2;\, r_1) = \pi(r_1 + r_2) + 2(r_2 - r_1)\arcsin\left(\frac{r_2 - r_1}{C}\right) + 2\sqrt{C^2 - (r_2 - r_1)^2} - L_b = 0, \qquad \text{(C.1.1)}$$

where L_b is the fixed belt length. A separate invocation solves for the center-to-center distance C itself at initialization, using the same equation evaluated at the boundary radii $(r_{1,\min},\, r_{2,\max})$ with C as the unknown; this is exactly the initialization step implied by Equation (3.1.13) before the ratio law Equation (3.6.2) can be evaluated over the shift range.

Bracketing strategy. The arcsin term in Equation (C.1.1) is real only when $|r_2 - r_1| \leq C$, so the natural domain for r_2 is the open interval $(r_1 - C,\, r_1 + C)$. Within this interval a sign change is guaranteed: at the lower boundary $r_2 \to r_1 - C$, the arcsin saturates to $-\pi/2$ and the square root vanishes, giving $g \to 2\pi r_1 - 2\pi C - L_b < 0$ for all physically valid parameters. At the upper boundary $r_2 \to r_1 + C$, by symmetry $g \to 2\pi r_1 + 2\pi C - L_b > 0$, satisfied whenever the belt length is less than the circumference subtended by the wider pulley pair — a requirement any real CVT must satisfy. A small safety margin $\varepsilon = 10^{-9}$ m is applied at both ends to keep arcsin away from its domain boundary where the function becomes numerically steep.

For center-to-center distance solving, the bracket is instead the interval $(\Delta r + \varepsilon,\ L_b/2)$, where $\Delta r = |r_{2,\text{eff}} - r_{1,\text{eff}}|$ is the effective-radius difference at the boundary configuration. The lower bound ensures arcsin is defined; the upper bound is a conservative geometric limit (a straight two-span belt cannot exceed L_b across a gap of $L_b/2$). Both brackets are validated by an explicit sign-change check before invoking the solver, so any failure to bracket immediately identifies an infeasible geometry rather than producing a silent wrong answer.

Brent's method. Both roots are found using `scipy.optimize.brentq`, an implementation of Brent's algorithm Brent (1973). The method combines three strategies: bisection (guaranteed bracket-halving), secant interpolation (superlinear convergence near smooth roots), and inverse quadratic interpolation (order-≈ 1.84 convergence when three previous iterates are well-conditioned). Whenever the faster interpolation steps would place the next iterate outside the current bracket, the algorithm falls back to bisection, preserving the worst-case $O(\log_2(1/\varepsilon))$ guarantee of pure bisection. Because the sign-change brackets described above are guaranteed for all valid CVT geometries, `brentq` is certain to converge. Both root solves use an absolute tolerance of 10^{-9} m, well below any physically meaningful length precision.

C.2 ODE Integration for Dynamic Simulation

The full CVT state vector $\mathbf{x} = (\omega_p, \omega_s, s, \dot{s}, v_b)$ is integrated using `scipy.integrate.solve_ivp`. The governing system is written as a first-order ODE system in Equation (3.6.13), with rotational sub-equations Equation (3.6.8) and Equation (3.6.9), axial dynamics Equation (3.6.11), belt-speed dynamics for v_b, and algebraic geometry closures Equation (3.6.2) and Equation (3.6.3). Although Equation (3.6.11) is second-order in s, introducing $(s, \dot{s})$ as separate states converts the complete model to first-order form for time integration. The time integrator is used with the `RK45` method — the Dormand–Prince explicit embedded Runge–Kutta pair Dormand and Prince (1980). The method propagates the state with a 5th-order formula and simultaneously computes a 4th-order companion estimate for local error control, using the same six function evaluations. The tableau is FSAL (first-same-as-last): the final stage of each accepted step is reused as the first stage of the next, so only five new evaluations are needed per step in steady operation.

Adaptive step size control. At each candidate step the per-component local error estimate $\hat{e}_i$ is compared to a mixed tolerance threshold. A step is accepted when

$$\max_i \frac{|\hat{e}_i|}{\mathrm{atol} + \mathrm{rtol} \cdot |y_i|} \leq 1, \tag{C.2.1}$$

and the next step size is scaled proportionally to $(\mathrm{tol}/\hat{e})^{1/5}$. The tolerances used in this work are $\mathrm{atol} = 10^{-6}$ and $\mathrm{rtol} = 10^{-4}$ (Table C.1). The absolute floor prevents spurious step rejection when a state variable passes through zero (e.g. $\dot{s} \approx 0$ at shift equilibrium or $\omega_p \approx 0$ at engine start), where a purely relative criterion would demand unreachable accuracy.

Parameter	**Value**	**Interpretation**
`atol`	10^{-6}	absolute floor: 1 µm on s, 1 µrad/s on ω
`rtol`	10^{-4}	relative tolerance: 0.01% of current state magnitude
`t_eval`	10 000 pts	uniform output grid over 30 s

Table C.1: ODE solver tolerance parameters.

Event detection. Phase transitions (shift engagement, boundary contact, back-shift onset) are located using `solve_ivp`'s event mechanism. These events correspond to qualitative changes in the dynamics around the bounded shift coordinate used in Equation (3.6.11) and the full state evolution in Equation (3.6.13). After each accepted step the solver evaluates the event functions on a degree-4 dense-output interpolant and applies a root-finding pass to detect any sign change within the step. When a crossing is found it is bracketed at sub-tolerance precision before being reported, so transitions are located with the same accuracy as the integration tolerance rather than being snapped to the coarse output grid.

Phase splitting at hard constraints. The shift bounds $s \in [0, s_{\max}]$ with $\dot{s} = 0$ at both endpoints are handled by partitioning the trajectory into separate `solve_ivp` calls rather than imposing inequality constraints inside the ODE. Each phase — normal shifting, clamped-at-maximum, and back-shifting — presents a smooth right-hand side to the integrator, allowing RK45 to operate at full 5th-order accuracy throughout. The terminal event of each phase supplies the initial condition of the next. This is numerically cleaner than directly clamping Equation (3.6.11) inside a single solve because it preserves a smooth right-hand side within each phase. Up to ten alternating full-shift / back-shift cycles are supported before the simulation terminates, preventing unbounded phase loops for unusual parameter regimes.

C.3 Slip-Law Regularization and Blend Parameter Selection

The traction law in Equation (3.5.9) is physically clear but discontinuous at $v_{\text{rel}} = 0$, which can induce step-size chatter near stick–slip transitions. A practical regularized formulation uses three ideas: a smooth saturation law, a clamped no-slip torque request, and a blending rule between those two limits.

First define a smooth saturation torque

$$\tau_{\text{sat}} = \tau_{\max} \tanh\left(\frac{v_{\text{rel}}}{v_{\text{smooth}}}\right), \tag{C.3.1}$$

where $v_{\text{smooth}} > 0$ sets the transition width in slip-speed units. For $|v_{\text{rel}}| \gg v_{\text{smooth}}$, this approaches the usual friction-limited value $\pm\tau_{\max}$, while near zero it removes the sharp sign discontinuity.

Next, clamp the no-slip torque request:

$$\tau_{\text{req}} = \text{clip}(\tau_{\text{ns}},\, -\tau_{\max},\, \tau_{\max})\,. \tag{C.3.2}$$

This is the largest admissible torque consistent with the no-slip request, without allowing the requested value to exceed traction capacity.

Then define the blend factor

$$\alpha = \text{clip}\left(\frac{|v_{\text{rel}}|}{v_{\text{blend}}},\, 0,\, 1\right), \tag{C.3.3}$$

with $v_{\text{blend}} > 0$ defining the width of the blending region, and use the transmitted torque

$$\tau_c = (1 - \alpha)\,\tau_{\text{req}} + \alpha\,\tau_{\text{sat}}. \tag{C.3.4}$$

This has a useful interpretation. For very small mismatch, $\alpha \approx 0$, so the model stays close to the clamped no-slip demand. For larger mismatch, $\alpha \to 1$, so the law transitions to the smooth Coulomb limit. In other words, the regularized model behaves like a requested no-slip torque near stick and like a friction-limited sliding law once the mismatch becomes appreciable.

How to choose v_{smooth}. Choose the smoothing scale small enough that $\tanh(v_{\text{rel}}/v_{\text{smooth}})$ is close to a sign law through most of the slip regime, but not so small that the solver again sees an effectively discontinuous transition. A practical workflow is:

1. Start with v_{smooth} around 0.5–2% of a representative relative-speed magnitude seen during actual slip.
2. Decrease it until predicted macro outputs (ratio trajectory, launch time, peak speeds) stop changing materially.
3. Increase it slightly if event localization or adaptive step-size behavior becomes noisy near $v_{\text{rel}} \approx 0$.

The blend scale v_{blend} should generally be chosen greater than or equal to v_{smooth}, so that the handoff from clamped demand to the smooth Coulomb limit is not sharper than the smoothing law itself.